

REGENERATIVE MEDICINE MARKET

GLOBAL FORECAST TO 2027

BY PRODUCT (CELL THERAPY [AUTOLOGOUS, ALLOGENEIC], GENE THERAPY, TISSUE
ENGINEERING [SCAFFOLDS, HYDROGELS]), APPLICATION (MUSCULOSKELETAL,
ONCOLOGY, CARDIOVASCULAR, OPHTHALMOLOGY, WOUND CARE), AND REGION

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
APAC	Asia Pacific
ARM	Alliance for Regenerative Medicine
ASCB	American Society for Cell Biology
ASGCT	American Society of Gene & Cell Therapy
BLA	Biologics License Application
CCRM	Centre for Commercialization of Regenerative Medicine
CDC	Centers for Disease Control and Prevention
cDGS	Congenital Disorder of Glycosylation Syndrome
CEO	Chief Executive Office
CIRM	California Institute of Regenerative Medicine
CNS	Central Nervous System
COVID-19	Coronavirus Disease
CROs	Contract Research Organizations
CVD	Cardiovascular Disorders
ESBB	European, Middle Eastern & African Society for Biopreservation & Biobanking
ESGCT	European Society of Gene & Cell Therapy
FDA	US Food and Drug Administration
HFEA	Human Fertilization and Embryology Authority
HTA	Human Tissue Authority
ICRO	International Cell Research Organization
IDF	International Diabetes Federation
IPO	Initial Public Offering
ISBER	International Society for Biological and Environmental Repositories
ISCT	International Society for Cellular Therapy
ISSCR	International Society for Stem Cell Research
M&A	Mergers and Acquisitions
MHRA	Medicines and Healthcare Products Regulatory Agency
MN	Million

NCBI	National Center for Biotechnology Information
NIH	National Institutes of Health
OCED	Organization for Economic Co-operation and Development
PDUA	Prescription Drug Free Act
R&D	Research & Development
R/R	Relapsed / Refractory
RMAT	Regenerative Medicine Advanced Therapy
RoAPAC	Rest of Asia Pacific
RoE	Rest of Europe
RoW	Rest of the World
SFLF	Stem for Life Foundation
SMEs	Subject-matter Experts
SSI	Surgical Site Infections
UK	United Kingdom
UKCBB	UK Cord Blood Bank, Ltd.
US	United States
USD	US Dollar
VP	Vice President
WHO	World Health Organization

1 INTRODUCTION

1.1 STUDY OBJECTIVES

- To define, describe, and forecast the regenerative medicine market based on product, application, and region
- To provide detailed information regarding the major factors influencing the market growth (such as drivers, restraints, challenges, and opportunities)
- To strategically analyze micromarkets¹ with respect to individual growth trends, prospects, and contributions to the overall regenerative medicine market
- To analyze market opportunities for stakeholders and provide details of the competitive landscape for key players
- To forecast the size of the market segments concerning five regions: North America, Europe, Asia Pacific, Latin America, and Middle East & Africa.
- To profile the key players and comprehensively analyze their product portfolios, market positions, and core competencies²
- To track and analyze competitive developments such as product launches, partnerships, expansions, collaborations, and acquisitions in the regenerative medicine market

1.2 MARKET DEFINITION

Regenerative medicine is defined as the process of regeneration or replacement of human tissues or cells to restore normal function. Regenerative medicine combines various biomedical approaches and clinical therapies involving gene therapy, cell therapy, and tissue engineering. Some examples of regenerative medicine include:

- Cell Therapies: The injection of stem cells or progenitor cells
- Immunomodulation Therapy: Regeneration by biologically active molecules administered alone or as secretions by infused cells
- Tissue Engineering: Transplantation of laboratory-grown organs and tissues

¹Micromarkets are defined as further segments and subsegments of the regenerative medicine market.

²Core competencies of companies are captured in terms of their key developments and key strategies adopted to sustain their positions in the market.

1.2.1 INCLUSIONS AND EXCLUSIONS

PARTICULAR	INCLUSION	EXCLUSION
Market	Therapeutic Products (such as Stem Cell Therapy Products, Gene Therapy Products), Cell Transplantation, and Tissue-engineered Products	- Tools and Services Excluded - Regenerative Product Banking - Primary Cell-based Products/Cell Lines
Product	- Cell Therapy - Gene Therapy - Tissue Engineered Products	- Products from Animal Sources/Derivation - Products for Veterinary Treatments
Application	- Musculoskeletal Disorders - Dermatology & Wound Care - Oncology - Ophthalmology - Cardiovascular Diseases - Other Applications	- Veterinary Applications - Cosmetic Applications - Research Applications

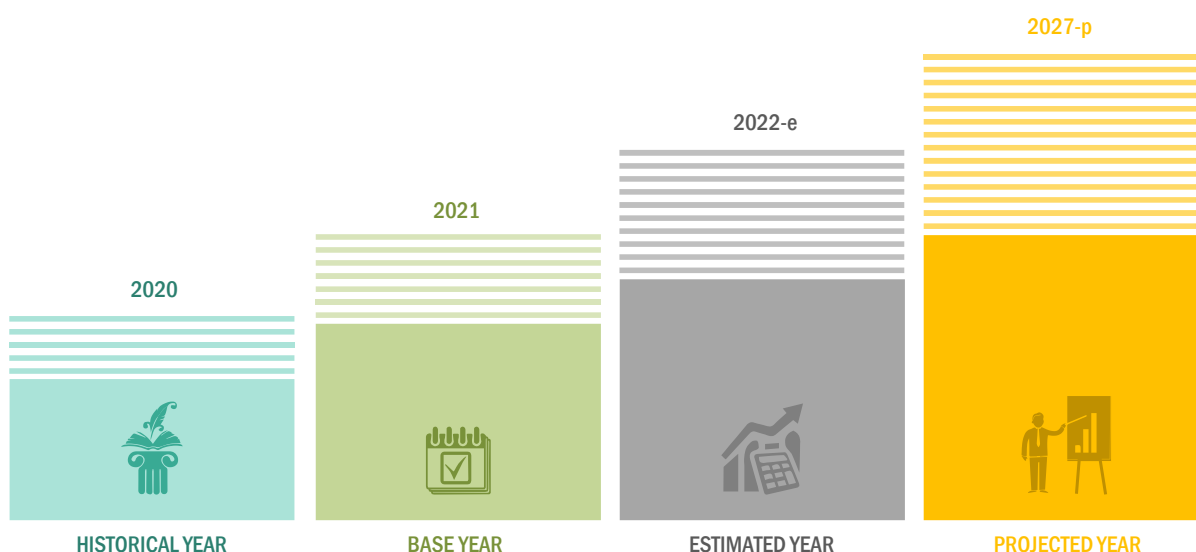
1.3 MARKET SCOPE

FIGURE 1 REGENERATIVE MEDICINE MARKET SEGMENTATION



Note: Other applications include autoimmune diseases, neurology diseases, respiratory diseases, and infectious diseases.

1.3.1 YEARS CONSIDERED



Note: e-Estimated; p-Projected

Note: For market estimates, the calendar year has been considered, whereas for company profiles, the fiscal year has been considered.

1.4 CURRENCY CONSIDERED

The currency used in this report is the United States Dollar (USD), with market size indicated in USD million/billion.

- Revenues of all companies were sourced from their annual reports
- For companies reporting their revenues in currencies other than USD, the average annual currency rate for the particular year was used to convert the value to USD

1.5 LIMITATIONS

- The market study does not cover the market size in terms of volume.
- Some companies in this market are privately owned, and their revenues are unavailable in the public domain. Hence, revenues for these companies are not included in this report.
- Company developments not reported in the public domain are not included.
- A limited number of industry experts participated in the primary research from Africa, the Middle East, and Latin America due to specific language barriers. In such cases, the geographic market size was derived based on weightages assigned to these markets based on qualitative insights from global industry experts, the analysis of demographic parameters, and the adoption trends for typical life science research technologies observed in these markets.

1.6 STAKEHOLDERS

- Healthcare Providers and Clinical Experts
- Pharmaceutical & Biotechnology Companies
- Stem Cell Banks/Sample Repositories
- CROs & CDMOs
- Research Institutes & Universities

1.7 SUMMARY OF CHANGES

Refinements in the segments of the global regenerative medicine market:

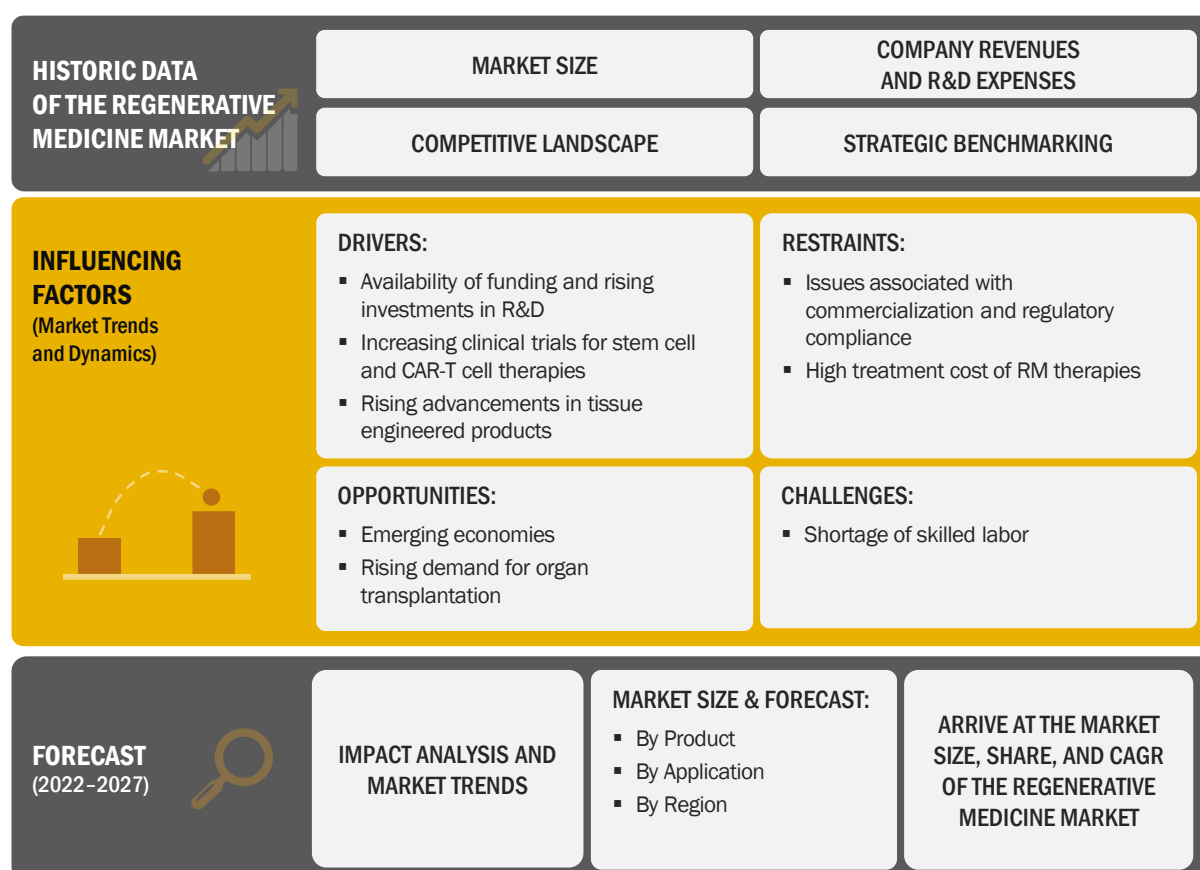
- Additional insights in the market overview, such as Porter's analysis, technology analysis, trends/disruptions impacting customers' businesses, and regulatory framework, among others, have been added to offer information regarding the regenerative medicine market from a macro perspective.
- The ecosystem analysis and regulatory scenario are added in the market overview chapter.
- The new edition of the report provides updated financial information till 2021 (depending on availability) for each listed company in a graphical representation. This will help analyze the present status of profiled companies in terms of their financial strength, profitability, key revenue-generating region/country, and business segment focus in terms of the highest revenue-generating segment.
- The new edition of the report also includes a detailed analysis of regenerative medicine cell therapy products, including stem cell therapy and cell transplantations.
- Recent developments help understand market trends and growth strategies adopted by players in the market. The number of deals (acquisitions, agreements, expansions) has increased in the last two years (2020-2022).
- Tracking product portfolios of prominent market players helps to analyze the key product categories in the regenerative medicine market. The new edition of the report provides an updated product portfolio of the companies profiled in the report.
- Additional company profiles have been added (a total of 25 companies profiled).
- Market evaluation framework, market share analysis, and competitive leadership mapping have been added in the competitive landscape chapter of the report.
- The competitive leadership mapping for key players and SMEs has been updated based on recent developments in the company evaluation matrix.

2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

This research study involved extensive secondary sources, directories, and databases (such as D&B Hoovers, Bloomberg Businessweek, Factiva, and Dun & Bradstreet) to identify and collect information useful for this technical, market-oriented, and commercial study of the regenerative medicine market. The primary sources are mainly industry experts from core and related industries and preferred suppliers, manufacturers, distributors, service providers, reimbursement providers, technology developers, and organizations related to all segments of the regenerative medicine industry's value chain. In-depth interviews were conducted with various primary respondents, including key industry participants, subject-matter experts (SMEs), C-level executives of key market players, and industry consultants, to obtain and verify critical qualitative and quantitative information.

FIGURE 2 RESEARCH DESIGN



2.1.1 SECONDARY DATA

The secondary sources referred to for this research study include publications from government and international sources. These include the World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), Regenerative Medicine Alliance of Canada, Federation of European Academies of Medicine (FEAM), IMAPAC Pte Ltd. Other sources include corporate filings such as annual reports, SEC filings, investor presentations, and financial statements; research journals; press releases; and trade, business, and professional associations.

Secondary research was mainly used to obtain key information about the industry's total pool of key players, market classification and segmentation according to industry trends to the bottom-most level, global market trends, and key developments by both public and private organizations in the regenerative medicine market.

Secondary research involved three major activities:

Background Study

- Building a basic understanding of the regenerative medicine market
- Analyzing the MarketsandMarkets data repository, followed by drawing and updating data tables using current information
- Identifying data gaps and key global players in each segment

Market Understanding

- Identifying stakeholders and key decision-makers in different regions
- Identifying the selection criteria of key decision-makers
- Analyzing the competitive landscape
- Analyzing major global players with their respective product portfolios
- Analyzing the strategies adopted by global players to position their product offerings
- Analyzing markets at the regional/country level

Trends

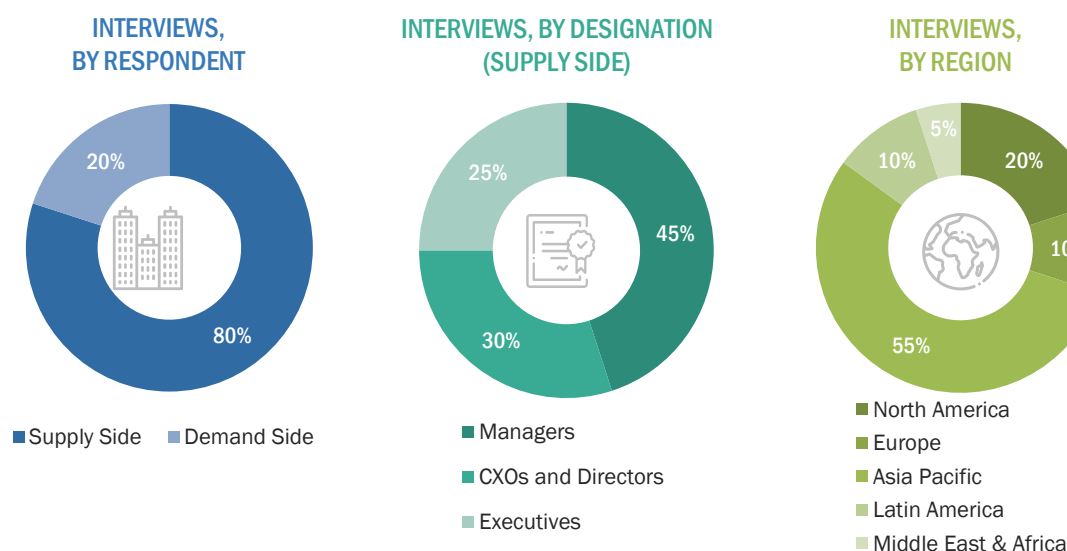
- Identifying and analyzing key products and industry trends to extract qualitative and quantitative information and include considerations in the final forecasts
- Identifying the major drivers, restraints, challenges, and opportunities in the market
- Identifying industry trends
- Analyzing the regulatory landscape

2.1.2 PRIMARY DATA

Extensive primary research was conducted after acquiring basic knowledge about the global regenerative medicine market scenario through secondary research. Several primary interviews were conducted with market experts from both the demand side (personnel from hospitals and clinics) and supply side (C-level and D-level executives, product managers, and marketing and sales managers of key manufacturers, distributors, and channel partners, among others, from tier 1 and tier 2 companies engaged in offering services) across five major regions—North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Approximately 40% of primary interviews were conducted with supply-side representatives, while demand-side participants accounted for the remaining. This primary data was collected through questionnaires, e-mails, online surveys, personal interviews, and telephonic interviews.

FIGURE 3 REGENERATIVE MEDICINE MARKET: BREAKDOWN OF PRIMARIES

SUPPLY-SIDE PARTICIPANTS		DEMAND-SIDE PARTICIPANTS
Novartis AG	MEDIPOST	California Institute for Regenerative Medicine (CIRM)
Stryker Corporation	STEMPEUTICS RESEARCH PVT LTD.	Harvard Medical School and Massachusetts General Hospital
Mesoblast Ltd	Nuo Therapeutics, Inc.	SriShankara Cancer Hospital and Research Center



* Others include sales managers, marketing managers, and product managers.

Note: The tier of companies is defined based on their total revenues for 2021. Tier 1= >USD 100 million, Tier 2= USD 10 million to USD 100 million, and Tier 3= <USD 10 million.

Primary research was used in this report to:

- Validate the market segmentation defined through a product portfolio assessment of the leading players in the regenerative medicine market
- Understand key industry trends and issues defining the strategic growth objectives of market players
- Gather both demand- and supply-side validation of the key factors affecting the market growth (such as market drivers, restraint, challenge, and opportunity)
- Validate assumptions for the market sizing and forecasting model used for this study
- Understand the market position of leading players in the enteral feeding formulas market and their market shares/rankings

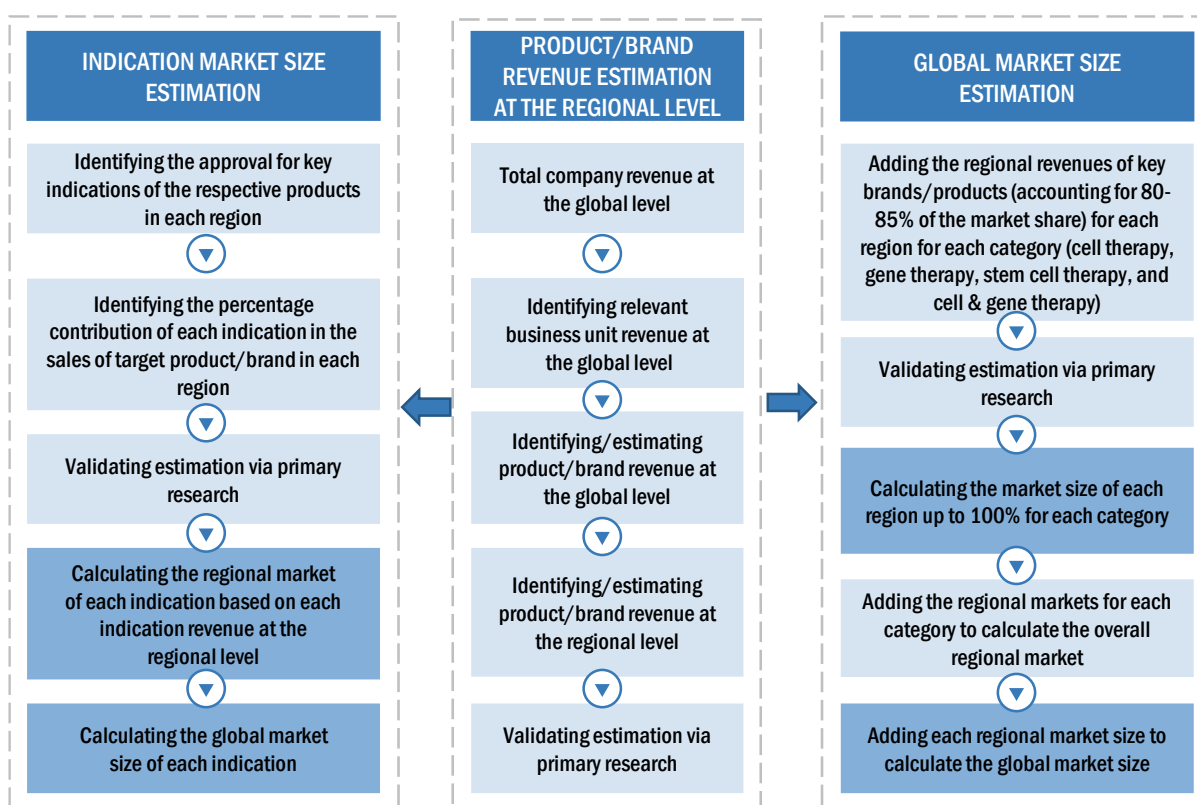
2.2 MARKET SIZE ESTIMATION

Top-down and bottom-up approaches were used to estimate and validate the regenerative medicine market size and the size of various other dependent submarkets. The research methodology used to estimate the market size includes the following details:

- The key players in the market were identified through secondary research, and their market shares in the respective regions were determined through primary and secondary research.
- The entire procedure includes the study of the annual and financial reports of the top market players and extensive interviews for key insights from industry leaders such as CEOs, VPs, directors, and marketing executives.
- All percentage shares, splits, and breakdowns were determined using secondary sources and verified through primary sources.
- All possible parameters that affect the markets covered in this research study have been accounted for, viewed in extensive detail, verified through primary research, and analyzed to get the final quantitative and qualitative data.
- The above-mentioned data was consolidated and added with detailed inputs and analysis from MarketsandMarkets and presented in this report.

2.2.1 BOTTOM-UP APPROACH

In this report, the size of the global regenerative medicine market was arrived at by using the revenue share analysis of leading players. For this purpose, major players in the market were identified, and their revenues from the regenerative medicine business were determined through various insights gathered during the primary and secondary research phases. Secondary research included the study of the annual and financial reports of the top market players, whereas primary research included extensive interviews with key opinion leaders, such as CEOs, directors, and marketing executives of key market players.

FIGURE 4 REGENERATIVE MEDICINE MARKET: BOTTOM-UP APPROACH


Note: This methodology was followed only for a few products for which information on the ASP and the number of installations was obtained from secondary and primary research.

2.2.2 TOP-DOWN APPROACH

The overall regenerative medicine market size was used in the top-down approach to estimate the sizes of other individual markets (mentioned in the market segmentation by product, application, and region) through percentage splits from secondary and primary research.

For the calculation of each type of specific market segment, the most appropriate, immediate parent market size was used to implement the top-down approach. The bottom-up approach was also implemented (wherever applicable) for data extracted from secondary research to validate the market segment revenues obtained.

FIGURE 5 REGENERATIVE MEDICINE MARKET: TOP-DOWN APPROACH

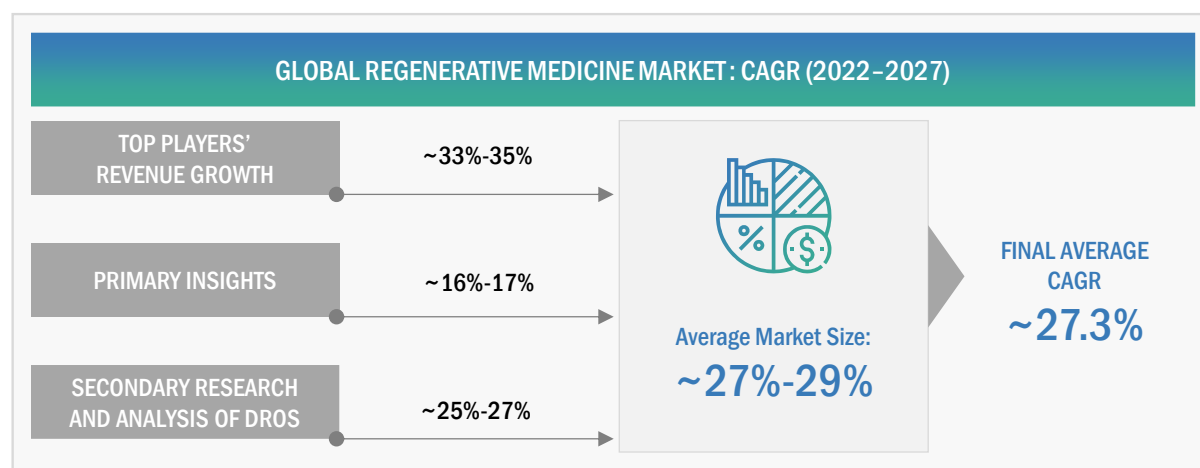

Market shares were then estimated for top companies to verify the total market revenue arrived at earlier in the bottom-up approach. The exact values of the overall parent market size and each individual market size were determined and confirmed in this study with the help of the data triangulation procedure and the validation of data through primaries. The data triangulation procedure implemented for this study is explained in the next section.

2.3 GROWTH RATE ASSUMPTIONS

The growth forecast model assessed the qualitative and quantitative factors affecting the market growth. These mainly included:

- Historic revenue growth trends of the leading players in the market.
- Demand-side factor assessment, including the incidence of cancer, wound care, heart diseases, and increasing demand for regenerative medicine market in these applications
- Impact assessment of the key factors influencing market growth—drivers, restraint, challenge, and opportunity.
- Impact assessment of the historic competitive trends driving market growth—product launches, strategic growth initiatives, innovation trends, and R&D initiatives.
- Impact assessment of regulatory guidelines/mandates on market growth over the forecast period. Only the defined regulatory guidelines related to the study period were considered. The impact of any future regulatory uncertainties on market growth is not accounted for.

FIGURE 6 REGENERATIVE MEDICINE MARKET: CAGR PROJECTIONS, 2022–2027



Source: MarketsandMarkets Analysis

After taking an average of the results from all approaches, the final CAGR for 2022–2027 was estimated at 27.3% for this report.

FIGURE 7 REGENERATIVE MEDICINE MARKET: GROWTH ANALYSIS OF DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

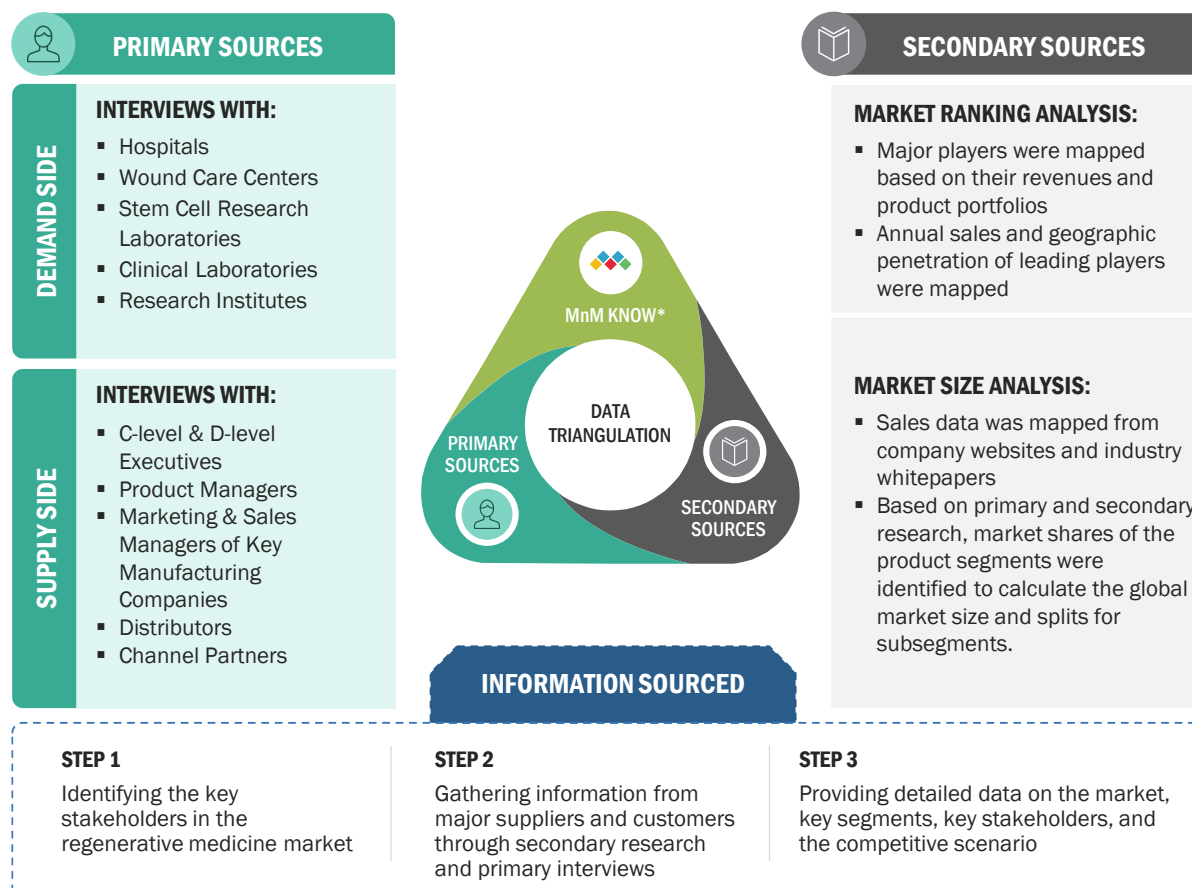
DRIVERS	IMPACT LEVEL
Availability of funding and rising investments in R&D	●
Increasing clinical trials for stem cell and CAR-T cell therapies	●
Rising advancements in tissue engineered products	●
RESTRAINTS	2 YEARS
Issues associated with commercialization and regulatory compliance	●
High treatment cost of RM therapies	●
OPPORTUNITIES	2 YEARS
Emerging economies	●
Rising demand for organ transplantation	●
CHALLENGES	2 YEARS
Shortage of skilled labor	●
IMPACT LEVEL: ● High ● Medium ● Low	

Source: Secondary Research and MarketsandMarkets Analysis

2.4 MARKET BREAKDOWN AND DATA TRIANGULATION

After arriving at the overall market size, the regenerative medicine market was split into segments and subsegments from the market size estimation process explained above. Data triangulation and market breakdown procedures were employed wherever applicable to complete the overall market engineering process and arrive at the exact statistics for all segments and subsegments. The data was triangulated by studying various factors and trends from both the demand and supply sides. Along with this, the regenerative medicine market was validated using both top-down and bottom-up approaches.

FIGURE 8 DATA TRIANGULATION METHODOLOGY



*MnM KNOW stands for MarketsandMarkets' "Knowledge Asset Management" framework. In this context, it stands for the existing market research knowledge repository that acts as an independent source to help us validate information gathered from primary and secondary sources. In this report, reference was taken from MnM related market reports such as the pharmaceutical membrane filtration market, sterile filtration market, depth filtration market, hollow-fiber filtration market, tangential flow filtration market reports and customization done on sterile filtration market.

2.5 PRIMARY INSIGHTS

FIGURE 9 MARKET VALIDATION FROM PRIMARY EXPERTS

FACTORS AFFECTING MARKET GROWTH	
 Founder, <i>Stem Cell Therapy Manufacturer</i>	 The stem cell industry is expected to grow at a high rate in the coming years. The high growth of this market can be attributed to the rising demand for cell therapies in the treatment of pulmonary & neurological disorders.
 Researcher, <i>Leading CRO</i>	 Increasing R&D activities, rising demand for personalized genome therapies, and the high incidence of cancer are the key factors driving the market growth.
 Sales Manager, <i>Big Pharma Company</i>	 Increasing clinical trials are being conducted for CAR-T cell therapies; this is expected to have a positive effect on the overall regenerative medicine industry.

MARKET DYNAMICS	TOP FIVE MARKET PLAYERS												
Emergence of Fasttrack and facilitated clinical approvals in the Asia Pacific market is expected to drive the overall market growth.	<table> <tr> <th>COMPANY</th><th>MARKET RANK (2021)</th></tr> <tr> <td>Novartis AG (Switzerland)</td><td>1</td></tr> <tr> <td>Biogen (US)</td><td>2</td></tr> <tr> <td>Gilead Sciences, Inc. (US)</td><td>3</td></tr> <tr> <td>Sarepta Therapeutics, Inc. (US)</td><td>4</td></tr> <tr> <td>Amgen, Inc. (US)</td><td>5</td></tr> </table>	COMPANY	MARKET RANK (2021)	Novartis AG (Switzerland)	1	Biogen (US)	2	Gilead Sciences, Inc. (US)	3	Sarepta Therapeutics, Inc. (US)	4	Amgen, Inc. (US)	5
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Novartis AG (Switzerland)	1												
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Sarepta Therapeutics, Inc. (US)	4												
Amgen, Inc. (US)	5												
–Regenerative Medicine Market Experts													

2.6 RESEARCH ASSUMPTIONS

- Market values have been rounded off to the nearest decimal place. Due to this, there may be slight differences between actual totals and the totals mentioned in market tables.
- Region-wise inflation/deflation for product pricing during 2022–2027 has not been considered.
- It has been assumed that dollar fluctuations will not affect the growth forecast to a significant extent.
- The revenues of private companies acquired from paid databases were assumed to align with actual company financials.
- A positive economic climate will continue in emerging countries until 2027.
- There will be no major changes in government regulations that will have a significant impact on the market.
- All market segments and subsegments listed in this report are considered to be mutually exclusive.
- Wherever the exact segmental market splits were unavailable, the historical trend and inputs from primary respondents have been considered.

2.7 RISK ANALYSIS

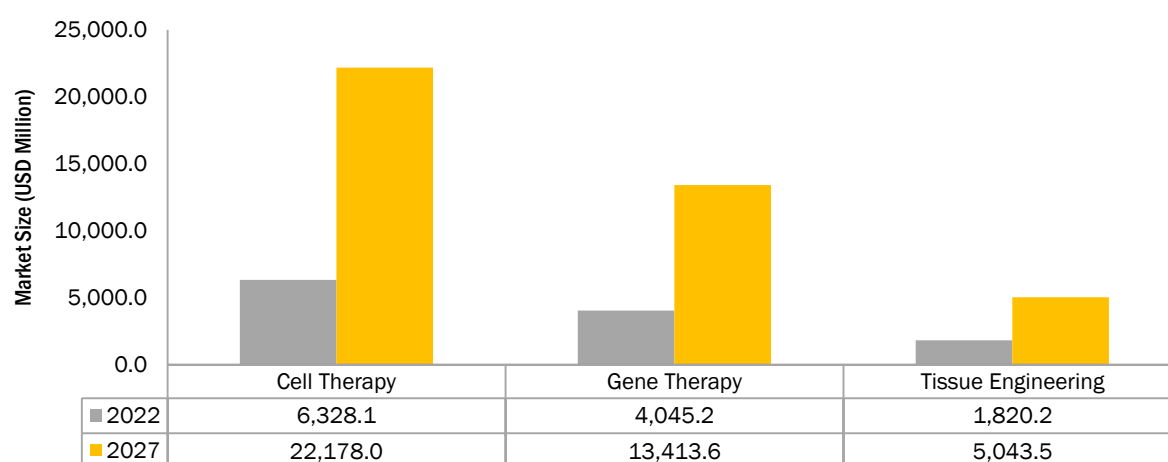
DESCRIPTION	RISK ANALYSIS
The growth rate for each segment is considered based on growth ranks.	Moderate
The percentage splits for the product segments are based on the matrix mapping of products and primary insights.	Low to Moderate

3 EXECUTIVE SUMMARY

The regenerative medicine market is projected to reach USD 40,635.04 million by 2027 from USD 12,193.44 million in 2022, at a CAGR of 27.2% during the forecast period. Market growth is driven by the availability of funding and rising investments in R&D, increasing clinical trials for stem cell and CAR-T cell therapies, and rising advancements in tissue-engineered products. However, the high treatment cost of RM therapies and issues associated with commercialization and regulatory compliance factors are expected to restrain the market growth during the forecast period.

This report segments the regenerative medicine market based on product, application, and region.

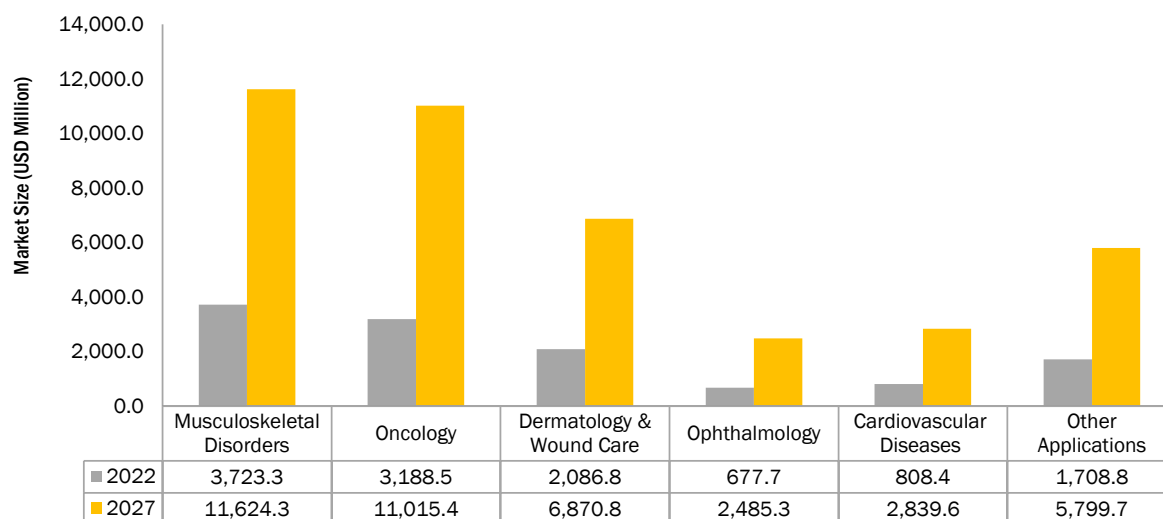
FIGURE 10 REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2022 VS. 2027 (USD MILLION)



Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

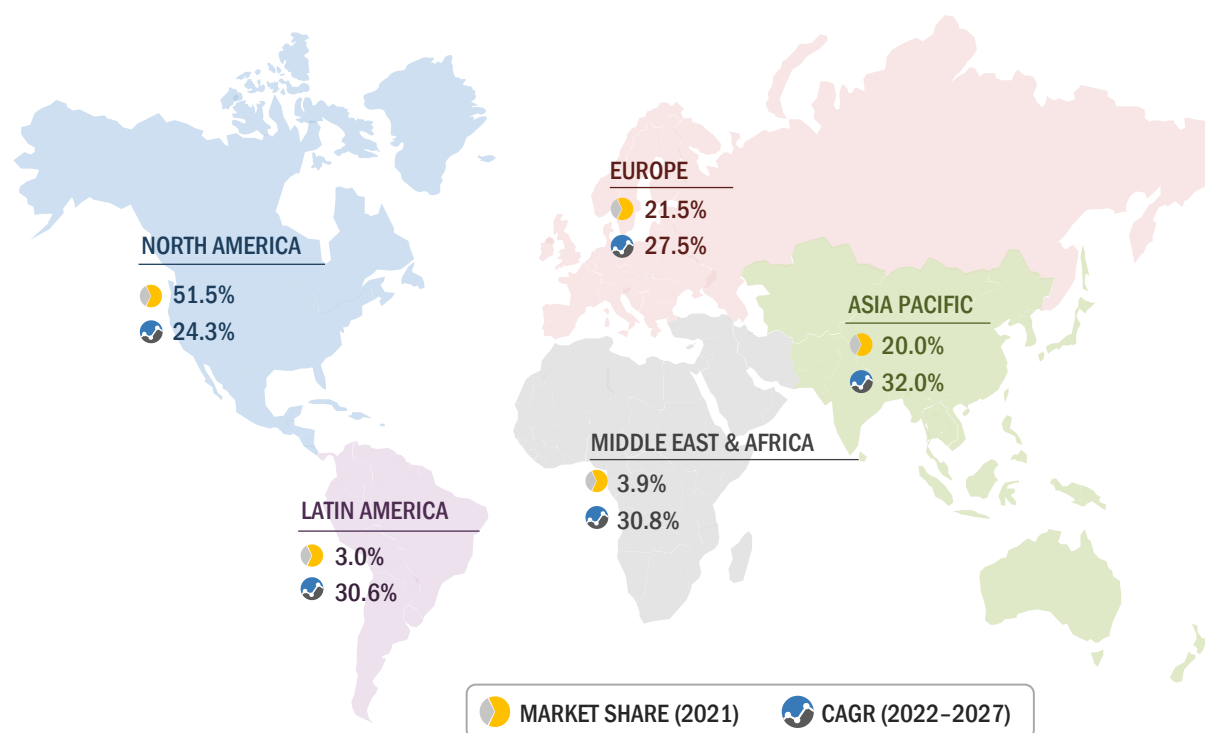
Based on product, the regenerative medicine market is segmented into cell therapy, gene therapy, and tissue engineering. The gene therapy segment accounted for the largest share of 50.8% of the regenerative medicine market in 2021. Technological advancements such as gene editing and CRISPR are expected to propel the commercialization of gene therapies, thereby contributing to overall market growth.

FIGURE 11 REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2022 VS. 2027 (USD MILLION)



Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

Based on applications, the regenerative medicine market is segmented into musculoskeletal disorders, wound care, oncology, ocular disorders, dental, and other applications. In 2021, the musculoskeletal disorders segment accounted for the largest market share at 30.9%. This can be attributed to the rising prevalence of orthopedic diseases, the growing geriatric population, the increasing number of stem cell research projects, the growing number of clinical research/trials, and the rich pipeline of stem cell products for treating musculoskeletal disorders.

FIGURE 12 GEOGRAPHICAL SNAPSHOT OF REGENERATIVE MEDICINE MARKET

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

The regenerative medicine market is segmented into five major regions: North America, Europe, Asia Pacific, Latin America, and the Middle East & Africa. In 2021, North America accounted for the largest share of 51.5% of the regenerative medicine market. The growth in the North American regenerative medicine market can be attributed to rising stem cell banking, tissue engineering, and drug discovery in the region; expansion of the healthcare sector; and the high adoption of stem cell therapy and cell immunotherapies for the treatment of cancer and chronic diseases.

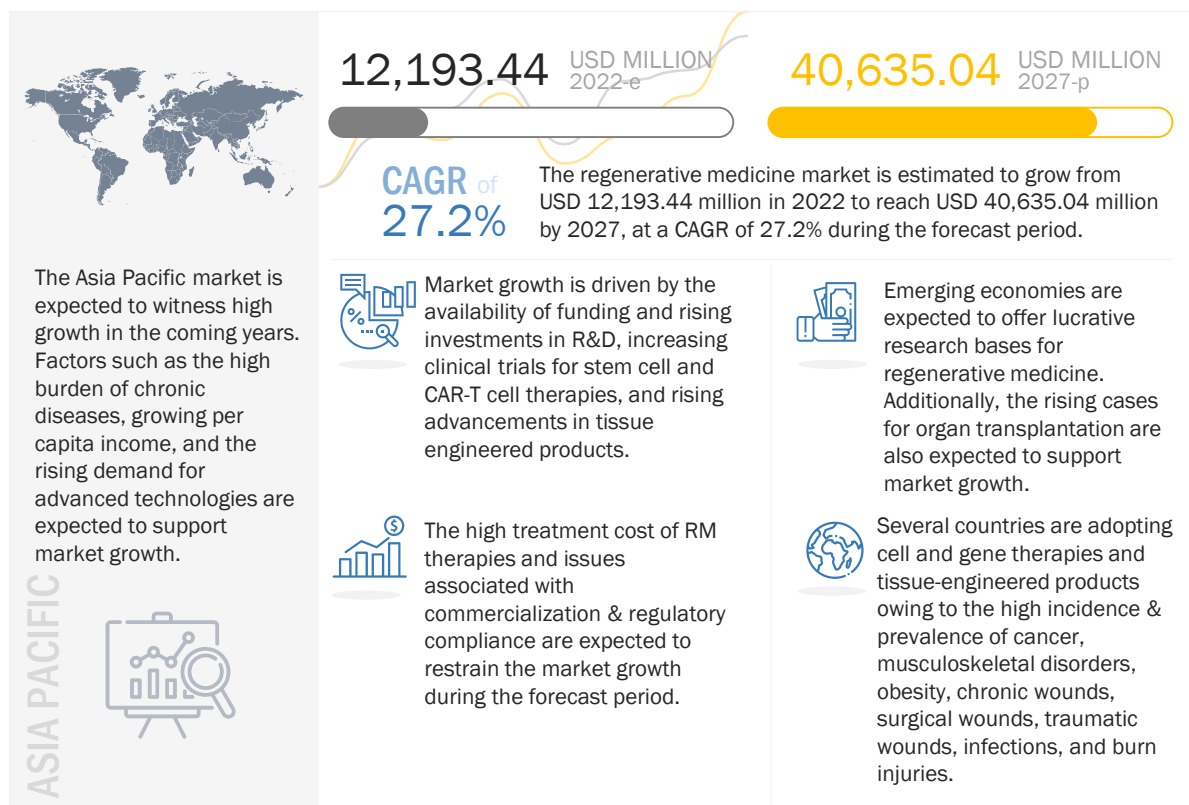
The Asia Pacific region is expected to grow at the fastest rate, with a CAGR of 32%; the region presents significant opportunities for the growth of the regenerative medicine market. Factors such as increasing research activities in Asian countries, the availability of funding for research, and the growing number of biobanks and research centers are driving the growth of the Asia Pacific regenerative medicine market.

The regenerative medicine market is moderately consolidated, with various players offering regenerative medicine products. Prominent players in this market are Novartis AG (Switzerland), Biogen (US), Sarepta Therapeutics, Inc. (US), Gilead Sciences, Inc. (US), Amgen, Inc. (US), Smith & Nephew plc (UK), MEDIPOST Co., Ltd. (South Korea), JCR Pharmaceuticals Co., Ltd (Japan), Takeda Pharmaceutical Company (Japan), CORESTEM Inc (South Korea), Vericel Corporation (US), MIMEDX (US), Organogenesis Inc. (US), Medtronic plc (US), Orthocell Ltd. (Australia), Mesoblast Ltd (Australia), BioRestorative Therapies (US), Pluri Inc, (Israel), Tegoscience (South Korea), Anterogen Co., Ltd. (South Korea), bluebird bio (US), APAC Biotech (India), Shenzhen Sibiono Genetech Co., Ltd. (China), Aspect Biosystems (Canada), and Athersys, Inc. (US)

4 PREMIUM INSIGHTS

4.1 REGENERATIVE MEDICINE MARKET OVERVIEW

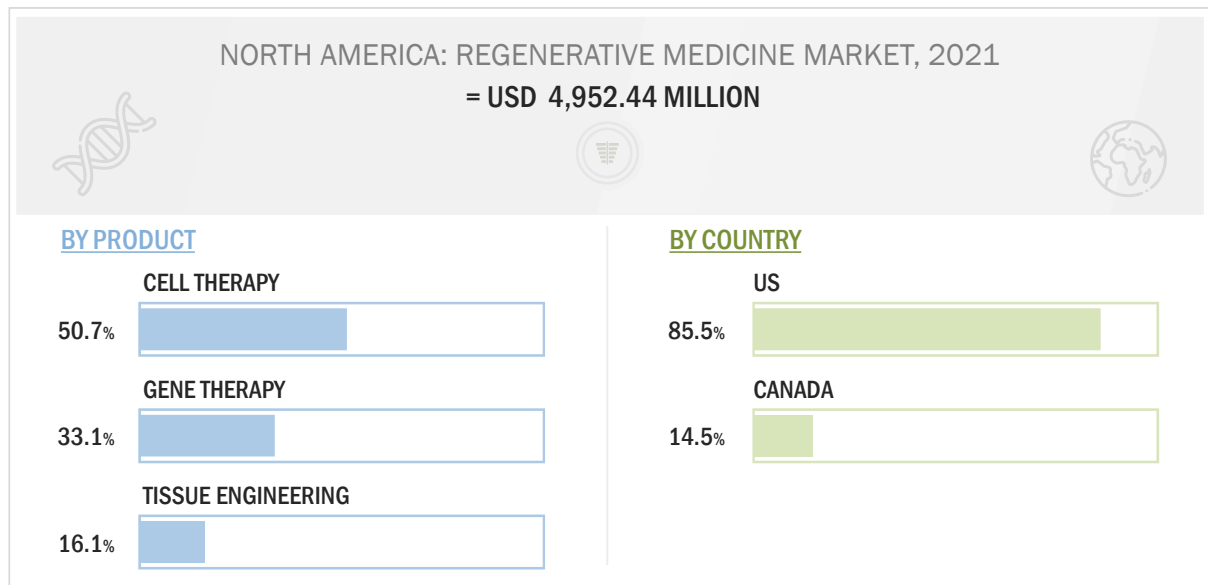
FIGURE 13 AVAILABILITY OF FUNDING FOR RESEARCH AND INCREASING CLINICAL TRIALS IN REGENERATIVE MEDICINE TO DRIVE MARKET



Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.2 NORTH AMERICA: REGENERATIVE MEDICINE MARKET

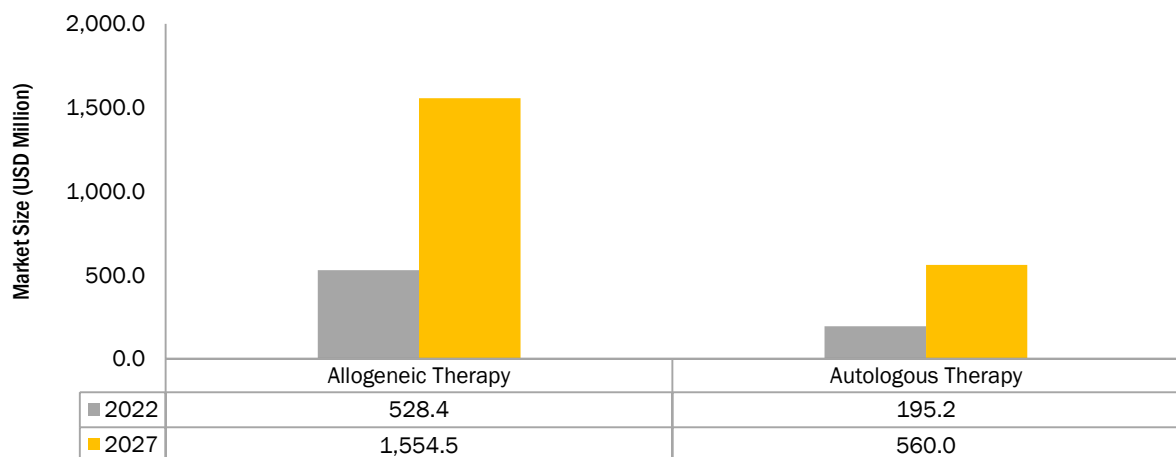
FIGURE 14 CELL THERAPY ACCOUNTED FOR LARGEST MARKET SHARE IN 2021



Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.3 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, 2022 VS. 2027 (USD MILLION)

FIGURE 15 ALLOGENEIC THERAPY TO GROW AT HIGHEST CAGR DURING FORECAST PERIOD



Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5 MARKET OVERVIEW

5.1 INTRODUCTION

The global regenerative medicine market is projected to reach USD 40,635.04 million by 2027 from USD 12,193.44 million in 2022, at a CAGR of 27.2% during the forecast period. Market growth is driven by the availability of funding and rising investments in R&D, increasing clinical trials for stem cell and CAR-T cell therapies, and rising advancements in tissue-engineered products. However, the high treatment cost of RM therapies and commercialization & regulatory are expected to restrain the market growth during the forecast period.

5.2 MARKET DYNAMICS

FIGURE 16 REGENERATIVE MEDICINE MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

 DRIVERS	▪ Availability of funding and rising investments in R&D ▪ Increasing clinical trials for stem cell and CAR-T cell therapies ▪ Rising advancements in tissue-engineered products
 RESTRAINTS	▪ Emerging economies to offer lucrative research bases for RM ▪ Rising cases for organ transplantation
 OPPORTUNITIES	▪ Issues associated with commercialization and regulatory compliance ▪ High treatment cost of RM therapies
 CHALLENGES	▪ Shortage of skilled professionals

Source: MarketsandMarkets Analysis

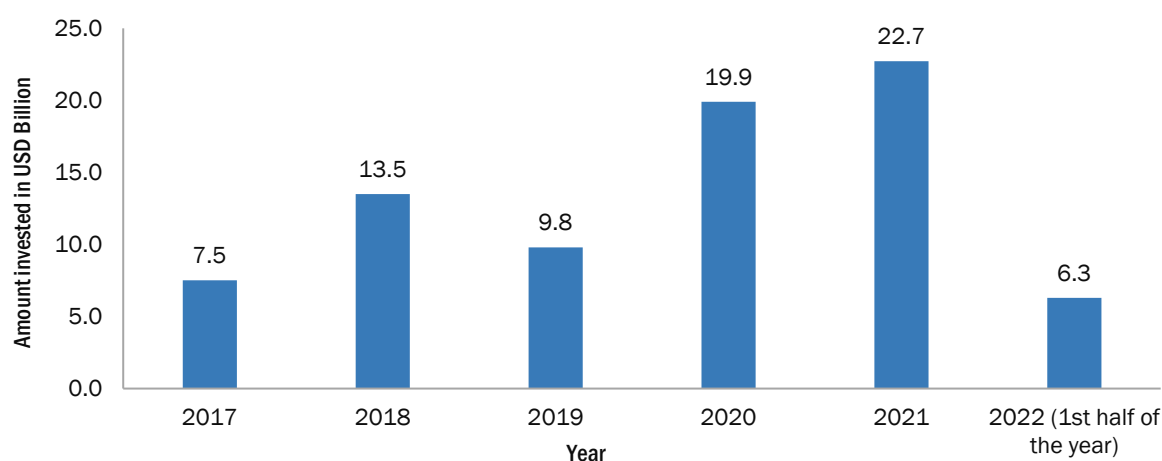
5.2.1 DRIVERS

5.2.1.1 Availability of funding and rising investments in R&D

The rising need for improved therapies for cancer, diabetes, skin diseases, musculoskeletal diseases, chronic wounds, burns, and CVD has resulted in increased research activities. Additionally, this has caused a surge in the availability of funding for regenerative medicine research. The new cell therapies, gene therapy, and tissue-engineered products are expensive in manufacturing & development; therefore, there is a rising need for funding.

According to the Alliance for Regenerative Medicine (ARM), in 2021, investments reached USD 22,700 million, with key investors like Amgen, Sanofi, and Gilead Sciences, easily surpassing USD 19,900 million in 2020. These rising investments have led to an improvement in the services offered to patients. The US had a total investment of USD 17.6 billion, followed by Europe and the Asia Pacific, with USD 3.3 billion and USD 2.2 billion, respectively. 2021 was a favorable year for new product approvals, with six new therapies available globally. In 2021, three new approvals for CAR-T products were registered. The availability of new therapeutic modalities that address the causes of specific diseases has driven unprecedented private investments in biotech and cell & gene therapy.

FIGURE 17 INVESTMENTS IN REGENERATIVE MEDICINE, 2017–2022 (USD BILLION)



Source: Alliance for Regenerative Medicine

TABLE 1 FUNDING INITIATIVES FROM CORPORATE PARTNERSHIPS IN 2022 (USD MILLION)

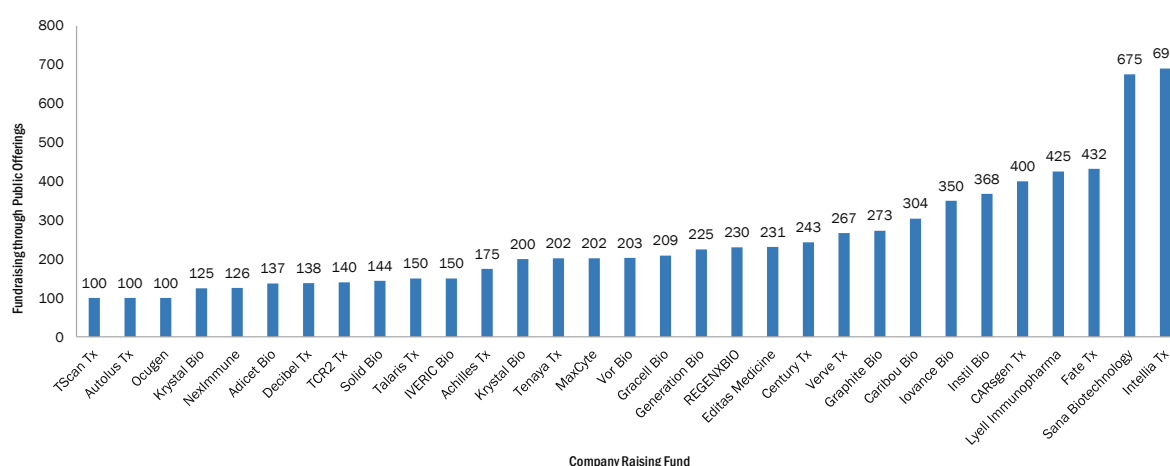
COMPANY (SIGNING)	COMPANY (SIGNING WITH)	FUNDING
Century Therapeutics Inc	Bristol-Myers Squibb Company	150
Homology Medicines	OXFORD BIOMEDICAL RESEARCH	130
Beam Therapeutics	Pfizer Inc.	300

Source: Alliance for Regenerative Medicine

TABLE 2 FUNDING FROM CORPORATE PARTNERSHIPS IN 2021 (USD MILLION)

COMPANY (SIGNING)	COMPANY (SIGNING WITH)	FUNDING
Vertex Pharmaceuticals	CRISPR Tx	900
REGENXBIO Inc.	AbbVie	370
Adaptimmune Therapeutics plc.	Genentech	150
Eli Lilly and Company	Precision BioSciences	135
Ensoma	Takeda Pharmaceutical Company	100

Source: Alliance for Regenerative Medicine

FIGURE 18 FUNDRAISING THROUGH PUBLIC OFFERINGS (IPO AND FOLLOW-ONS), 2021 (USD MILLION)

Source: Alliance for Regenerative Medicine

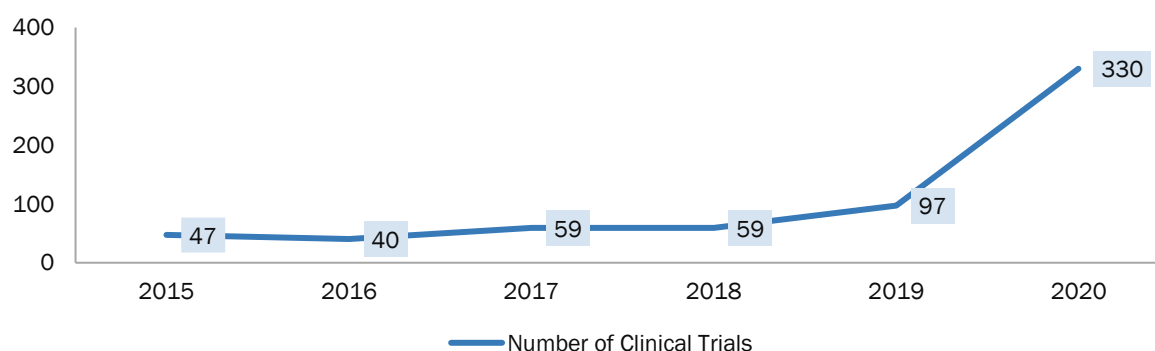
Additionally, investments targeted toward research activities to develop advanced cell therapy technologies are expected to drive market growth. In March 2022, the National Heart Lung and Blood Institute granted USD 12 million to a researcher at the Indiana University School of Medicine. The grant aims at exploring gene therapy applications for potentially safer curative treatments for hemophilia. In October 2020, the Government of Canada granted USD 20.3 million through the Stem Cell Network's competitive research funding program, which supports the country's most promising stem cell research to regenerative medicine and genomics research to support Canadians' health. The funding will support nine translational projects, and four clinical trials across the country to bring new therapies to the clinic and foster continued growth in Canada's regenerative medicine sector. This development includes supporting 200 researchers and trainees at 27 Canadian research institutions and direct partnerships with several emerging biotechnology companies.

5.2.1.2 Increasing clinical trials for stem cell and CAR-T cell therapies

Several companies are focusing on developing regenerative medicine due to the rising number of clinical trials conducted on stem cell therapies. The increasing number of clinical trials has largely taken place owing to the rising government and public & private investments for R&D in cell-based therapies, supportive regulatory framework, and accelerated approval provisions by the FDA in regenerative medicine R&D.

The COVID-19 pandemic has also triggered the demand for stem cell-based therapies to treat infectious diseases. The surge during the pandemic has accelerated the overall number of clinical trials between 2019 and 2020. For instance, the clinical trials based on stem cells increased exponentially by 240%, from 97 to 330. Most trials are in Phase 2 stages with a major focus on pulmonary and neurological disorders. (Source: ClinicalTrial.gov) Most of these clinical trials are focused on placental/umbilical cord-derived stem cells, followed by adipose tissue-derived therapies. As of 2022, more than 5,000 clinical trials have been registered by ClinicalTrials.gov in stem cell research. Stem cell therapies are expanding across applications associated with brain & nerves and cardiovascular disorders, thus expected to drive market growth in the coming years.

FIGURE 19 NUMBER OF STEM CELL THERAPY CLINICAL TRIALS (2015–2020)



Source: ClinicalTrial.gov

Chimeric Antigen Receptor (CAR)-T cell therapy has emerged as a revolutionary treatment modality in cancer treatment. CAR-T cell-based treatment has produced remarkable clinical responses with certain B-cell leukemia or lymphoma subsets. However, challenges persist in line with the therapeutic efficacy of CAR-T cells in solid tumors and hematological malignancies. Advancements targeted to mitigate these challenges are expected to drive market growth.

5.2.1.3 Rising advancements in tissue-engineered products

Incorporating scaffolds, growth factors, and cell-based products in therapeutic interventions are expected to propel the demand for tissue-engineered products. Due to tissue engineering, new approaches for regenerative medicine and disease models have emerged. The most important pivotal developments in tissue engineering include induced pluripotent stem cell technologies (iPSC), 3D bioprinting, and organ-on-a-chip technologies.

Robust investments in tissue engineering are expected to drive the uptake of therapeutics that possess commercial importance and mitigate disease simultaneously. Below are some of the prominent advancements in the field of tissue engineering:

Scaffolds:

Biomaterial-based three-dimensional structures are emerging as promising materials to act as scaffolds for seeding cells for in vivo and in vitro experiments. One material that has gained traction is polyhydroxyalkanoates (PHAs) which are thermoplastic biopolyesters. Factors such as biodegradability, enhanced biocompatibility, thermo-processability, diverse mechanical properties, and non-toxicity make PHAs the preferred TE material. Additionally, this material's customization through chemical modification, biological, and physical blending with other materials make them an attractive TE material. Scaffolds are used for several tissue engineering applications such as bone graft substitutes, cardiovascular patches, and stents for nerve repair and in implantology as valves and sutures. Advancements directed toward modifying biomaterials in tissue engineering are expected to drive the growth of the regenerative medicine market.

Hydrogels:

Hydrogels offer a novel alternative to 2D cell culture by designing scaffolds in tissue engineering for three-dimensional cell culture. Advancements in hydrogel engineering are presently being targeted toward improving the mechanical properties of these materials. Recent research studies have demonstrated new procedures to improve the mechanical properties of hydrogels by incorporating nanomaterials, such as graphene and its derivatives. Several techniques have been developed to manufacture hydrogels, including microscale-ordered particles (1–500 µm), called microgels or microparticle-based hydrogels (HMP). Such advancements are expected to propel the adoption of hydrogels in bone cement in orthopedics, wound dressings, and tissue scaffolds in regenerative medicine, further propelling the market growth.

Such advancements in the field of tissue engineering are expected to positively impact the tissue engineering market and facilitate the translation of regenerative medicine from research to clinical applications.

5.2.2 RESTRAINTS

5.2.2.1 Issues associated with commercialization and regulatory compliance

Regenerative medicine is an emerging interdisciplinary field aiming to replace or regenerate human cells, tissues, or organs to restore or establish normal functions. According to the Alliance of Regenerative Medicine (ARM) annual report, more than 1000 clinical trials investigating RMs for treating various diseases are underway as of 2022. A small number of regenerative medicines have received market authorization worldwide relative to a large number of clinical trials currently being conducted. Regulatory issues are one of the key restraints for the market authorization of regenerative medicine.

Despite the rapid growth in regenerative medicine research and clinical trials, only a few regenerative medicines have been approved globally. Regulatory hurdles constitute one of the biggest challenges for the successful clinical translation and commercialization of regenerative medicines. Regenerative medicines, especially non-orphan regenerative medicines, are associated with a lower approval rate and take longer to approve than other chemical products.

Market authorization requirements, such as compliance with good manufacturing procedures (GMPs) and conducting pivotal studies with robust study design, are difficult to fulfill. The requirements are difficult to fulfill and have failed to consider the unique characteristic of regenerative medicines. Inconsistency in regulatory requirements across countries remains a major hurdle for manufacturers to navigate regarding market authorization processes. Differences have been recorded in the classification of regenerative medicine across various regions. For instance, Tisagenlecleucel (Kymriah) has been approved as cellular therapy in Japan and Australia, whereas Kymriah is classified as a gene therapy product across Europe and

the US. This has impacted the classification of regenerative medicines into key categories, particularly cell and gene therapy (CGT), thus restraining the market growth to a certain extent.

5.2.2.2 High treatment cost of RM therapies

Stem cell therapy is still highly specialized and has not been adopted as the mainstream treatment modality, affecting reimbursement policies. Additionally, the field is further restricted by stringent regulations. Despite the comparatively better efficacy of stem cell therapies than traditional treatments, the response in demand for these therapies is less than the anticipation. This can be attributed to the high costs of these therapies and the difficulty of obtaining reimbursements and coverage. As a result, multiple patients cannot avail of these therapies or get delayed treatments owing to delayed reimbursements, leading to failure, death, or worsening of the condition.

Due to the high costs associated with gene and cell therapies, coupled with limited or no coverage/reimbursement scenario, consumers are demotivated to avail of these treatments. Therefore, the high cost of RM therapies factor is expected to restrain the growth of the regenerative medicine market during the forecast period.

Stem Cell Therapy: Costs range between USD 5000 to USD 50,000. Significant variations in stem cell treatment costs are attributed to the following:

- The type of stem cells administered
- Quantity of cells are administered
- The quality of the cells (viable, ethical source of cells)
- Location of labs and manufacturing facilities
- The source of stem cells - umbilical cord tissue, adipose tissue, and bone marrow

Gene Therapy: Gene therapies presently commercialized (Luxturna and Zolgensma) are single-dose treatments that insert a new gene to replace a malfunctioning or missing gene. The inserted gene is anticipated to provide lasting functionality in the targeted cells. Thus, there is an unavoidable consideration for gene therapies that increases prices. Luxturna's treatment cost ranges between USD 800,000 to USD 850,000. Whereas, Zolgensma costs around USD 2 million.

5.2.3 OPPORTUNITIES

5.2.3.1 Emerging economies

The Asia Pacific region is active in regenerative medicine research across gene therapy, cell therapy, and tissue engineering. The Asia Pacific region includes various countries such as Taiwan, Indonesia, Malaysia, Singapore, Thailand, Vietnam, Philippines, Australia, and New Zealand. These emerging markets offer the potential to conduct credible research on regenerative medicine therapies. The unmet healthcare needs in these countries are expected to propel research for regenerative medicine technologies. For instance:

New Zealand:

In 2012, Osiris Therapeutics received approval for its first-in-class stem cell therapy Prochymal (remestemcel-L), in New Zealand to treat graft-versus-host disease. Regulatory approaches in New Zealand have also seen changes in licensing and conditional marketing similar to other Asian countries, which is expected to offer lucrative opportunities to regional market players to expand their business. Osiris submitted a New Medicine Application (NMA) to Medsafe, New Zealand's medical regulatory agency, in May 2011. It was granted Priority Review in June 2011 to provide an expedited review for new drugs which offer a significant clinical advantage over current treatment options.

Taiwan:

Taiwan is one of the first countries in the Asia Pacific region to have an experimental regulatory science (RS) model. Taiwan set up the Center for Drug Evaluation (CDE) to regularize health technology assessments, clinical trial protocols, and new drug applications as early as 1998. In the past decade, this innovative experimental model of regulatory science in Asia has had many achievements.

Taipei Medical University (TMU) Center for Cell Therapy and Regeneration Medicine leads Taiwan's Regeneration Technology Development program and has a collaborative network with universities from different countries, including Taiwan. TMU states that, as opposed to controversies around cell therapies elsewhere, Taiwan's government has supported TMU research into mesenchymal stem cells. The nation's progressive regulatory environment also supports research teams as they develop new applications and technologies.

Thailand:

Stem cell tourism is an emerging area of medical tourism activity. Thailand is one of the leading countries in international stem cell tourism, along with the US, China, India, and Mexico. In November 2020, Thanyapura Sports and Health Resort (Phuket) partnered with a stem cells company, StemCells21, to offer premium stem cell treatments in Thailand. The partnership will offer premium stem cell treatments for various degenerative diseases, immune disorders, injuries, and anti-aging.

5.2.3.2 Rising cases of organ transplantation

Regenerative medicine is emerging as a novel approach to organ transplantation. Treatment options for end-stage organ disease include support of organ function through respiratory devices or ventricular pumps, dialysis to manage disease progression, and, eventually, transplantation. In recent years, genetic editing and cellular reprogramming advancements have led to supplying de novo generated cells and tissues as an alternative to transplantation.

The development of organ decellularization techniques has directed research activities towards developing acellular scaffolds for applications in regenerative medicine, including for recellularization purposes. An interesting initiative in this field is the heterotopic implantation of decellularized hearts with the intent to allow the recipient body to repopulate the scaffold with endothelial and stromal cell types. Other advancements in this field are the generation of allogenic hydrogels for applications in tissue engineering, including 3D bioprinting.

The benefits of regenerative medicine research in recent years have witnessed considerable progress and led to the development of organ-like structures known as organoids conceived from adult and pluripotent stem cells for virtually all tissue types. Kidney organoids, with a surprising level of complexity, can be created within a few weeks from an undefined clump of pluripotent stem cells.

Regenerative medicine is a rapidly evolving tool that will offer lucrative growth opportunities in organ transplantation, thus expected to drive market growth in the coming years.

5.2.4 CHALLENGES

5.2.4.1 Shortage of skilled professionals

As regenerative medicine progresses, the skills required are also expected to evolve. Regenerative medicine biomanufacturing is experiencing rapid growth, leading to a gap between the skill sets required by laborers in the workforce and an inability to hire for those skills in the market. Despite robust funding in the cell and gene therapy (CGT) market, manufacturing these therapies has become challenging due to the shortage of skilled personnel. A limited workforce and restricted knowledge associated with manufacturing novel and emerging therapies contribute to the slow commercialization of CGTs.

Commercial companies and industry organizations have launched initiatives by introducing academic institutions to deal with the shortage of skilled staff via mentoring programs and training courses. For instance, BioCentriq—a CDMO for CGT—has become a part of a ‘talent factory’ with access to students from a Master’s program in cell & gene therapy at the New Jersey Institute of Technology. In November 2021, Biocentriq collaborated with Terumo Blood and Cell Technologies (US) to accelerate the adoption of automated manufacturing, which can reduce the human input required for culturing cells.

5.3 PORTER’S FIVE FORCES ANALYSIS

Porter’s Five Forces Analysis is a framework that determines the competitive intensity in the regenerative medicine market. It helps with decision-making related to the entry or exit of players from the market. This section analyzes the regenerative medicine market from five different perspectives: the degree of competition within the industry, the threat of new entrants, the bargaining power of suppliers, the bargaining power of buyers, and the threat of substitutes.

TABLE 3 REGENERATIVE MEDICINE MARKET: PORTER’S FIVE FORCES ANALYSIS

PORTER’S FIVE FORCES	IMPACT
Degree of Competition	Low
Bargaining Power of Suppliers	Moderate
Bargaining Power of Buyers	High
Threat of Substitutes	Moderate
Threat of New Entrants	Low

5.3.1 THREAT OF NEW ENTRANTS

Many players are established in the regenerative medicine market, indicating the competitive environment, which restricts the entry of new businesses. Established companies have a competitive advantage over new entrants, as they operate efficiently due to the benefits of economies of scale, which they have already achieved. This enables established players to offer their products at lower prices, grabbing the market share of new entrants. Though new entrants in the market can produce systems efficiently and charge low prices for their products, already established players operating in the market can pose a hindrance to them.

New entrants in the market have lesser experience as compared to established players. It becomes difficult for start-ups to set up large manufacturing facilities; most start-up companies go for small scale. These high entry barriers pose a competitive disadvantage for new entrants, limiting competition from smaller companies in the global regenerative medicine market.

These entrants must improve their brand value to compete effectively, and for this, they require heavy investment. The expansion of business also requires substantial investments, which, in turn, increases the overall cost of production capacity. Initial capital requirements represent an entry barrier, depending upon the type of product a new player is attempting to manufacture. Hence, the threat of new entrants in the regenerative medicine market is low.

5.3.2 THREAT OF SUBSTITUTES

The threat of substitutes for regenerative medicine is moderate, as there are other kinds of treatments compared to regenerative medicine therapies where regulatory approval is comparatively lesser stringent and provides better results in efficacy and cost.

5.3.3 BARGAINING POWER OF BUYERS

The bargaining power of buyers refers to the pressure that buyers exert on regenerative medicine therapy providers. Few companies in the market provide FDA/other regulatory-approved therapies. Business-oriented buyers prefer end-to-end providing served firms for efficient quality checks and improving compliance, which has been approved and commercialized by all regulatory bodies. Hence, the bargaining power of buyers in this market is high.

5.3.4 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers refers to the extent to which raw material suppliers can pressurize manufacturers of regenerative medicine. Suppliers of raw materials and consumables can influence the prices of final products. The availability of major constituents of drug development is not uniform across the industry. It depends upon the level of vertical integration. This leaves the industry vulnerable to an exchange rate risk. It also affects the ability of competitors to source raw materials at a cheaper rate.

Moreover, the cost of materials also depends on the demand and supply balance. Following the supply shortage and higher demand, the cost of raw materials may increase, again increasing the supplier's bargaining power. Apart from this, the cost of the raw materials is effective, but the cost required for developing, qualifying, validating, and implementing is quite high for regenerative medicine manufacturing instruments. The overall bargaining power of suppliers in this market is moderate.

5.3.5 INTENSITY OF COMPETITIVE RIVALRY

The competitive rivalry within the industry refers to the extent to which market players exert pressure on each other. If the competition among existing players in an industry is intense, it drives down prices and decreases the industry's overall profitability.

The market for regenerative medicine is moderately consolidated, with many small to medium manufacturers competing with established players. There is a wide variety of regenerative medicine therapy providers who offer safe and cost-effective therapies.

The key competitors such as Novartis AG (US), Amgen, Inc. (US), Biogen (US), and Sarepta Therapeutics (US) possess substantial market share, are targeting attractive market segments, and pricing their products accordingly. The presence of many companies has increased the intensity of rivalry in this market. Therefore, the intensity of competitive rivalry in the regenerative medicine therapy market is low.

5.4 TECHNOLOGY ANALYSIS

Regenerative medicine technologies revolve around tissue engineering, gene therapy, and cell therapy to augment and repair cells or tissues and metabolic processes in the body. Regenerative technologies are emerging as attractive treatment modalities, and associated advancements are set to propel market growth.

Scientists are researching gene therapy for various illnesses, including severe combination immunodeficiencies, hemophilia, Parkinson's disease, cancer, and HIV. A "vector" is a carrier that can transfer a gene to a cell. Although the viruses used in gene therapy have undergone modifications to make them safe, certain hazards are still involved.

FIGURE 20 KEY STEPS IN GENE THERAPY



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

Genetic material must initially enter the diseased person's cells for any gene therapy to be effective. This may occur in one of two ways:

In Vivo Delivery

Direct genetic material delivery to an organ, either intravenously (via an IV) or locally, is known as in vivo gene therapy. In vivo gene therapy treats a defective or absent gene by directly introducing functioning copies of the gene into the target cells using a vector. Gene therapy can be delivered in vivo, as has been demonstrated in numerous scientific fields. Some approved gene therapies transport genetic material inside the body. Targeted in vivo gene therapy will advance if scientists work to improve gene delivery techniques.

Ex Vivo Delivery

Ex vivo gene therapy involves taking certain cells from an individual, genetically modifying them in a lab, and then re-injecting them into the individual. Ex vivo gene therapy replaces target cells with defective or missing genes by genetically altering a patient's stem cells.

Hematopoietic stem cells (HSCs) are currently the target of ex vivo gene therapy, which treats genetic illnesses affecting tissues & organs (that can be reached by blood cells) and blood & immunological diseases.

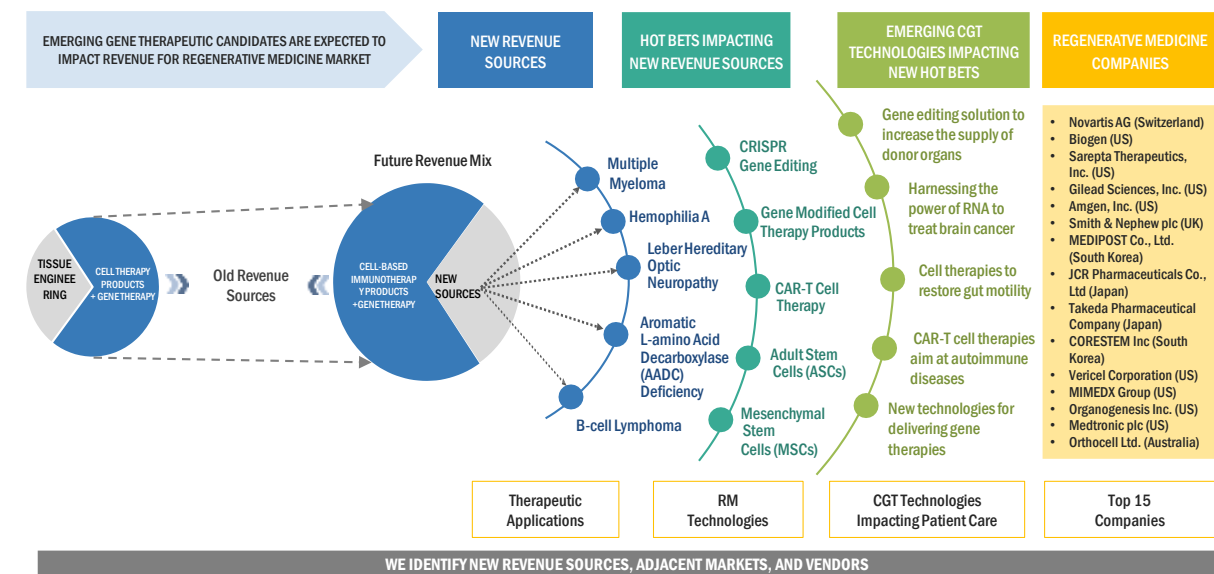
TABLE 4 **STEM CELL THERAPIES AND GENE THERAPIES**

PARAMETERS	STEM CELL THERAPY	GENE THERAPY
Nature	Transplantation of human cells to replace or repair damaged tissue and/or cells	Uses genetic material or DNA to manipulate a patient's cells for the treatment of an inherited or acquired disease
Stem Cell type	Autologous cells & allogeneic cells	Somatic gene therapy and Germline gene therapy
Risk	Complexities	Safety
Function	Regeneration of cells (Regenerative Therapies)	Genetic editing or repair (Genetic Engineering)
Type of Cell Source	Cord blood stem cells, Induced pluripotent stem cells (iPSCs), Adult stem cells, and Embryonic stem cells	Chimeric Antigen Receptors (CAR) T-cell Therapies, T-cell Receptor (TCR) Therapies, Natural Killer (NK) Cell Therapies, Tumor-Infiltrating Lymphocytes (TILs), Marrow-derived Lymphocytes (MILs), Gamma-delta T-cells, and Dendritic Vaccines
Application	Neurodegeneration disorders, blood disorders, musculoskeletal disorders, diabetes, and cardiovascular diseases	Oncology, inherited disorders, and rare disorders
Commercialization	FDA-approved products available	FDA-approved products available

Source: Press Releases and MarketsandMarkets Analysis

5.5 TRENDS/DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES

Decades of research on potential cell and gene therapy applications have led to new therapeutic avenues such as Leber hereditary optic neuropathy (LHON), different types of myeloma, and Aromatic L-amino Acid Decarboxylase (AADC) Deficiency. Eye-related therapies have become an attractive investment area for stem cell therapy manufacturers. This trend is particularly observed for macular degeneration, with increased clinical trials using stem cell transplants to treat macular degeneration. These clinical trials have shown noticeable vision improvement in several participants. Whereas the advent of CRISPR, demand for CAR-T cell therapy, and adult stem cells could act as new revenue pockets.

FIGURE 21 REVENUE SHIFT TOWARD REGENERATIVE MEDICINE COMPANIES


Source: MarketsandMarkets analysis

5.6 SUPPLY CHAIN ANALYSIS

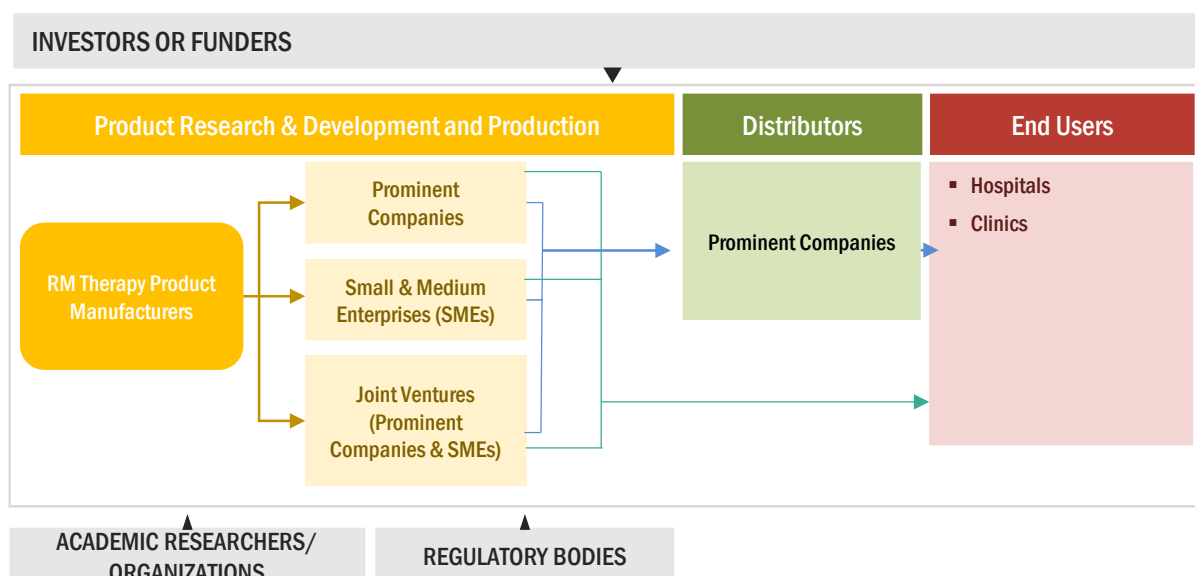
Prominent companies providing regenerative medicine, small and medium-sized enterprises (SMEs), distributors, and end customers (research institutes and contract research organizations) are the key stakeholders in the supply chain of regenerative medicine. Investors/funders are the major influencers.

Prominent companies

Prominent companies are well-established, financially stable players. They have been operating in the industry for several years and have a diversified product portfolio, proprietary technologies, strong global distribution networks, and offices at strategic locations. There are prominent companies in this market, as business monetary investments differ from process to process. For instance, large-scale investments are necessary to make this business profitable. These players include Smith & Nephew plc., Novartis AG, Stryker Corporation, 3M Group, Medtronic plc., Abbvie, and Takeda Pharmaceutical Company Ltd.

Small and Medium-sized Enterprises (SMEs)

SMEs are companies with a narrow product range and weaker financial strength and distribution networks than prominent companies. Some of these companies strategically enter into partnerships or joint ventures with prominent companies to increase their shares and visibility in the market. This enables them to enhance their distribution networks, gain access to new technologies, and acquire other companies' technical/operational know-how. Prominent companies increasingly acquire SMEs to enhance their product offerings and geographical presence. Some SMEs operating in the regenerative medicine market include Vericel Corporation, MIMEDX Group, Organogenesis, MEDIPOST Co. Ltd., Zimmer Biomet, Bioventus, Amgen, Inc., and Orthocell Ltd., among others.

FIGURE 22 SUPPLY CHAIN ANALYSIS: REGENERATIVE MEDICINE MARKET


Source: MarketsandMarkets Analysis

5.7 VALUE CHAIN ANALYSIS

The regenerative medicine market has grown significantly in the last decade. The value chain of regenerative medicine begins with R&D. Extensive R&D is required to develop therapies. R&D activities are divided into in-house and outsourced tasks. In-house tasks focus on critical activities that deal with fundamental analysis and the electronic interpretation of testing parameters. During the R&D phase, basic research is conducted, and prototypes are built, adding <10% value to the product. The raw material, manufacturing, and assembly phases claim most of the addition (around 60– 70%). Around 10% to 20% of the value is added during marketing and sales activities, including geographic distribution and access. However, the gross profit earned by every stakeholder depends on the region and product type.

COLLECTION & PRODUCT DEVELOPMENT

The R&D phase comprises collection and product development, process regulatory approval, and sustainable engineering. In the regenerative medicine market, various players have invested significantly in R&D activities, which has led to the launch of advanced or unique products in the market.

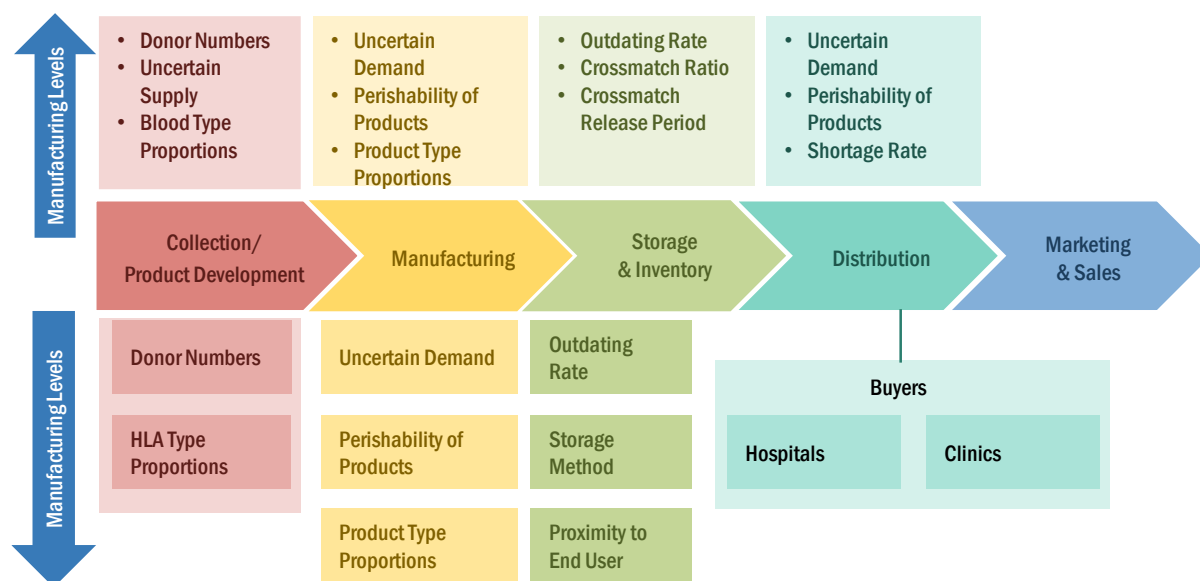
STORAGE & MANUFACTURING

The storage and manufacturing phase involves storing blood products and manufacturing CGT products. Companies in this market have been involved in expansion activities, primarily to set up new sites to produce regenerative medicine therapy products.

DISTRIBUTION & MARKETING

This is an essential procedure in the value chain of any product. Players have entered into partnerships to distribute regenerative medicine therapies to reach a larger customer base.

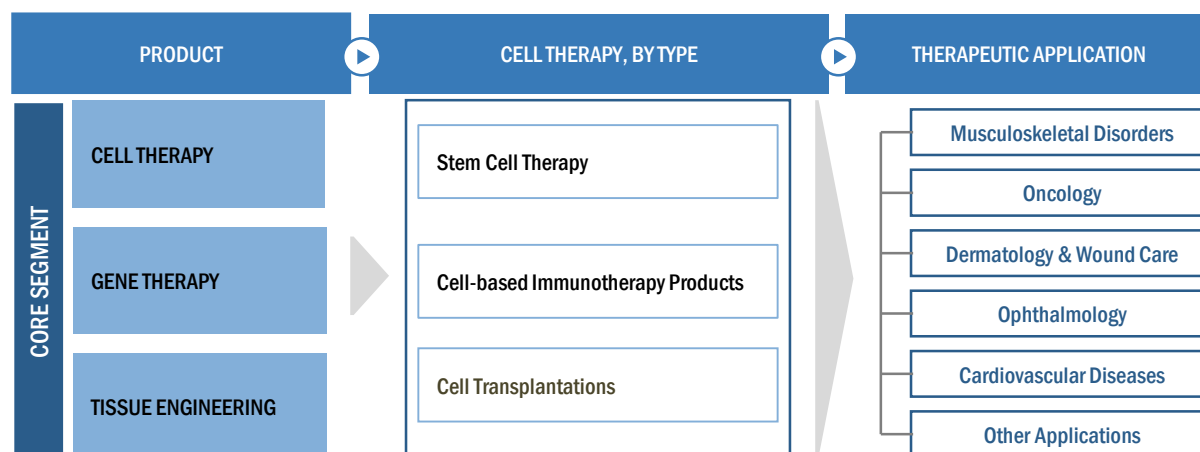
FIGURE 23 VALUE CHAIN ANALYSIS: REGENERATIVE MEDICINE MARKET



Source: MarketsandMarkets analysis

5.8 ECOSYSTEM ANALYSIS

FIGURE 24 REGENERATIVE MEDICINE MARKET: ECOSYSTEM ANALYSIS



Source: Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5.9 KEY CONFERENCES AND EVENTS IN 2022

5.9.1 REGENERATIVE MEDICINE CONFERENCES IN 2022

TABLE 5 REGENERATIVE MEDICINE CONFERENCES IN 2022

COUNTRY/REGION	YEAR & MONTH	CONFERENCE	DESCRIPTION
UK	August 2022	4th Global Conference on Tissue Engineering and Regenerative Medicine, Stem Cell Research	This event focused on cell therapy involving introducing cells directly into the body for healing. The therapeutic function propagates in a single cell. For regenerative medicine, cell therapy aims to establish a long-term transplant with the ability to conduct the organ's functions.
UAE	November 2022	International Conference on Regenerative Medicine and Therapeutic Applications ICRMTA	This event focused on the principles of regenerative medicine, the biologic and molecular basis for regenerative medicine, cells and tissue development, biomaterials for regenerative medicine, and therapeutic applications
Japan	November 2022	International Conference on Developmental Biology and Regenerative Medicine ICDBRM	Developmental biology, development, differentiation, growth mechanisms, molecular genetics of development, and control of gene expression
UK	December 2022	International Conference on Tissue Engineering & Regenerative Medicine	This event will focus on recent research regarding stem cell and regenerative medicine and their analysis. This conference will consist of keynote talks, session talks, oral presentations, poster presentations, and B2B networking.

Source: Org. Websites, Expert Interviews, and MarketsandMarkets Analysis

5.10 REGULATORY ANALYSIS

5.10.1 FDA APPROVALS

Although the approval of products for clinical use is challenging, the therapeutic efficacy of cell and tissue replacement and stem cell therapies have resulted in a rise in the number of regulatory-approved diagnostic products. In 2022, 3 new therapies were approved by the FDA and EU. Moreover, 1,220 clinical trials are going on in the year 2022.

5.10.2 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

TABLE 6 NORTH AMERICA: REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
US	US Food and Drug Administration	Government Regulatory Body	It is a cabinet-level department of the US government concerned with US policies regarding drugs and food safety. The FDA regulates food, drugs, biologics, medical devices, electronic products, cosmetics, veterinary products, and tobacco products.
US	Center for Biologics Evaluation and Research (CBER)	Center within US FDA	It is a part of the US FDA. CBER regulates biological products for human use under applicable federal laws, including the Public Health Service Act and the Federal Food, Drug, and Cosmetic Act.
US	Alliance of Advanced Biomedical Engineering (AABME)	Part of The American Society of Mechanical Engineers	The Alliance of Advanced Biomedical Engineering (AABME) is the American Society of Mechanical Engineers (ASME) initiative designed to stimulate biomedical innovation by bringing together and providing resources to the biomedical engineering community.

Source: US Food and Drug Administration, Center for Biologics Evaluation and Research (CBER), and Alliance of Advanced Biomedical Engineering (AABME)

TABLE 7 EUROPE: REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY/ REGION	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
Netherlands	European Medicines Agency (EMA)	Government Regulatory Body	The EMA is a decentralized agency of the European Union (EU) responsible for the scientific evaluation, supervision, and safety monitoring of medicines in the EU.
UK	Medicines and Healthcare products Regulatory Agency (MHRA)	Government Regulatory Body	The MHRA is a government regulatory agency of the Department of Health and Social Care in the UK, responsible for regulating the safety of medicines and medical devices.

Source: European Medicines Agency (EMA) and Medicines and Healthcare products Regulatory Agency (MHRA)

TABLE 8 ASIA PACIFIC: REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
India	Central Drugs Standard Control Organization (CDSCO)	Government Regulatory Body	The CDSCO is a national regulatory body for cosmetics, pharmaceuticals, and medical devices in India.
Japan	Pharmaceuticals and Medical Devices Agency (PMDA)	Government Regulatory Body	The PMDA is a regulatory agency in Japan that works with the Ministry of Health, Labour, and Welfare. It ensures the safety, efficacy, and quality of pharmaceuticals and medical devices.
India	National Centre for Cell Science (NCCS)	Autonomous Organization	The NCCS, an autonomous organization aided by the Department of Biotechnology, Government of India, was established to facilitate cell biology research in India.
South Korea	Ministry of Food and Drug Safety (MFDS)	Government Regulatory Body	The Ministry of Food and Drug Safety (MFDS), formerly known as the Korea Food & Drug Administration (KFDA), oversees the safety and efficacy of drugs.
Australia	TGA	Government Regulatory Body	This government authority is responsible for evaluating, assessing, and monitoring products defined as therapeutic goods. TGA regulates medicines, medical devices, and biologicals.

Source: Central Drugs Standard Control Organization (CDSCO), Pharmaceuticals and Medical Devices Agency (PMDA), National Centre for Cell Science (NCCS), and Ministry of Food and Drug Safety (MFDS)

TABLE 9 REST OF THE WORLD: REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
Saudi Arabia	Saudi Food and Drug Authority (SFDA)	Government Regulatory Body	The SFDA is a government regulatory body in Saudi Arabia. Its objectives are to ensure the safety of food and drugs for humans and animals, the safety of biological & chemical substances, and the safety of electronic devices related to human health.
Egypt	Egyptian Drug Authority (EDA)	Government Regulatory Body	The EDA is a pharmaceutical regulatory body in Egypt. It is a part of the Ministry of Health (MOH) in Egypt and is responsible for protecting public health by regulating the safety and quality of pharmaceutical products.

Source: Saudi Food and Drug Authority (SFDA) and Egyptian Drug Authority (EDA)

5.11 PRICING ANALYSIS

TABLE 10 PRICES OF REGENERATIVE MEDICINES

PRODUCT	PRICE PER TREATMENT (USD)
Zolgensma	4 million-5 million
Luxturna	\$850,000- \$950,000 per a one-time treatment
Kymriah	\$475,000-575,000 for one treatment course
Zynteglo	1-2 million
Stemchymal	USD 5,862-58,620
Gendicine and Oncorine	USD 387 per injection
Hearticellgram	USD 169,202
Cupistem	USD 3,000- 5,000 per treatment
Cartistem (Korea)	20,000-40,000
Hearticellgram-AMI (Korea)	15,000-20,000 for one shot
NEURONATA-R (Korea)	55,000- 70,000 annually
Stempeucel (India)	2,000-3,000/treatment
Temcell HS (Japan)	7,000-8,000 per bag
Allostem (US)	500-3,500
Ostecel Plus (US)	400-5,500
Trinity Evolution(US)	500-5,500

Source: Company website and MarketsandMarkets Analysis

5.11.1 REGENERATIVE MEDICINE MARKET: AVERAGE SELLING PRICE OF TREATMENT

Regenerative medicine costs vary depending on the number of products administered during the treatment. The average cost of regenerative medicine products ranges between USD 5,000 to USD 50,000 for one treatment. It is also imperative to consider that the cost of regenerative medicine products will vary depending on the type of treatment.

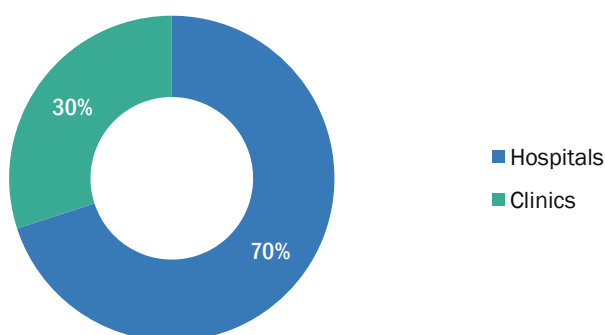
Regenerative medicine has gained more traction with time, as customers are looking at therapies like stem cell injection for regular use. The reimbursement policies and insurance claims have also helped attract customers towards these therapies, which has substantially affected the cost. However, a major reason for the perception change of stem cell treatments is that stem cell clinics sell unproven and dangerous offerings. Additionally, some universities and large medical centers also offer stem cell offerings that are not proven clinically. These increasing malpractices affect patients who may pay hefty amounts for baseless treatments; they may even pay above USD 100,000.

Valid and clinically-proven gene therapy prices will not likely be reduced owing to the high cost of gene delivery technologies and raw materials, such as collecting samples from patients, and the requirement of multiple products & procedures for an impactful treatment. Regulation is also a major factor in making the production of gene therapies commercially viable. As the industry matures, regulatory authorities could coordinate measures and gain regulatory approvals for several RM therapies by standardizing elements of the production process.

5.12 KEY STAKEHOLDERS AND BUYING CRITERIA

5.12.1 KEY STAKEHOLDERS IN BUYING PROCESS

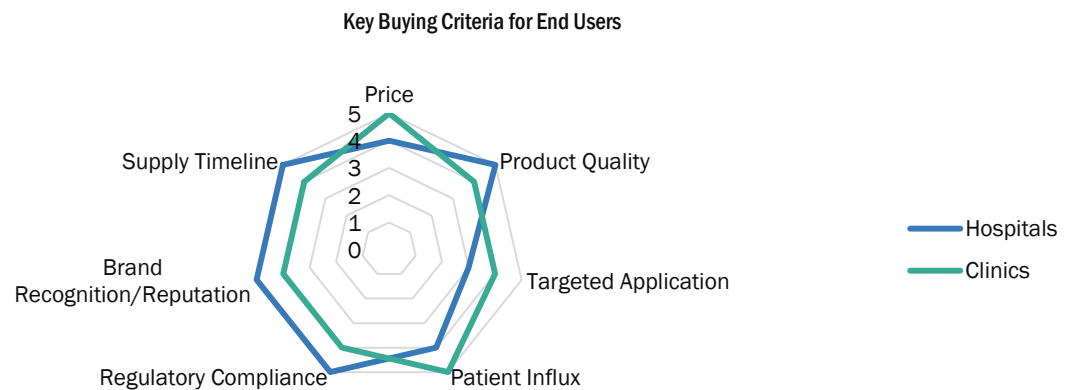
FIGURE 25 INFLUENCE OF STAKEHOLDERS ON BUYING PROCESS OF REGENERATIVE MEDICINE PRODUCTS



Note: Healthcare providers include hospitals, clinics, or cell & regenerative medicine therapy centers.

5.12.2 BUYING CRITERIA FOR REGENERATIVE MEDICINE PRODUCTS

FIGURE 26 KEY BUYING CRITERIA FOR END USERS



Note: Respondents rated each criterion on a scale of one (least important) to ten (most important).

6 REGENERATIVE MEDICINE MARKET, BY PRODUCT

KEY FINDINGS

- The cell therapy segment accounted for the largest share of 51% of the regenerative medicine market in 2021. This segment is projected to reach USD 22,177.95 million by 2027 from USD 6,328.08 million in 2022, at a CAGR of 28.5% during the forecast period.
- The gene therapy segment accounted for the second-largest share of 33% of the regenerative medicine market in 2021. This segment is projected to register the highest CAGR of 27.1% during the forecast period.
- The market for tissue engineering is projected to reach a value of USD 5,043.50 million by 2027 from an estimated value of USD 1,820.20 million in 2022, at a CAGR of 22.6% during the forecast period.

6.1 INTRODUCTION

Based on product, the regenerative medicine market is segmented into gene therapy, cell therapy, and tissue engineering products. The cell therapy segment accounted for the largest share of 51% of the regenerative medicine market in 2021. The high adoption of stem cell therapy is a key contributor to the growth of this segment. The growing applications of hematopoietic stem cell transplantations (HSCT) are expected to drive the growth of the cell therapy market.

TABLE 11 REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022– 2027)
Cell Therapy	3,665.10	4,881.86	6,328.08	8,034.10	10,331.59	13,307.08	17,166.10	22,177.95	28.5%
Gene Therapy	2,449.07	3,190.36	4,045.17	5,133.15	6,518.92	8,285.22	10,538.11	13,413.58	27.1%
Tissue Engineering	1,269.84	1,544.06	1,820.20	2,311.10	2,819.38	3,431.73	4,166.40	5,043.50	22.6%
Total	7,384.01	9,616.28	12,193.44	15,478.36	19,669.89	25,024.04	31,870.62	40,635.04	27.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.2 GENE THERAPY

6.2.1 STRONG PRODUCT PIPELINE DUE TO BROADENING APPLICATIONS IN DISEASE TREATMENT TO DRIVE MARKET

Gene therapy is a treatment technique that uses genes to treat diseases. This branch of medicine focuses on altering a cell's genetic makeup to have a therapeutic impact or to treat a disease by replacing or repairing damaged genetic material. FDA approval for gene treatments initiated with products such as LUXTURN A (for RPE65 mutation-induced blindness) from Spark Therapeutics. The gene therapy market has a robust product pipeline for the upcoming two years. Companies are submitting new clinical trial data for gene therapy candidates for FDA approval. BioMarin Pharmaceutical Inc. submitted new data in 2021 on its gene therapy candidate – Roctavian.

Gene therapies are closely being studied across several applications, including pediatric spinal muscular atrophy, biallelic RPE65 mutation-associated retinal dystrophy, lymphoma, TDT-triggered hemoglobin (Hb) reduction, head & neck squamous cell carcinoma, GVHD after haploidentical bone marrow transplantation, and severe combined immunodeficiency due to ADA deficiency. The benefits of gene therapy treatment in such applications are expected to propel the growth of this segment in the coming years.

TABLE 12 REGENERATIVE MEDICINE MARKET FOR GENE THERAPY, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,283.84	1,640.22	2,039.37	2,537.23	3,158.48	3,934.10	4,902.83	6,113.24	24.6%
Europe	525.04	687.58	876.71	1,119.10	1,430.05	1,829.41	2,342.85	3,003.67	27.9%
Asia Pacific	472.69	639.55	840.29	1,102.78	1,445.81	1,893.78	2,478.50	3,241.31	31.0%
Latin America	74.33	98.84	127.85	165.41	214.06	277.11	358.84	464.85	29.5%
Middle East & Africa	93.17	124.17	160.94	208.64	270.51	350.82	455.08	590.51	29.7%
Total	2,449.07	3,190.36	4,045.17	5,133.15	6,518.92	8,285.22	10,538.11	13,413.58	27.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 13 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR GENE THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	1,104.06	1,402.28	1,733.25	2,143.58	2,652.52	3,284.03	4,067.95	5,041.37	23.8%
Canada	179.77	237.94	306.12	393.65	505.97	650.07	834.89	1,071.87	28.5%
Total	1,283.84	1,640.22	2,039.37	2,537.23	3,158.48	3,934.10	4,902.83	6,113.24	24.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 14 EUROPE: REGENERATIVE MEDICINE MARKET FOR GENE THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	140.64	179.61	223.21	277.51	345.15	429.45	534.50	665.45	24.4%
France	90.23	119.17	153.23	197.23	254.11	327.73	423.11	546.81	29.0%
UK	63.01	85.62	113.12	149.44	197.41	260.79	344.54	455.25	32.1%
Rest of Europe	231.16	303.18	387.15	494.92	633.38	811.44	1,040.69	1,336.15	28.1%
Total	525.04	687.58	876.71	1,119.10	1,430.05	1,829.41	2,342.85	3,003.67	27.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 15 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR GENE THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	150.03	192.72	239.63	296.44	364.66	445.72	540.79	650.50	22.1%
China	102.81	140.69	187.03	248.43	329.72	437.32	579.71	768.12	32.6%
India	69.56	99.13	136.85	188.27	258.24	353.29	482.18	656.72	36.8%
South Korea	83.35	113.91	151.17	200.43	265.56	351.64	465.38	615.67	32.4%
Rest of Asia Pacific	66.94	93.11	125.62	169.21	227.62	305.82	410.43	550.30	34.4%
Total	472.69	639.55	840.29	1,102.78	1,445.81	1,893.78	2,478.50	3,241.31	31.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.3 CELL THERAPY

Good Manufacturing Practice (GMP) refers to the regulations or guidelines set up by various agencies which control the authorization and licensing for the manufacturing & sales of food, drug products, and active pharmaceutical products. Companies increasingly focus on gaining GMP approvals for stem cell therapy production facilities. These approvals are likely to enhance the quality and efficiency of production facilities and increase overall market growth for stem cell therapy products. In May 2020, Takara Bio, Inc. completed the establishment of a new GMP facility—the Center for Gene and Cell Therapy Processing II (CGCPII)—in Shiga, Japan. In September 2022, Charles River Laboratories, International Inc. received regulatory approval, in the form of Good Manufacturing Practice (GMP) certification from the European Medicines Agency (EMA), for commercial manufacturing of allogeneic cell therapy products. The GMP certification of Charles River's Memphis contract development and manufacturing (CDMO) facility complements an existing GMP license for Investigational Medicinal Product (IMP) production. The Memphis site can manufacture and ship drug products for Europe distribution.

TABLE 16 REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	2,304.90	3,065.05	3,966.38	5,026.82	6,452.73	8,295.79	10,681.30	13,773.05	28.3%
Cell-based Immunotherapy Products	1,360.20	1,816.81	2,361.69	3,007.28	3,878.86	5,011.29	6,484.80	8,404.90	28.9%
Total	3,665.10	4,881.86	6,328.08	8,034.10	10,331.59	13,307.08	17,166.10	22,177.95	28.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 17 REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,927.15	2,511.35	3,183.45	3,862.06	4,795.58	5,959.75	7,412.55	9,226.73	23.7%
Europe	787.84	1,051.97	1,366.95	1,777.34	2,312.43	3,010.60	3,922.21	5,113.39	30.2%
Asia Pacific	702.39	977.83	1,322.09	1,785.67	2,409.59	3,248.99	4,377.92	5,895.87	34.9%
Latin America	109.93	151.02	201.73	269.39	359.70	480.23	641.13	855.92	33.5%
Middle East & Africa	137.79	189.68	253.86	339.64	454.28	607.50	812.29	1,086.04	33.7%
Total	3,665.10	4,881.86	6,328.08	8,034.10	10,331.59	13,307.08	17,166.10	22,177.95	28.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 18 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	1,657.32	2,147.10	2,705.72	3,236.10	3,976.12	4,887.86	6,011.56	7,396.95	22.3%
Canada	269.83	364.25	477.72	625.96	819.46	1,071.89	1,400.99	1,829.78	30.8%
Total	1,927.15	2,511.35	3,183.45	3,862.06	4,795.58	5,959.75	7,412.55	9,226.73	23.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 19 EUROPE: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/ Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	211.07	274.90	348.22	441.08	558.69	707.62	896.18	1,134.86	26.7%
France	135.41	182.36	238.97	313.34	411.09	539.64	708.82	931.62	31.3%
UK	94.55	130.98	176.36	237.32	319.18	429.11	576.70	774.88	34.5%
Rest of Europe	346.82	463.73	603.39	785.60	1,023.48	1,334.24	1,740.51	2,272.03	30.4%
Total	787.84	1,051.97	1,366.95	1,777.34	2,312.43	3,010.60	3,922.21	5,113.39	30.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 20 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	225.07	294.71	373.35	470.35	588.92	732.37	903.70	1,105.05	24.2%
China	152.11	215.14	295.51	405.37	555.44	760.30	1,039.80	1,420.95	36.9%
India	102.90	151.57	216.16	307.13	434.93	614.05	864.61	1,214.50	41.2%
South Korea	123.30	174.13	238.73	326.90	447.14	611.02	834.26	1,138.25	36.7%
Rest of Asia Pacific	99.01	142.29	198.34	275.91	383.16	531.26	735.55	1,017.11	38.7%
Total	702.39	977.83	1,322.09	1,785.67	2,409.59	3,248.99	4,377.92	5,895.87	34.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.3.1 STEM CELL THERAPY

Stem cell-based products are somatic cellular therapies involving stem cell lines to treat various diseases. These stem cells can be procured from adipose tissue-derived mesenchymal stem cells, bone marrow-derived mesenchymal stem cells, and placental or cord-derived stem cells. Stem cell therapies have several therapeutic applications, but exhaustive research and clinical evidence are required before commercializing stem cell therapy products.

Adipose-derived stem cell tissues are a type of mesenchymal stem cell that can be obtained easily from adipose tissues. It also possesses many of the same regenerative properties as other MSCs do. These cells are multipotent and are easy to isolate & harvest. Unlike bone marrow stem cells present in a low quantity in the bone marrow, adipose stem cells can be harvested in high quantities using minimally invasive procedures such as liposuction. Moreover, these tissues have an abundance of MSCs and chondrogenic potentials compared to bone marrow stem cells.

Various companies have commercialized MSC-based therapies, contributing to this segment's revenue growth. Prochymal is a stem cell therapy currently marketed by Smith+Nephew (developed by Osiris Therapeutics). Prochymal is derived from human mesenchymal stromal cells and is designed to treat GvHD patients. Bone marrow-derived MSC products available in the market include TEMCELL HS Inj by JCR Pharmaceuticals Co. Ltd. (Japan); Cellgram-AMI by Pharmicell Co. Ltd. (South Korea); Stravix, Cartiform, Truskin, and Bio4 by Osiris Therapeutics Inc., (US); and Osteocel cellular bone graft by NuVasive Inc., (US).

TABLE 21 REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	827.87	1,104.86	1,434.97	1,825.31	2,351.88	3,035.18	3,923.13	5,078.64	28.8%
Cell Transplantations	1,477.02	1,960.19	2,531.42	3,201.52	4,100.86	5,260.61	6,758.17	8,694.40	28.0%
Total	2,304.90	3,065.05	3,966.38	5,026.82	6,452.73	8,295.79	10,681.30	13,773.05	28.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 22 REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,212.52	1,578.28	1,998.39	2,421.60	3,003.49	3,728.31	4,631.78	5,758.68	23.6%
Europe	495.41	660.36	856.61	1,111.87	1,444.11	1,876.85	2,440.89	3,176.61	30.0%
Asia Pacific	441.40	613.06	826.95	1,114.25	1,499.96	2,017.59	2,712.02	3,643.36	34.5%
Latin America	69.04	94.58	125.98	167.75	223.34	297.31	395.76	526.81	33.1%
Middle East & Africa	86.53	118.76	158.46	211.36	281.83	375.73	500.85	667.58	33.3%
Total	2,304.90	3,065.05	3,966.38	5,026.82	6,452.73	8,295.79	10,681.30	13,773.05	28.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 23 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	1,042.77	1,349.43	1,698.62	2,029.32	2,490.60	3,058.28	3,757.16	4,617.84	22.1%
Canada	169.75	228.85	299.77	392.28	512.89	670.03	874.62	1,140.85	30.6%
Total	1,212.52	1,578.28	1,998.39	2,421.60	3,003.49	3,728.31	4,631.78	5,758.68	23.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 24 EUROPE: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	132.76	172.66	218.40	276.24	349.40	441.90	558.85	706.66	26.5%
France	85.16	114.50	149.81	196.12	256.88	336.67	441.52	579.37	31.1%
UK	59.45	82.22	110.51	148.44	199.29	267.46	358.82	481.27	34.2%
Rest of Europe	218.04	290.98	377.90	491.07	638.54	830.82	1,081.71	1,409.31	30.1%
Total	495.41	660.36	856.61	1,111.87	1,444.11	1,876.85	2,440.89	3,176.61	30.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 25 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	141.48	184.87	233.71	293.82	367.13	455.60	561.00	684.57	24.0%
China	95.60	134.91	184.90	253.07	345.98	472.52	644.77	879.13	36.6%
India	64.66	95.02	135.19	191.62	270.69	381.26	535.53	750.43	40.9%
South Korea	77.47	109.12	149.23	203.82	278.07	379.01	516.15	702.40	36.3%
Rest of Asia Pacific	62.20	89.14	123.92	171.92	238.09	329.21	454.56	626.84	38.3%
Total	441.40	613.06	826.95	1,114.25	1,499.96	2,017.59	2,712.02	3,643.36	34.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.3.1.1 Cell Transplantations

6.3.1.1.1 Potential of Allo-SCT in treating AML to drive segment growth

Allogeneic hematopoietic cell transplantation (HCT) is gaining demand as a therapeutic strategy for many patients suffering from hematologic malignancies and other marrow failures or immune system disorders. More than 15,000 HCTs are performed in the US each year, which is expected to increase over time. In addition, allogeneic stem cell transplant (allo-SCT) has emerged as an important treatment strategy in adults suffering from Acute Myeloid Leukemia (AML). Allo-SCT is gaining demand attributed to its augmented anti-leukemic activity consequent upon dose intensification and the genesis of a potent graft-versus-leukemia effect. Moreover, increasing donor availability and the advent of reduced-intensity conditioning (RIC) regimens have intensified the preference for allo-SCT as a therapeutic strategy. Allo-SCT is a key component of the treatment algorithm in patients with AML.

TABLE 26 REGENERATIVE MEDICINE MARKET FOR CELL TRANSPLANTATIONS, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	775.92	1,008.39	1,274.78	1,542.29	1,909.83	2,366.92	2,935.76	3,644.13	23.4%
Europe	317.56	422.39	546.74	708.14	917.74	1,190.16	1,544.46	2,005.59	29.7%
Asia Pacific	283.45	392.61	528.13	709.65	952.64	1,277.78	1,712.70	2,294.29	34.1%
Latin America	44.41	60.64	80.50	106.84	141.77	188.10	249.56	331.09	32.7%
Middle East & Africa	55.68	76.16	101.27	134.61	178.88	237.65	315.68	419.30	32.9%
Total	1,477.02	1,960.19	2,531.42	3,201.52	4,100.86	5,260.61	6,758.17	8,694.40	28.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 27 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR CELL TRANSPLANTATIONS, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	667.25	862.13	1,083.53	1,292.45	1,583.74	1,941.66	2,381.61	2,922.56	22.0%
Canada	108.67	146.26	191.25	249.84	326.09	425.26	554.15	721.57	30.4%
Total	775.92	1,008.39	1,274.78	1,542.29	1,909.83	2,366.92	2,935.76	3,644.13	23.4%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 28 EUROPE: REGENERATIVE MEDICINE MARKET FOR CELL TRANSPLANTATIONS, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	85.03	110.38	139.36	175.94	222.11	280.38	353.91	446.67	26.2%
France	54.57	73.22	95.61	124.90	163.27	213.55	279.47	365.98	30.8%
UK	38.11	52.59	70.53	94.54	126.65	169.59	227.02	303.82	33.9%
Rest of Europe	139.85	186.20	241.25	312.76	405.72	526.64	684.06	889.12	29.8%
Total	317.56	422.39	546.74	708.14	917.74	1,190.16	1,544.46	2,005.59	29.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 29 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR CELL TRANSPLANTATIONS, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	90.79	118.33	149.22	187.13	233.23	288.71	354.60	431.61	23.7%
China	61.37	86.38	118.07	161.18	219.76	299.33	407.35	553.93	36.2%
India	41.53	60.86	86.34	122.04	171.91	241.44	338.18	472.53	40.5%
South Korea	49.78	69.91	95.33	129.81	176.57	239.94	325.79	442.01	35.9%
Rest of Asia Pacific	39.98	57.13	79.17	109.49	151.16	208.36	286.78	394.21	37.9%
Total	283.45	392.61	528.13	709.65	952.64	1,277.78	1,712.70	2,294.29	34.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.3.1.2 Stem Cell Therapy Products

Stem cell therapy is an evolving technology that can be a great alternative to traditional treatments. Presently, the stem cell therapy market is evolving, with only a few stem cell-based products commercially available in the market. The stem cell industry has many companies focused on developing stem cell therapy products for various therapeutic applications (such as cardiovascular diseases, musculoskeletal diseases, wounds and injuries, and neurological diseases). However, many stem cell-based products are currently in the clinical research phase.

Prominent players that contribute to the stem cell therapy market include Smith+Nephew (UK), MEDIPOST Co., Ltd. (South Korea), Anterogen Co., Ltd. (South Korea), CORESTEM (South Korea), Pharmicell Inc. (South Korea), NuVasive, Inc. (US), RTI Surgical (US), AlloSource (US), JCR Pharmaceuticals Co., Ltd. (Japan), Takeda Pharmaceutical Company Limited (Japan), Holostem Therapie Avanzate Srl (Italy), Orthofix (US), Regrow Biosciences Pvt Ltd. (India), and STEMPEUTICS RESEARCH PVT LTD. (India).

TABLE 30 REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	436.59	569.89	723.61	879.32	1,093.66	1,361.39	1,696.02	2,114.55	23.9%
Europe	177.85	237.97	309.87	403.74	526.37	686.69	896.43	1,171.03	30.5%
Asia Pacific	157.95	220.45	298.81	404.60	547.32	739.81	999.31	1,349.07	35.2%
Latin America	24.64	33.95	45.48	60.91	81.57	109.21	146.20	195.72	33.9%
Middle East & Africa	30.85	42.60	57.19	76.75	102.96	138.09	185.17	248.28	34.1%
Total	827.87	1,104.86	1,434.97	1,825.31	2,351.88	3,035.18	3,923.13	5,078.64	28.8%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 31 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	375.52	487.30	615.09	736.87	906.86	1,116.62	1,375.55	1,695.27	22.5%
Canada	61.08	82.60	108.52	142.44	186.80	244.77	320.47	419.28	31.0%
Total	436.59	569.89	723.61	879.32	1,093.66	1,361.39	1,696.02	2,114.55	23.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 32 EUROPE: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	47.73	62.28	79.04	100.31	127.29	161.52	204.94	259.99	26.9%
France	30.59	41.28	54.20	71.21	93.61	123.13	162.04	213.39	31.5%
UK	21.34	29.62	39.97	53.90	72.64	97.87	131.80	177.45	34.7%
Rest of Europe	78.19	104.79	136.65	178.31	232.82	304.17	397.65	520.20	30.7%
Total	177.85	237.97	309.87	403.74	526.37	686.69	896.43	1,171.03	30.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 33 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	50.69	66.54	84.49	106.69	133.90	166.89	206.40	252.96	24.5%
China	34.23	48.53	66.82	91.89	126.22	173.18	237.41	325.20	37.2%
India	23.13	34.16	48.84	69.58	98.78	139.81	197.35	277.89	41.6%
South Korea	27.69	39.21	53.90	74.01	101.50	139.06	190.36	260.39	37.0%
Rest of Asia Pacific	22.21	32.01	44.75	62.43	86.93	120.86	167.79	232.63	39.1%
Total	157.95	220.45	298.81	404.60	547.32	739.81	999.31	1,349.07	35.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.3.1.2.1 Autologous Therapy

6.3.1.2.1.1 Low risk of post-treatment complications to support segment growth

The stem cell is first collected from the patient, cultured using ex vivo techniques for a required time period, and subsequently used for treating the donor in autologous stem cell therapies. Autologous stem cell therapy follows a service-based model in which a single patient can be treated from a single cell source only at a time, making therapy time-consuming, costly, and unviable for production scale-up.

The growth of the autologous segment is attributed to the low risk of post-treatment complications compared to allogeneic stem cell therapy, the increasing number of clinical trials based on autologous types, and an increasing preference for autologous stem cell therapy in high-income countries.

One limiting factor associated with autologous stem cell therapies is the need to collect a sufficient number of cells suitable to undergo the manufacturing process with a required proliferation rate. Lack of ex-vivo durability is another factor that limits the adoption of these stem cell therapies. Despite the above, a gradual increase in the adoption of autologous in the coming years is expected due to minimal risks associated with immune reaction and disease transmission.

TABLE 34 REGENERATIVE MEDICINE MARKET FOR AUTOLOGOUS THERAPY, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	118.62	154.27	195.17	236.31	292.84	363.20	450.82	560.01	23.5%
Europe	48.38	64.59	83.84	108.88	141.49	183.98	239.39	311.70	30.0%
Asia Pacific	45.77	63.95	86.54	116.97	157.94	213.08	287.25	386.98	34.9%
Latin America	6.73	9.26	12.38	16.55	22.12	29.56	39.50	52.78	33.6%
Middle East & Africa	8.43	11.62	15.58	20.88	27.97	37.46	50.16	67.15	33.9%
Total	227.94	303.70	393.51	499.59	642.36	827.28	1,067.12	1,378.62	28.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 35 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR AUTOLOGOUS THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	102.02	131.90	165.88	197.98	242.75	297.78	365.45	448.70	22.0%
Canada	16.60	22.37	29.30	38.33	50.10	65.42	85.37	111.31	30.6%
Total	118.62	154.27	195.17	236.31	292.84	363.20	450.82	560.01	23.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 36 EUROPE: REGENERATIVE MEDICINE MARKET FOR AUTOLOGOUS THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	12.98	16.88	21.36	27.03	34.20	43.27	54.73	69.23	26.5%
France	8.32	11.20	14.66	19.19	25.14	32.96	43.23	56.74	31.1%
UK	5.81	8.04	10.82	14.54	19.53	26.22	35.19	47.22	34.3%
Rest of Europe	21.28	28.47	37.00	48.12	62.62	81.54	106.24	138.52	30.2%
Total	48.38	64.59	83.84	108.88	141.49	183.98	239.39	311.70	30.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 37 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR AUTOLOGOUS THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	13.80	18.09	22.94	28.92	36.24	45.11	55.70	68.17	24.3%
China	9.32	13.20	18.12	24.84	34.00	46.50	63.53	86.72	36.8%
India	6.30	9.30	13.26	18.82	26.63	37.56	52.85	74.16	41.1%
South Korea	10.29	14.64	20.07	27.50	37.62	51.41	70.21	95.80	36.7%
Rest of Asia Pacific	6.05	8.72	12.15	16.90	23.45	32.50	44.96	62.13	38.6%
Total	45.77	63.95	86.54	116.97	157.94	213.08	287.25	386.98	34.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.3.1.2.2 Allogeneic Therapy

6.3.1.2.2.1 Favorable scaling up of production processes to support segment growth

Stem cells used for allogeneic stem cell therapy are obtained from sources other than the target patient, usually from a donor. In allogeneic stem cell therapy, one cell source can produce many doses, thus making allogeneic therapy economically viable and less time-consuming. Moreover, it is easier to ramp up allogeneic therapy production processes than autologous therapy. More than ten approved commercialized allogeneic stem cell therapies are available globally.

Some examples of commercialized allogeneic stem cell therapies include Bio4 (Smith+Nephew), CARTISTEM (MEDIPOST Co., Ltd.), and hUCB-MS - allogeneic human Umbilical Cord Blood-derived Mesenchymal Stem Cell (MEDIPOST).

TABLE 38 REGENERATIVE MEDICINE MARKET FOR ALLOGENEIC THERAPY, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	317.97	415.62	528.44	643.01	800.82	998.19	1,245.20	1,554.54	24.1%
Europe	129.46	173.38	226.03	294.86	384.88	502.70	657.04	859.33	30.6%
Asia Pacific	112.18	156.49	212.27	287.63	389.38	526.73	712.07	962.09	35.3%
Latin America	17.91	24.69	33.10	44.36	59.45	79.65	106.70	142.94	34.0%
Middle East & Africa	22.42	30.98	41.61	55.87	74.99	100.63	135.01	181.13	34.2%
Total	599.94	801.16	1,041.45	1,325.72	1,709.51	2,207.90	2,856.02	3,700.02	28.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 39 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR ALLOGENEIC THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	273.50	355.40	449.22	538.89	664.12	818.84	1,010.10	1,246.57	22.6%
Canada	44.48	60.22	79.22	104.11	136.71	179.35	235.11	307.97	31.2%
Total	317.97	415.62	528.44	643.01	800.82	998.19	1,245.20	1,554.54	24.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 40 EUROPE: REGENERATIVE MEDICINE MARKET FOR ALLOGENEIC THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	34.75	45.40	57.68	73.28	93.09	118.25	150.20	190.76	27.0%
France	22.27	30.08	39.55	52.02	68.47	90.17	118.81	156.65	31.7%
UK	15.53	21.58	29.16	39.36	53.12	71.65	96.61	130.23	34.9%
Rest of Europe	56.91	76.32	99.65	130.19	170.20	222.63	291.41	381.68	30.8%
Total	129.46	173.38	226.03	294.86	384.88	502.70	657.04	859.33	30.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 41 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR ALLOGENEIC THERAPY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	36.89	48.45	61.55	77.77	97.65	121.78	150.70	184.79	24.6%
China	24.90	35.33	48.70	67.06	92.22	126.69	173.89	238.48	37.4%
India	16.83	24.86	35.59	50.76	72.15	102.25	144.51	203.73	41.8%
South Korea	17.40	24.57	33.83	46.51	63.88	87.65	120.16	164.59	37.2%
Rest of Asia Pacific	16.16	23.29	32.60	45.53	63.48	88.36	122.82	170.50	39.2%
Total	112.18	156.49	212.27	287.63	389.38	526.73	712.07	962.09	35.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.3.2 CELL-BASED IMMUNOTHERAPY PRODUCTS

6.3.2.1 Adoption of T-cell therapy in cancer treatment to support market growth

Cell-based immunotherapy uses the patient's immune system cells as a therapeutic agent. This process is carried out by isolating cells from the patient's body, followed by alteration/modification, expansion, and finally re-infused into the patient.

Some of the strategies currently being investigated in this field are mentioned below:

- Adoptive T cell therapy: This therapy focuses on increasing tumor-specific T cells: Different T cells have been used for adoptive therapy, including tumor-infiltrating lymphocytes (TILs) expanded ex vivo or peripheral blood T cells genetically engineered ex vivo to either express a cancer-specific T cell receptor (TCR) or a chimeric antigen receptor (CAR; CAR-T cells) against a cancer epitope.
- Natural killer (NK) cell therapy aims to activate innate immune responses. Different from events required for T cell activation, NK cell activation is governed by the interaction of NK receptors with target cells, independent of antigen processing and presentation. Natural killer (NK) cell is a specialized immune effector cell type that plays a critical role in immune activation against abnormal cells.

The majority of the cell-based approaches are still in the experimental phase compared to other cancer immunotherapy strategies; the first two adoptive T cell therapies (Ex vivo expanded peripheral blood T cells and Ex vivo – genetically engineered T cells) have already been approved in the US, and partially approved across the Europe region. Kymriah (Novartis AG) and Yescarta (Gilead Sciences) are approved CAR-T cell therapies targeting CD19 B-cell lymphomas.

TABLE 42 REGENERATIVE MEDICINE MARKET FOR CELL-BASED IMMUNOTHERAPY PRODUCTS, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	714.64	933.07	1,185.06	1,440.45	1,792.09	2,231.44	2,780.77	3,468.05	24.0%
Europe	292.43	391.60	510.33	665.47	868.32	1,133.75	1,481.32	1,936.77	30.6%
Asia Pacific	260.99	364.77	495.15	671.42	909.63	1,231.40	1,665.90	2,252.51	35.4%
Latin America	40.88	56.44	75.75	101.65	136.37	182.93	245.37	329.11	34.2%
Middle East & Africa	51.26	70.92	95.40	128.29	172.45	231.77	311.44	418.46	34.4%
Total	1,360.20	1,816.81	2,361.69	3,007.28	3,878.86	5,011.29	6,484.80	8,404.90	28.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 43 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR CELL-BASED IMMUNOTHERAPY PRODUCTS, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	614.55	797.67	1,007.10	1,206.78	1,485.52	1,829.58	2,254.40	2,779.11	22.5%
Canada	100.08	135.39	177.96	233.68	306.57	401.86	526.37	688.93	31.1%
Total	714.64	933.07	1,185.06	1,440.45	1,792.09	2,231.44	2,780.77	3,468.05	24.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 44 EUROPE: REGENERATIVE MEDICINE MARKET FOR CELL-BASED IMMUNOTHERAPY PRODUCTS, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	78.31	102.24	129.82	164.84	209.29	265.72	337.33	428.19	27.0%
France	50.25	67.86	89.16	117.22	154.20	202.96	267.30	352.26	31.6%
UK	35.10	48.77	65.86	88.88	119.89	161.65	217.89	293.61	34.8%
Rest of Europe	128.78	172.74	225.49	294.53	384.94	503.42	658.80	862.71	30.8%
Total	292.43	391.60	510.33	665.47	868.32	1,133.75	1,481.32	1,936.77	30.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 45 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR CELL-BASED IMMUNOTHERAPY PRODUCTS, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	83.60	109.84	139.64	176.53	221.79	276.77	342.69	420.48	24.7%
China	56.51	80.23	110.61	152.30	209.46	287.78	395.03	541.82	37.4%
India	38.24	56.55	80.98	115.52	164.23	232.79	329.08	464.08	41.8%
South Korea	45.83	65.00	89.50	123.08	169.07	232.01	318.11	435.85	37.2%
Rest of Asia Pacific	36.81	53.15	74.42	104.00	145.07	202.04	280.99	390.28	39.3%
Total	260.99	364.77	495.15	671.42	909.63	1,231.40	1,665.90	2,252.51	35.4%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

6.4 TISSUE ENGINEERING (TE)

6.4.1 HIGH DEMAND FOR TE PRODUCTS IN DERMATOLOGY AND WOUND CARE APPLICATIONS TO DRIVE MARKET

The field of tissue engineering has gained considerable traction in regenerative medicine, with TE products emerging as promising alternatives for traditional wound care. The average treatment cost associated with tissue engineering is lower than traditional treatment approaches, making it a viable therapeutic approach compared to skin transplantations, which cost between USD 18,000 to USD 28,000. As a result, replacing damaged skin with skin cell sheets has become a prominent and preferred treatment option for patients. This trend is expected to support the growth of the tissue engineering segment in the regenerative medicine market.

Currently, more than 100 TE products have been commercialized in the field of regenerative medicine market. Some of the commercialized TE products include GraftJacket (Smith+Nephew), Restore (Smith+Nephew), TissueMend (TEI Biosciences), and Apligraf (Organogenesis, Inc.) (Advantages offered by TE products such as faster tissue regeneration, better biocompatibility, easy availability, and high affordability, have attracted investments from venture capitalists. Research activities aim to explore different types of tissue engineering-based regeneration products, particularly wound healing, bone healing, and burn treatment.

TABLE 46 REGENERATIVE MEDICINE MARKET FOR TISSUE ENGINEERING, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	673.78	800.87	921.53	1,228.54	1,521.04	1,882.61	2,329.35	2,881.02	25.6%
Europe	274.11	332.01	389.17	453.42	524.35	600.76	680.25	758.54	14.3%
Asia Pacific	238.87	305.76	379.81	470.18	579.95	712.57	871.85	1,061.76	22.8%
Latin America	36.90	46.80	57.57	70.57	86.16	104.74	126.66	152.25	21.5%
Middle East & Africa	46.18	58.61	72.11	88.39	107.89	131.05	158.29	189.93	21.4%
Total	1,269.84	1,544.06	1,820.20	2,311.10	2,819.38	3,431.73	4,166.40	5,043.50	22.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 47 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR TISSUE ENGINEERING, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	579.55	684.99	783.78	1,065.90	1,330.53	1,661.55	2,075.75	2,594.15	27.0%
Canada	94.23	115.87	137.75	162.64	190.51	221.06	253.60	286.87	15.8%
Total	673.78	800.87	921.53	1,228.54	1,521.04	1,882.61	2,329.35	2,881.02	25.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 48 EUROPE: REGENERATIVE MEDICINE MARKET FOR TISSUE ENGINEERING, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
UK	73.61	87.20	99.95	113.85	128.73	144.24	159.79	174.51	11.8%
Germany	47.16	57.68	68.28	80.35	93.87	108.71	124.47	140.47	15.5%
France	32.88	41.31	50.15	60.45	72.23	85.42	99.72	114.52	18.0%
Rest of Europe	120.46	145.83	170.80	198.77	229.51	262.41	296.27	329.03	14.0%
Total	274.11	332.01	389.17	453.42	524.35	600.76	680.25	758.54	14.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 49 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR TISSUE ENGINEERING, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	78.07	92.41	105.19	118.21	130.86	142.30	151.40	156.75	8.3%
China	51.34	67.46	85.89	108.95	137.63	173.12	216.73	269.89	25.7%
India	34.68	47.39	62.64	82.28	107.41	139.32	179.54	229.76	29.7%
South Korea	41.50	54.28	68.97	87.30	110.06	138.15	172.59	214.48	25.5%
Rest of Asia Pacific	33.28	44.23	57.12	73.45	93.99	119.69	151.59	190.88	27.3%
Total	238.87	305.76	379.81	470.18	579.95	712.57	871.85	1,061.76	22.8%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

7 REGENERATIVE MEDICINE MARKET, BY APPLICATION

KEY FINDINGS

- In 2021, the musculoskeletal disorders segment accounted for the largest share of 31% of the regenerative medicine market. This segment is projected to reach USD 11,624.26 million by 2027 from USD 3,723.29 million in 2022, at a CAGR of 25.6% during the forecast period.
- The oncology segment accounted for the second-largest share of 26% of the regenerative medicine market in 2021. This segment is projected to register a CAGR of 28.1% during the forecast period.
- The regenerative medicine market for the ophthalmology segment is projected to reach USD 2,485.32 million by 2027 from USD 677.68 million in 2022, at the highest CAGR of 29.7% during the forecast period.

7.1 INTRODUCTION

Based on application, the regenerative medicine market is segmented into musculoskeletal disorders, dermatology & wound care, oncology, cardiovascular diseases, ophthalmology, cell transplantation, and other applications. In 2021, the musculoskeletal disorders segment accounted for the largest market share at 31%. The large share of this segment can be attributed to the increasing prevalence of orthopedic diseases, the rising number of stem cell research projects, and the rich pipeline of stem cell products for treating musculoskeletal disorders.

TABLE 50 NUMBER OF CLINICAL TRIALS FOR REGENERATIVE MEDICINE PRODUCTS, BY APPLICATION, 2020 VS. 2021

Application	2020	2021
Musculoskeletal Disorders	54	67
Infectious Diseases	73	85
Hematology	57	64
Oncology	554	641
Cardiovascular Diseases	66	53
Central Nervous System Disorders	94	94
Metabolic Disorder	37	34
Dermatology	24	27
Genitourinary & Sex Hormones	25	19
Ophthalmology	51	51
Respiratory	20	22
Immunology	39	47
Gastroenterology	32	29

Note: The number of clinical trials for other applications is not mentioned in the above list.

Source: Alliance for Regenerative Medicine (ARM)

TABLE 51 REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	2,307.37	2,971.02	3,723.29	4,669.25	5,859.49	7,357.95	9,245.45	11,624.26	25.6%
Oncology	1,903.68	2,496.80	3,188.53	4,076.53	5,217.73	6,685.96	8,577.01	11,015.35	28.1%
Dermatology & Wound Care	1,269.64	1,649.60	2,086.76	2,642.64	3,350.22	4,251.88	5,402.06	6,870.85	26.9%
Ophthalmology	394.99	524.32	677.68	876.87	1,135.87	1,472.99	1,912.29	2,485.32	29.7%
Cardiovascular Diseases	480.04	631.24	808.37	1,036.58	1,330.96	1,711.18	2,202.89	2,839.58	28.6%
Other Applications	1,028.27	1,343.29	1,708.81	2,176.49	2,775.63	3,544.09	4,530.91	5,799.68	27.7%
Total	7,384.01	9,616.28	12,193.44	15,478.36	19,669.89	25,024.04	31,870.62	40,635.04	27.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.2 MUSCULOSKELETAL DISORDERS

7.2.1 RISING RESEARCH ON MUSCULOSKELETAL REGENERATION AND PREVALENCE OF ORTHOPEDIC DISORDERS TO DRIVE MARKET

Musculoskeletal regeneration research is majorly focused on the clinical implementation and improvements of treatments for patients with conditions of the tendons, nerves, muscles, cartilage, meniscus, and bones. This segment includes CNS-related diseases and orthopedic diseases. Human stem cells derived from the bone marrow are used for bone and cartilage repair to treat bone fractures and non-unions, regenerate articular cartilage in arthritic joints, and heal ligaments or tendons.

Researchers are focusing on clinical stem cell studies to treat patients with spinal cord injuries. The increasing prevalence of orthopedic diseases, the increasing number of stem cell research projects, and the rich pipeline of stem cell products for treating musculoskeletal disorders under both autologous and allogeneic cell therapies are the key drivers for this growing market segment. According to WR Hambrecht & Co., a leading investment banking firm, the number of surgical procedures using orthobiologics and bone graft substitutes will continue to increase in the coming years. According to the Centers for Medicare and Medicaid Services, as of 2020, approximately 900,000 surgeries (which require bone regeneration) are performed annually in the US. Spinal fusion (about 450,000 surgeries are performed annually in the US) was the most common surgery. Also, according to NIH, approximately 2.2 million bone graft procedures are being performed each year, with the number of operative procedures for repairing bony defects estimated to grow by approximately 13% annually.

CNS disorders include developmental, degenerative, and traumatic neurological conditions in the brain and spinal cord. Disorders associated with the central nervous system, such as Parkinson's disease (PD), Huntington's disease (HD), multiple sclerosis (MS), and stroke, are caused by a loss or dysfunction in neurons and glial cells in the brain or spinal cord. Tissue regeneration strategies, such as cell regeneration or drug delivery of therapeutic molecules to the central nervous system, have also demonstrated promising results in experimental animal models. However, this progress has been difficult to translate clinically. In recent years, neurons and glial cells have successfully been generated from stem cells, such as embryonic stem cells and neural stem cells, to develop stem cell-based brain transplantation therapies. The increasing incidence of neurological disorders and the growing focus of players on the development of stem cell therapeutics for the treatment of incurable CNS disorders are the key factors driving the growth of this application segment.

The emergence of iPS cells and increasing initiatives towards developing cellular and acellular products are other key factors supporting the growth of this market segment. Also, the growing cases of osteoarthritis in the geriatric population, rising accident cases and spinal surgeries, and extensive investment in research & development for regenerative musculoskeletal medicine are expected to provide growth opportunities for this segment.

Vericel Corporation (US), Organogenesis (US), Zimmer Biomet (US), AbbVie (US), Medtronic plc (US), and Aspect Biosystems (Canada) are some of the key players in this market.

TABLE 52 REGENERATIVE MEDICINE MARKET FOR MUSCULOSKELETAL DISORDERS, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,221.27	1,545.30	1,902.76	2,344.22	2,889.61	3,563.67	4,397.04	5,427.71	23.3%
Europe	494.92	638.93	802.99	1,010.13	1,271.90	1,603.01	2,022.21	2,553.39	26.0%
Asia Pacific	437.86	585.65	760.97	987.76	1,280.99	1,659.89	2,149.29	2,781.10	29.6%
Latin America	68.11	89.35	113.96	145.33	185.29	236.19	301.00	383.53	27.5%
Middle East and Africa	85.21	111.80	142.61	181.82	231.71	295.19	375.91	478.52	27.4%
Total	2,307.37	2,971.02	3,723.29	4,669.25	5,859.49	7,357.95	9,245.45	11,624.26	25.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 53 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR MUSCULOSKELETAL DISORDERS, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	1,050.63	1,321.86	1,618.38	1,982.48	2,429.71	2,979.26	3,654.75	4,485.27	22.6%
Canada	170.64	223.44	284.38	361.73	459.89	584.41	742.29	942.44	27.1%
Total	1,221.27	1,545.30	1,902.76	2,344.22	2,889.61	3,563.67	4,397.04	5,427.71	23.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 54 EUROPE: REGENERATIVE MEDICINE MARKET FOR MUSCULOSKELETAL DISORDERS, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
UK	133.15	168.02	206.29	253.37	311.29	382.56	470.26	578.17	22.9%
Germany	85.22	111.06	140.89	178.91	227.40	289.28	368.34	469.44	27.2%
France	59.35	79.48	103.48	134.69	175.28	228.08	296.77	386.15	30.1%
Rest of Europe	217.19	280.37	352.32	443.16	557.94	703.10	886.84	1,119.63	26.0%
Total	494.92	638.93	802.99	1,010.13	1,271.90	1,603.01	2,022.21	2,553.39	26.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 55 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR MUSCULOSKELETAL DISORDERS, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	140.70	177.53	216.80	263.35	318.04	381.55	454.31	536.16	19.9%
China	94.93	129.17	170.74	225.45	297.40	391.99	516.30	679.59	31.8%
India	64.11	90.67	124.23	169.59	230.78	313.15	423.82	572.25	35.7%
South Korea	76.67	103.78	136.45	179.21	235.12	308.20	403.65	528.26	31.1%
Rest of Asia Pacific	61.46	84.50	112.75	150.16	199.64	265.00	351.22	464.85	32.8%
Total	437.86	585.65	760.97	987.76	1,280.99	1,659.89	2,149.29	2,781.10	29.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.3 ONCOLOGY

7.3.1 INCREASING STEM CELL TRANSPLANTS TO DRIVE MARKET

There has been increasing use of regenerative therapy, such as cell therapy and tissue engineering scaffolds, to combat various forms of cancer. As of 2020, over 60,000 stem cell transplants were performed annually worldwide to treat cancer. According to Statista, global oncology spending has increased from USD 187 billion to USD 208 billion in 2022. Many government and non-government bodies are investing significantly in cancer research to develop regenerative medicine, including tissue engineering, cell therapy, and gene therapy. These organizations also focus on developing cancer models to effectively treat cancer patients.

Some of the notable developments in regenerative medicine for oncology applications are :

- Smart Immune (France), a clinical-stage T-cell medicine company, announced a research collaboration with Memorial Sloan Kettering Cancer Center (US) to expand its ProTcell pipeline with a next-generation CAR-ProTcell platform designed to address exhaustion of T-cell therapy for research in hematological and solid cancer diseases.
- Immatix (Germany), a clinical-stage biopharmaceutical company, is actively developing T cell-redirecting cancer immunotherapies. Bristol Myers Squibb (US) signed a license, development, and commercialization agreement for Immatix' T-cell receptor Bispecific candidate, IMA401, regenerative therapy for Recurrent and/or Refractory Solid Tumors.
- Bristol Myers Squibb (US) announced the first disclosure of results from a prespecified interim analysis of the pivotal TRANSFORM study, a global, randomized, multicenter, phase III study evaluating Breyanzi (lisocabtagene maraleucel; liso-cel), a CD19-directed CAR T-cell therapy, as a second-line treatment in adults with relapsed or refractory large B-cell lymphoma. The trial showed a complete or partial response, with 66% of patients achieving a complete response.

Such notable collaborations and various pipeline products with the extensive commercialized products below are expected to boost market growth for oncology applications.

TABLE 56 INDICATIVE LIST OF COMMERCIALIZED REGENERATIVE CELL THERAPIES FOR ONCOLOGY APPLICATIONS

COMPANY	BRAND NAME	TYPE	INDICATION	DATE OF APPROVAL	AVAILABILITY
BLUEBIRD BIO	Abecma	Cell-based Immunotherapy (CAR T –cell Therapy)	A chimeric antigen receptor (CAR) T-cell therapy for adults with relapsed or refractory multiple myeloma.	Approved by the US FDA in March 2021. Approved by Health Canada in May 2021. Approved by the EMA in August 2021. Approved for use in Japan in February 2022.	North America, Europe, Asia Pacific
BRISTOL MYERS SQUIBB	Breyanzi	Cell-based Immunotherapy (CAR T –cell Therapy)	Breyanzi is a CAR T-cell therapy for adults with relapsed or refractory large B-cell lymphoma.	The US FDA approves it for adult patients with relapsed or refractory diffuse large B-cell lymphoma (February 2021) and r/r large B-cell lymphoma as second-line treatment (June 2022). The European Commission approves it in adult patients with r/r large B-cell lymphoma, primary mediastinal large B-cell lymphoma, and follicular lymphoma grade 3B (April 2022).	North America & Europe
NOVARTIS AG	Kymriah	Cell-based Immunotherapy	For the treatment of acute lymphoblastic leukemia, chronic lymphoid leukemia, and diffuse large B-cell lymphoma up to the age of 25	August 2017 (FDA) For relapsed/refractory (r/r) large B-cell lymphoma, as approved by the US FDA in May 2018 Approved by the EC in August 2018 and by Health Canada in September 2018	North America & Europe
KITE PHARMA/ GILEAD SCIENCES	Yescarta	Cell-based Immunotherapy (CAR T –cell Therapy)	For the treatment of B-cell malignancies such as non-Hodgkin lymphoma, acute lymphoblastic leukemia, mantle cell	Approved by the US FDA in October 2017 Approved by the EC in August 2018	North America & Europe

			lymphoma, chronic lymphoid leukemia, and diffuse large B-Cell lymphoma		
DENDREON	Provenge	Cell-based Immunotherapy	Autologous cellular immunotherapy for the treatment of asymptomatic or minimally symptomatic metastatic castrate-resistant (hormone-refractory) prostate cancer	Biologics License Application (BLA) approved by the FDA in 2010	North America
GC PHARMA	Immunocell-LC	Cell-based Immunotherapy	A medicine for anticancer cell immunotherapy made with T-lymphocyte incubated and activated after extraction from the blood of a patient	Conditionally approved in South Korea in June 2007	Asia Pacific

Source: Alliance for Regenerative Medicine (AMR)

TABLE 57 REGENERATIVE MEDICINE MARKET FOR ONCOLOGY, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,000.38	1,282.83	1,600.88	1,998.96	2,497.45	3,121.93	3,904.56	4,885.78	25.0%
Europe	409.24	538.09	688.82	882.71	1,132.38	1,454.19	1,869.42	2,405.76	28.4%
Asia Pacific	365.17	501.04	668.11	890.36	1,185.97	1,579.17	2,102.19	2,797.95	33.2%
Latin America	57.19	77.49	102.13	134.63	177.52	234.14	308.92	407.73	31.9%
Middle East and Africa	71.70	97.36	128.60	169.86	224.41	296.53	391.92	518.14	32.1%
Total	1,903.68	2,496.80	3,188.53	4,076.53	5,217.73	6,685.96	8,577.01	11,015.35	28.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 58 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR ONCOLOGY, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	860.29	1,096.70	1,360.53	1,688.74	2,097.24	2,605.86	3,239.38	4,028.70	24.2%
Canada	140.09	186.12	240.35	310.22	400.21	516.06	665.19	857.07	29.0%
Total	1,000.38	1,282.83	1,600.88	1,998.96	2,497.45	3,121.93	3,904.56	4,885.78	25.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 59 EUROPE: REGENERATIVE MEDICINE MARKET FOR ONCOLOGY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
UK	109.60	140.52	175.30	218.77	273.12	341.08	426.09	532.41	24.9%
Germany	70.33	93.25	120.37	155.53	201.15	260.42	337.47	437.75	29.5%
France	49.11	67.00	88.88	117.88	156.33	207.32	274.94	364.67	32.6%
Rest of Europe	180.20	237.31	304.27	390.53	501.77	645.37	830.92	1,070.92	28.6%
Total	409.24	538.09	688.82	882.71	1,132.38	1,454.19	1,869.42	2,405.76	28.4%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 60 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR ONCOLOGY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	116.96	150.87	188.38	233.99	289.00	354.65	432.01	521.68	22.6%
China	79.08	110.22	149.27	201.97	273.09	369.00	498.35	672.77	35.1%
India	53.51	77.68	109.27	153.20	214.16	298.58	415.32	576.51	39.5%
South Korea	64.12	89.28	120.77	163.25	220.50	297.66	401.63	541.70	35.0%
Rest of Asia Pacific	51.50	72.98	100.41	137.94	189.23	259.28	354.89	485.29	37.0%
Total	365.17	501.04	668.11	890.36	1,185.97	1,579.17	2,102.19	2,797.95	33.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.4 DERMATOLOGY & WOUND CARE

7.4.1 RISING USE OF WOUND CARE BIOLOGICS IN ESTHETICS TO DRIVE MARKET

Several regenerative medicine products are available for treating acute and chronic wounds, such as diabetic foot ulcers, pressure ulcers, venous leg ulcers, burns, surgical wounds, and traumatic wounds. Many of these wound care biologics are substitutes for skin structures and promote healing and wound closure. It is estimated that in the US, 3% of the population is older than 65 and has open wounds. Many cutaneous pathologies, such as burns, diabetic ulcers, psoriasis, and vitiligo, are being treated by monotherapy and combination therapy with regenerative medicine from advanced research and development in the last two decades using various cell types and scaffolds.

Along with regenerative medicine for wound care, this science is rapidly utilized for aesthetic applications such as skin rejuvenation, hair restoration, skin pigmentation, anti-aging, and scar healing. For instance:

- Use of Adipose stem cells (ASCs) for skin rejuvenation and skin remodeling, owing to its capability of differentiating various cells, promoting angiogenesis, and secreting numerous cytokines.
- Use of MSC-exosomes exhibiting anti-inflammatory, anti-aging, and wound healing properties that are highly considered for aesthetic applications.
- Platelet-rich plasma (PRP) contains a significant additional level of concentrated platelets obtained through centrifuging whole blood for treating vitiligo.
- Autologous skin cells are harvested and prepared using novel systems to treat scars and burn wounds.

Regenerative medicine is also being commercialized, and below is a list of a few commercialized regenerative medicines for wound healing applications.

TABLE 61 INDICATIVE LIST OF REGENERATIVE MEDICINE PRODUCTS COMMERCIALIZED FOR TREATMENT OF ACUTE & CHRONIC WOUNDS

PRODUCT	MAJOR SUBSTANCES	ORIGIN	CONTRAINDICATIONS
Epicel (Vericel Corporation)	Cultured epidermal autografts	Single-cell suspension of skin	Skin replacement for patients with deep dermal or full-thickness burns
Apligraf (Organogenesis)	Collagen + Glycosaminoglycan Allogeneic + Fibroblasts Allogeneic + Keratinocytes	Bovine Tendon + Neonatal Foreskin	Infected wounds, known allergies to bovine collagen, and hypersensitivity to agarose
Dermagraft (Advanced BioHealing)	Polyglactin Mesh + Fibroblasts	Synthetic Mesh + Neonatal Foreskin	Infected wounds
Hyalograft (Fidia Advanced Biopolymers)	Hyaluronic Acid + Allogeneic Fibroblasts	Streptococcus Fermentation + Neonatal Foreskin	Hypersensitivity
Oasis (Smith & Nephew)	Collagen Matrix	Porcine Small Bowel Submucosa	Known allergy to porcine collagen

Source: Company Websites

TABLE 62 REGENERATIVE MEDICINE MARKET FOR DERMATOLOGY & WOUND CARE, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	665.24	846.52	1,048.32	1,299.02	1,610.62	1,998.08	2,480.07	3,079.89	24.1%
Europe	273.47	355.58	450.15	570.48	723.75	919.17	1,168.61	1,487.31	27.0%
Asia Pacific	244.56	331.61	437.02	575.65	757.96	997.74	1,313.13	1,728.08	31.6%
Latin America	38.33	51.35	66.94	87.28	113.84	148.54	193.89	253.21	30.5%
Middle East and Africa	48.06	64.55	84.33	110.21	144.05	188.35	246.36	322.34	30.8%
Total	1,269.64	1,649.60	2,086.76	2,642.64	3,350.22	4,251.88	5,402.06	6,870.85	26.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 63 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR DERMATOLOGY & WOUND CARE, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	571.97	723.66	891.00	1,097.68	1,353.06	1,668.74	2,059.10	2,541.99	23.3%
Canada	93.27	122.86	157.32	201.34	257.56	329.34	420.97	537.90	27.9%
Total	665.24	846.52	1,048.32	1,299.02	1,610.62	1,998.08	2,480.07	3,079.89	24.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 64 EUROPE: REGENERATIVE MEDICINE MARKET FOR DERMATOLOGY & WOUND CARE, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
UK	73.07	92.80	114.67	141.76	175.33	216.92	268.47	332.36	23.7%
Germany	46.95	61.60	78.70	100.63	128.81	165.05	211.70	271.81	28.1%
France	32.83	44.28	58.08	76.16	99.86	130.95	171.73	225.23	31.1%
Rest of Europe	120.62	156.90	198.71	251.92	319.74	406.24	516.71	657.91	27.1%
Total	273.47	355.58	450.15	570.48	723.75	919.17	1,168.61	1,487.31	27.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 65 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR DERMATOLOGY & WOUND CARE, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	78.31	99.79	123.09	151.07	184.36	223.54	269.07	321.08	21.1%
China	52.95	72.93	97.60	130.51	174.39	232.89	310.88	414.83	33.6%
India	35.84	51.42	71.49	99.07	136.90	188.69	259.48	356.12	37.9%
South Korea	42.95	59.12	79.06	105.65	141.10	188.35	251.31	335.22	33.5%
Rest of Asia Pacific	34.51	48.35	65.77	89.35	121.22	164.28	222.40	300.84	35.5%
Total	244.56	331.61	437.02	575.65	757.96	997.74	1,313.13	1,728.08	31.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.5 CARDIOVASCULAR DISEASES

7.5.1 INCREASING USE OF MSCS FOR CLINICAL RESEARCH ON IHD TO SUPPORT MARKET GROWTH

According to WHO, cardiovascular diseases are the leading cause of death globally. In 2020, approximately 19.1 million deaths were attributed to CVD globally. Cardiac regeneration is a broad effort to repair irreversibly damaged heart tissue with cutting-edge science, including stem cell and cell-free therapy. Regenerative medicine can be used for CVD, such as acute myocardial infarction, coronary artery disease, coronary microvascular disease, congestive heart failure, chronic ischemic heart disease, post-infarct heart failure, dilated cardiomyopathy, and heart failure. Regenerative medicine is expected to curb cardiovascular diseases that devices, medications, and other approaches cannot cure.

Ischemic heart disease is one of the major causes of mortality globally. As of 2020, 244.1 million people were living with ischemic heart disease (IHD). Cell therapy is an innovative and emerging therapeutic alternative to treat cardiovascular diseases by cell administration aimed at cardiac regeneration and repair. Regenerative medicine is used to treat IHD by repairing the lost part of the heart by using stem cells. Research is underway for treating IHD with cell and gene therapy.

Multiple clinical trials and research studies are being conducted for regenerative medicine in cardiovascular disease applications. For instance:

- The Center for Regenerative Biotherapeutics has developed cell-free therapies aligned with current standards of care for clinical trials. These novel therapies aim to deliver regenerative therapy after surgical stent implantation to reduce injury and protect the heart against inflammation.
- CHART-1 is designed as a large randomized, sham-controlled multicentre study to validate cardiopoietic stem cell therapy. The CHART-1 clinical trial is undergoing to examine the therapeutic impact of lineage-directed stem cells as a strategy to achieve cardiac regeneration in heart failure populations.

- Mayo Clinic researchers are developing a novel gel carrying the newest generation of gene therapy technology to help the heart form new blood vessels. These biological bypass grafts will be offered to patients who cannot benefit from stenting or coronary bypass surgery (coronary artery bypass grafting)
- The Center for Regenerative Biotherapeutics is creating a cell-based therapy designed to improve the integrity and health of the heart's smallest blood vessels. The therapy will help patients whose chest pain persists despite a negative angiogram and optimal medical therapy.

TABLE 66 REGENERATIVE MEDICINE MARKET FOR CARDIOVASCULAR DISEASES, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	251.40	322.05	401.50	500.86	625.17	780.77	975.62	1,219.71	24.9%
Europe	103.27	136.20	174.89	224.80	289.25	372.56	480.36	619.99	28.8%
Asia Pacific	92.56	127.95	171.87	230.72	309.55	415.13	556.54	745.92	34.1%
Latin America	14.55	19.94	26.56	35.38	47.12	62.76	83.60	111.36	33.2%
Middle East and Africa	18.26	25.10	33.55	44.82	59.87	79.96	106.78	142.60	33.6%
Total	480.04	631.24	808.37	1,036.58	1,330.96	1,711.18	2,202.89	2,839.58	28.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 67 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR CARDIOVASCULAR DISEASES, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	216.16	275.23	341.05	422.83	524.51	650.97	808.32	1,004.17	24.1%
Canada	35.24	46.82	60.46	78.03	100.66	129.79	167.29	215.54	28.9%
Total	251.40	322.05	401.50	500.86	625.17	780.77	975.62	1,219.71	24.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 68 EUROPE: REGENERATIVE MEDICINE MARKET FOR CARDIOVASCULAR DISEASES, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
UK	27.60	35.43	44.24	55.27	69.08	86.36	107.99	135.08	25.0%
Germany	17.73	23.56	30.48	39.47	51.16	66.37	86.19	112.03	29.7%
France	12.40	16.97	22.59	30.05	39.98	53.19	70.75	94.13	33.0%
Rest of Europe	45.54	60.24	77.58	100.00	129.03	166.64	215.42	278.75	29.2%
Total	103.27	136.20	174.89	224.80	289.25	372.56	480.36	619.99	28.8%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 69 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR CARDIOVASCULAR DISEASES, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	29.59	38.38	48.19	60.18	74.72	92.16	112.83	136.92	23.2%
China	20.03	28.11	38.31	52.17	70.98	96.50	131.11	178.03	36.0%
India	13.57	19.85	28.14	39.74	55.96	78.57	110.05	153.79	40.5%
South Korea	16.28	22.87	31.20	42.53	57.92	78.82	107.18	145.66	36.1%
Rest of Asia Pacific	13.09	18.73	26.03	36.09	49.97	69.08	95.37	131.52	38.3%
Total	92.56	127.95	171.87	230.72	309.55	415.13	556.54	745.92	34.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.6 OPHTHALMOLOGY

7.6.1 REGENERATION OF EYE TISSUES FOR AMD TO SUPPORT MARKET GROWTH

Tissue and stem cell therapies show significant development in the repair of ocular tissues. Researchers focus on developing revolutionary methods to regenerate eye tissues for various diseases, including age-related macular degeneration (AMD), glaucoma, corneal disease, and ocular cancer. Age-related macular degeneration (AMD) is a leading cause of vision loss in people over 65 years and is currently estimated to affect 11 million Americans and 170 million individuals worldwide.

The growing number of clinical-stage cell therapies for ophthalmology is expected to drive market growth. For instance:

TABLE 70 LIST OF PRODUCTS UNDER PIPELINE FOR REGENERATIVE MEDICINE IN OPHTHALMOLOGY

INDICATION	SPONSOR	PRODUCT	CELL TYPE	DELIVERY	PHASE
Dry age-related macular degeneration	Regenerative Patch Technologies	CPCB-RPE1	Human embryonic stem cell-derived RPE cells on a parylene membrane	Subretinal implantation	Phase 2a
Dry age-related macular degeneration	NIH	iPSC-derived RPE/PLGA	Autologous iPSC-derived RPE on a biodegradable poly lactic-co-glycolic acid (PLGA) scaffold	Subretinal transplantation	Phase 1/2 a
Dry age-related macular degeneration	Astellas	ASP7317 (MA09-hRPE)	Human embryonic stem cell-derived RPE cells	Subretinal injection	Phase 1/2a
Retinitis pigmentosa	jCyte	jCell (hRPC)	Retinal progenitor cells	Intravitreal injection	Phase 2b
Retinitis pigmentosa	Cedars-Sinai Medical Center	CNS10-NPC	Neural progenitor cells	Transplantation into subretinal space	Phase 1
Retinitis pigmentosa	ReNeuron	hRPC	Human retinal progenitor cells	Subretinal injection	Phase 1/2a
Limbal stem cell deficiency	The University of California, Los Angeles (UCLA) (Sophie Deng)	cLSC	Autologous limbal stem cells	Transplantation	Phase 1
Limbal stem cell deficiency	Holostem Therapie Avanzate s.r.l.	Holoclar	Autologous expanded limbal stem cells	Transplantation	Phase 4
Limbal stem cell deficiency	Massachusetts Eye and Ear Infirmary	Cultivated autologous limbal epithelial cell (CALEC) expanded on an amniotic membrane	Autologous limbal epithelial cells	Surgical transplantation	Phase 1/2

Source: National Center for Biotechnology Information (NCBI), Alliance for Regenerative Medicine (ARM), and MarketsandMarkets Analysis

TABLE 71 REGENERATIVE MEDICINE MARKET FOR OPHTHALMOLOGY, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	206.65	266.99	335.65	422.15	531.19	668.68	842.08	1,060.86	25.9%
Europe	84.99	113.17	146.67	190.24	246.93	320.77	417.03	542.61	29.9%
Asia Pacific	76.27	106.56	144.62	196.07	265.56	359.40	486.07	657.01	35.4%
Latin America	12.00	16.64	22.41	30.16	40.57	54.55	73.32	98.52	34.5%
Middle East and Africa	15.07	20.96	28.33	38.25	51.61	69.59	93.78	126.33	34.8%
Total	394.99	524.32	677.68	876.87	1,135.87	1,472.99	1,912.29	2,485.32	29.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 72 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR OPHTHALMOLOGY, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	177.68	228.16	285.07	356.32	445.57	557.38	697.49	873.11	25.1%
Canada	28.98	38.83	50.58	65.83	85.62	111.30	144.59	187.75	30.0%
Total	206.65	266.99	335.65	422.15	531.19	668.68	842.08	1,060.86	25.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 73 EUROPE: REGENERATIVE MEDICINE MARKET FOR OPHTHALMOLOGY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
UK	22.70	29.41	37.05	46.68	58.84	74.16	93.49	117.86	26.0%
Germany	14.59	19.57	25.55	33.38	43.63	57.08	74.73	97.91	30.8%
France	10.20	14.10	18.95	25.44	34.14	45.81	61.44	82.40	34.2%
Rest of Europe	37.49	50.09	65.13	84.74	110.32	143.72	187.36	244.43	30.3%
Total	84.99	113.17	146.67	190.24	246.93	320.77	417.03	542.61	29.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 74 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR OPHTHALMOLOGY, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	24.37	31.94	40.49	51.05	63.96	79.59	98.28	120.25	24.3%
China	16.50	23.40	32.22	44.30	60.84	83.45	114.36	156.59	37.2%
India	11.18	16.54	23.68	33.78	48.02	68.04	96.13	135.48	41.7%
South Korea	13.42	19.06	26.28	36.19	49.76	68.34	93.75	128.50	37.4%
Rest of Asia Pacific	10.79	15.63	21.94	30.74	42.98	59.97	83.54	116.19	39.6%
Total	76.27	106.56	144.62	196.07	265.56	359.40	486.07	657.01	35.4%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.7 OTHER APPLICATIONS

Other applications in this segment include autoimmune diseases, obstetrics and gynecology disorders, hernia repair, immunology diseases, respiratory diseases, lymphatic diseases, neurology, and infectious diseases.

The growth of the other applications segment is associated with the increasing incidence of neurological disorders, the rising prevalence of respiratory diseases, and the growing number of gynecological and hemorrhoid surgeries.

Neurology: Neurological diseases in this application segment include Amyotrophic Lateral Sclerosis, or ALS, which affects approximately one to two per 100,000 people in the US annually. In 2022 there was a breakthrough in regenerative medicine for slowing the progression of ALS. The treatment to slow the progression of this disease was demonstrated by clinical models showing evidence that mesenchymal stem cell transplantation may delay the disease onset and progression and increase lifespan. Such breakthrough medicine will most likely demonstrate positive growth in the market.

Respiratory: Regenerative medicine can provide modern and innovative therapies for people with lung diseases, such as chronic obstructive pulmonary disease (COPD), pulmonary fibrosis, cystic fibrosis, pulmonary arterial hypertension, and bronchiolitis obliterans. The regenerative treatment for respiratory diseases includes using iPS cells that could be differentiated into patient-specific pulmonary epithelial cells and delivered back to patients via cell therapy to repopulate acellular lung scaffolds for implantation for patients with end-stage lung diseases like emphysema, pulmonary hypertension, or pulmonary fibrosis. However, this segment is not yet growing at its full potential due to stringent regulatory approvals, the absence of any proven benefits through clinical trials, and the high treatment cost.

TABLE 75 REGENERATIVE MEDICINE MARKET FOR OTHER APPLICATIONS, BY REGION, 2020–2027 (USD MILLION)

Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	539.83	688.76	855.24	1,062.62	1,321.08	1,643.35	2,045.36	2,547.04	24.4%
Europe	221.10	289.59	369.31	471.50	602.63	771.07	987.69	1,266.54	28.0%
Asia Pacific	197.54	270.35	359.60	478.07	635.31	844.01	1,121.05	1,488.88	32.9%
Latin America	30.97	41.90	55.15	72.59	95.59	125.90	165.89	218.66	31.7%
Middle East and Africa	38.84	52.68	69.51	91.71	121.03	159.75	210.92	278.57	32.0%
Total	1,028.27	1,343.29	1,708.81	2,176.49	2,775.63	3,544.09	4,530.91	5,799.68	27.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 76 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR OTHER APPLICATIONS, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	464.21	588.77	726.72	897.52	1,109.08	1,371.23	1,696.22	2,099.22	23.6%
Canada	75.62	99.99	128.51	165.10	212.00	272.11	349.14	447.81	28.4%
Total	539.83	688.76	855.24	1,062.62	1,321.08	1,643.35	2,045.36	2,547.04	24.4%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 77 EUROPE: REGENERATIVE MEDICINE MARKET FOR OTHER APPLICATIONS, BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
UK	59.18	75.54	93.82	116.58	144.92	180.21	224.17	278.94	24.3%
Germany	37.99	50.16	64.49	82.99	106.91	137.87	177.96	229.96	29.0%
France	26.54	36.07	47.67	62.99	83.23	109.97	145.33	192.08	32.1%
Rest of Europe	97.39	127.82	163.34	208.94	267.57	343.02	440.22	565.56	28.2%
Total	221.10	289.59	369.31	471.50	602.63	771.07	987.69	1,266.54	28.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 78 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR OTHER APPLICATIONS,
BY COUNTRY, 2020–2027 (USD MILLION)

Country/Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Japan	63.24	81.32	101.22	125.36	154.37	188.88	229.40	276.22	22.2%
China	42.77	59.45	80.29	108.34	146.10	196.90	265.24	357.15	34.8%
India	28.95	41.92	58.83	82.29	114.76	159.63	221.54	306.84	39.1%
South Korea	34.70	48.21	65.09	87.80	118.35	159.45	214.72	289.05	34.7%
Rest of APAC	27.88	39.44	54.17	74.28	101.73	139.16	190.15	259.61	36.8%
Total	197.54	270.35	359.60	478.07	635.31	844.01	1,121.05	1,488.88	32.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8 REGENERATIVE MEDICINE MARKET, BY REGION

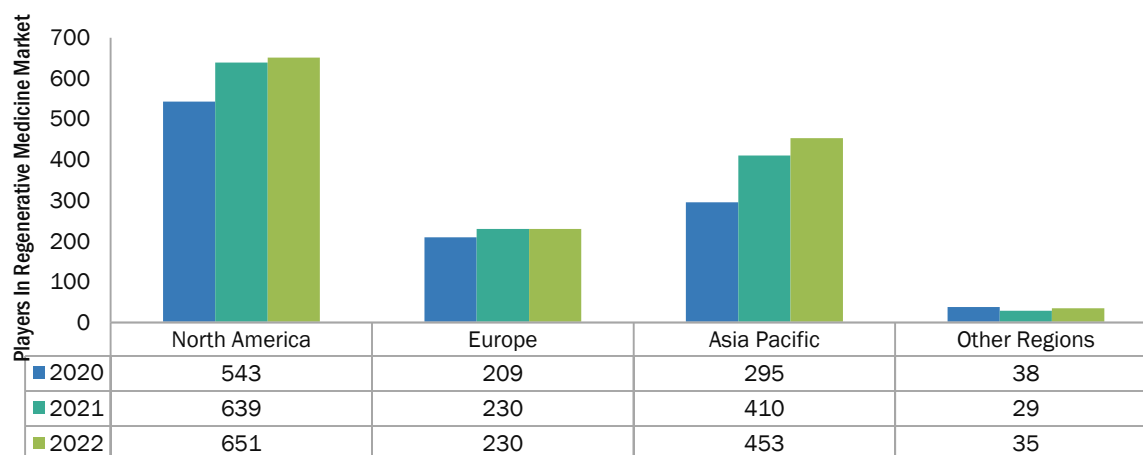
KEY FINDINGS

- North America accounted for the largest share of 51.50% of the regenerative medicine market in 2021. This market is projected to reach a value of USD 18,220.99 million by 2027 from an estimated value of USD 6,144.35 million in 2022, at a CAGR of 24.3% during the forecast period.
- The regenerative medicine market in the Asia Pacific is estimated to grow from USD 2,542.20 million in 2022 to reach USD 10,198.95 million by 2027.
- Asia Pacific is the fastest-growing region in the regenerative medicine market, projected to grow at a CAGR of 32.0% during the forecast period. Factors such as the high burden of chronic diseases, growing per capita income, and the rising demand for new technologies are expected to support market growth.
- Europe accounted for the second-largest global regenerative medicine market share in 2021. This market is expected to be valued at USD 2,632.83 million in 2022 and is projected to reach USD 8,875.59 million in 2027, at a CAGR of 27.5% during the forecast period.

8.1 INTRODUCTION

The regenerative medicine market has been segmented into five major regions: North America, Europe, Asia Pacific, Latin America, and the Middle East & Africa. In 2021, North America accounted for the largest share of 51.5% of the global regenerative medicine market, followed by Europe and the Asia Pacific, with shares of 21.54% and 20.00%, respectively. The growth in the North American regenerative medicine market can be attributed to rising stem cell research, expansion of the healthcare sector, and the high adoption of stem cell therapy and cell immunotherapies for treating cancer and other chronic diseases.

FIGURE 27 NUMBER OF PLAYERS IN REGENERATIVE MEDICINE MARKET, BY REGION, 2020–2022



Source: Alliance for Regenerative Medicine

TABLE 79 NUMBER OF CLINICAL TRIALS FOR REGENERATIVE MEDICINE (2022)

Region	Phase I	Phase II	Phase III	Total
US	166	285	54	505
Europe	60	236	82	378
Rest of the World	157	164	16	337
Total	383	685	152	1,220

Note: The rest of the World includes Canada, the Asia Pacific, Latin America, and the Middle East & Africa

Source: Alliance for Regenerative Medicine (ARM)

TABLE 80 REGENERATIVE MEDICINE MARKET, BY REGION, 2020–2027 (USD MILLION)

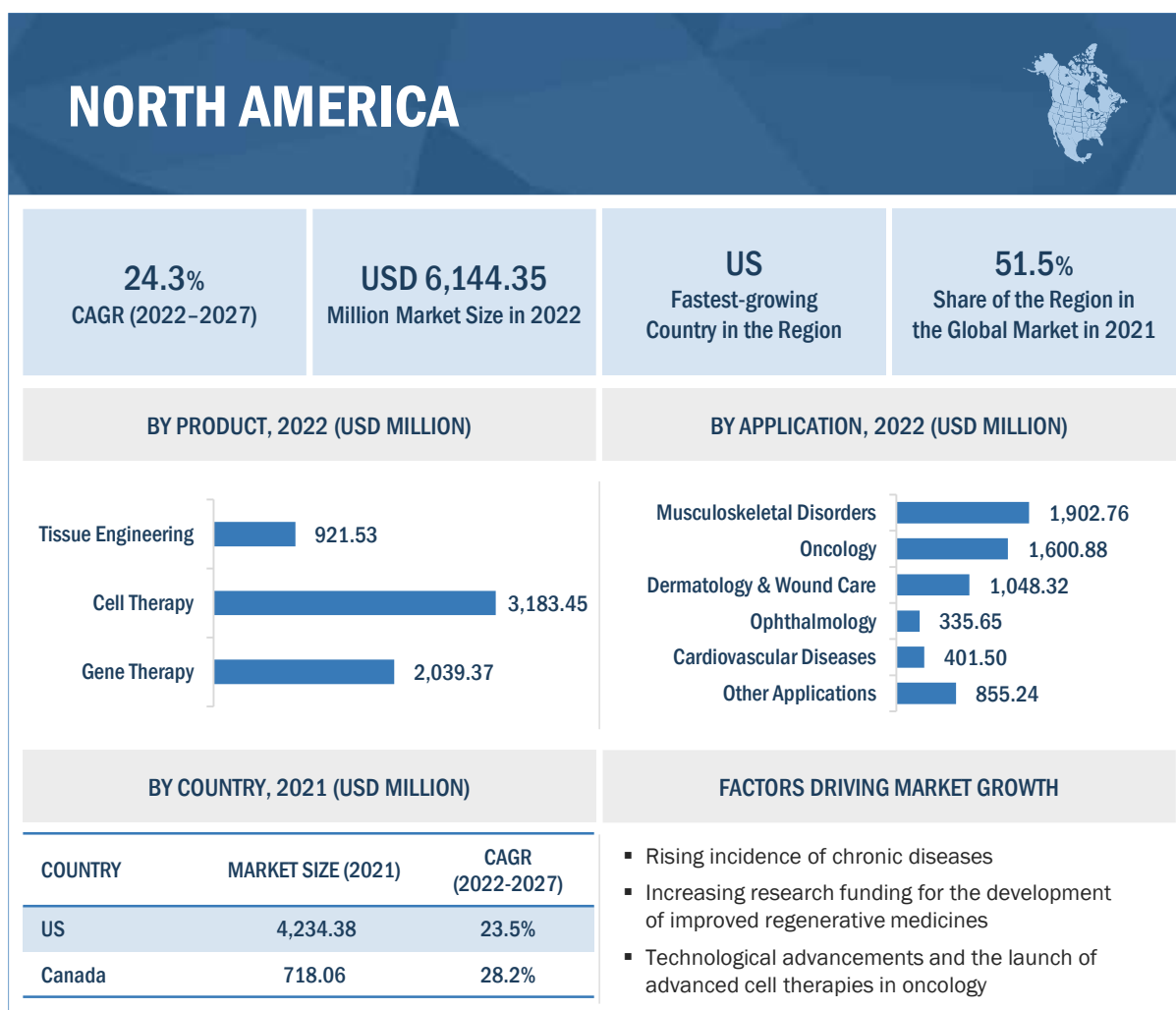
Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	3,884.77	4,952.44	6,144.35	7,627.82	9,475.10	11,776.46	14,644.74	18,220.99	24.3%
Europe	1,586.99	2,071.56	2,632.83	3,349.86	4,266.84	5,440.78	6,945.31	8,875.59	27.5%
Asia Pacific	1,413.96	1,923.15	2,542.20	3,358.63	4,435.34	5,855.35	7,728.27	10,198.95	32.0%
Latin America	221.15	296.66	387.14	505.37	659.93	862.08	1,126.63	1,473.02	30.6%
Middle East & Africa	277.14	372.47	486.92	636.67	832.68	1,089.37	1,425.67	1,866.49	30.8%
Total	7,384.01	9,616.28	12,193.44	15,478.36	19,669.89	25,024.04	31,870.62	40,635.04	27.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.2 NORTH AMERICA

The regenerative medicine market in North America is expected to grow rapidly in the coming years. There has been a tremendous increase in research targeted toward developing cell-based therapies in the past decade. Increasing research activities in regenerative medicine and growing awareness of personalized medicine have also resulted in the establishment of stem cell banking and tissue engineering. North America has public and private banks that store a limited number of cells and tissues. Several tissue banks have been established across the North American region, which differs in the purpose, size, and nature of the tissues collected. Over the years, North America has witnessed significant growth in stem cell banking for cells, tissues, and organs, triggered by advancements in medical research, an increasing number of clinical trials, and the growing biopharmaceutical industry.

Moreover, there has also been a rise in demand for advanced technology-based treatments like stem cell technology, biologics, tissue engineering, and biomaterials. According to the Heart Disease & Stroke Statistical Fact Sheet (2020), an estimated 40,000 infants are affected yearly by congenital heart defects in the United States. In the first year of life, roughly 25% of these, or 2.4 (per 1,000 live births), require therapy. Additionally, the Centers for Disease Control and Prevention (CDC) predicts that by the year 2030, there will be 12.1 million more cases of atrial fibrillation in the United States. With increased research-driven activities and demand for biologics in North America, the demand for regenerative medicine is expected to rise significantly.

FIGURE 28 NORTH AMERICA: REGENERATIVE MEDICINE MARKET SNAPSHOT

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 81 NORTH AMERICA: REGENERATIVE MEDICINE MARKET, BY COUNTRY, 2020–2027 (USD MILLION)

Country	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	3,340.93	4,234.38	5,222.75	6,445.58	7,959.17	9,833.44	12,155.26	15,032.47	23.5%
Canada	543.83	718.06	921.60	1,182.25	1,515.94	1,943.02	2,489.48	3,188.52	28.2%
Total	3,884.77	4,952.44	6,144.35	7,627.82	9,475.10	11,776.46	14,644.74	18,220.99	24.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 82 NORTH AMERICA: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	1,283.84	1,640.22	2,039.37	2,537.23	3,158.48	3,934.10	4,902.83	6,113.24	24.6%
Cell Therapy	1,927.15	2,511.35	3,183.45	3,862.06	4,795.58	5,959.75	7,412.55	9,226.73	23.7%
Tissue Engineering	673.78	800.87	921.53	1,228.54	1,521.04	1,882.61	2,329.35	2,881.02	25.6%
Total	3,884.77	4,952.44	6,144.35	7,627.82	9,475.10	11,776.46	14,644.74	18,220.99	24.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 83 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Cell Therapy	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	1,212.52	1,578.28	1,998.39	2,421.60	3,003.49	3,728.31	4,631.78	5,758.68	23.6%
Cell-based Immunotherapy Products	714.64	933.07	1,185.06	1,440.45	1,792.09	2,231.44	2,780.77	3,468.05	24.0%
Total	1,927.15	2,511.35	3,183.45	3,862.06	4,795.58	5,959.75	7,412.55	9,226.73	23.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 84 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	436.59	569.89	723.61	879.32	1,093.66	1,361.39	1,696.02	2,114.55	23.9%
Cell Transplantations	775.92	1,008.39	1,274.78	1,542.29	1,909.83	2,366.92	2,935.76	3,644.13	23.4%
Total	1,212.52	1,578.28	1,998.39	2,421.60	3,003.49	3,728.31	4,631.78	5,758.68	23.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 85 NORTH AMERICA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogenic Therapies	317.97	415.62	528.44	643.01	800.82	998.19	1,245.20	1,554.54	24.1%
Autologous Therapies	118.62	154.27	195.17	236.31	292.84	363.20	450.82	560.01	23.5%
Total	436.59	569.89	723.61	879.32	1,093.66	1,361.39	1,696.02	2,114.55	23.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 86 NORTH AMERICA: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	1,221.27	1,545.30	1,902.76	2,344.22	2,889.61	3,563.67	4,397.04	5,427.71	23.3%
Oncology	1,000.38	1,282.83	1,600.88	1,998.96	2,497.45	3,121.93	3,904.56	4,885.78	25.0%
Dermatology & Wound Care	665.24	846.52	1,048.32	1,299.02	1,610.62	1,998.08	2,480.07	3,079.89	24.1%
Ophthalmology	206.65	266.99	335.65	422.15	531.19	668.68	842.08	1,060.86	25.9%
Cardiovascular Diseases	251.40	322.05	401.50	500.86	625.17	780.77	975.62	1,219.71	24.9%
Other Applications	539.83	688.76	855.24	1,062.62	1,321.08	1,643.35	2,045.36	2,547.04	24.4%
Total	3,884.77	4,952.44	6,144.35	7,627.82	9,475.10	11,776.46	14,644.74	18,220.99	24.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.2.1 US

8.2.1.1 Rising number of pipeline products for regenerative medicine to drive market

The growth factors affecting the US market are increasing requirements for organ transplants, rising R&D in regenerative medicine, a growing pool of patients, technological innovation, rising healthcare expenditure, and the strong healthcare infrastructure in the country. The rising healthcare spending is associated with the growth in spending on private health insurance, physician and clinical services, hospital care, and the continuous growth in Medicaid and prescription drug spending. This has attracted manufacturers and biotechnology companies to invest in emerging fields, including the regenerative medicine market.

In February 2022, the FDA approved ciltacabtagene autoleucel (brand name CARVYKTI) for treating adult patients with relapsed or refractory multiple myeloma after four or more prior lines of therapy, including a proteasome inhibitor, an immunomodulatory agent, and an anti-CD38 monoclonal antibody. This is expected to open new channels for growth and development in the coming years.

TABLE 87 PRODUCTS AWAITING APPROVALS IN THE US (2023)

COMPANY	PRODUCT	THERAPY	INDICATION
Atara Biotherapeutics	Tab-Cel	Cell Therapy	EBV-associated Post-transplant Lymphoproliferative Disease
CRISPR Therapeutics & Vertex Pharmaceuticals	CTX001	CRISPR Therapy	Sickle cell disease
Adaptimmune Therapeutics PLC	Afami Cell	Cell Therapy	Advanced synovial sarcoma
Orchard Therapeutics	Libmeldy	Gene Therapy	Metachromatic leukodystrophy
bluebird bio, Inc.	Bb1111	Gene Therapy	Sickle cell disease
CARsgen	CT053	CAR-T therapy	R/R multiple myeloma

Source: Alliance for Regenerative Medicine (ARM)

TABLE 88 US: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	1,104.06	1,402.28	1,733.25	2,143.58	2,652.52	3,284.03	4,067.95	5,041.37	23.8%
Cell Therapy	1,657.32	2,147.10	2,705.72	3,236.10	3,976.12	4,887.86	6,011.56	7,396.95	22.3%
Tissue Engineering	579.55	684.99	783.78	1,065.90	1,330.53	1,661.55	2,075.75	2,594.15	27.0%
Total	3,340.93	4,234.38	5,222.75	6,445.58	7,959.17	9,833.44	12,155.26	15,032.47	23.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 89 US: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	1,042.77	1,349.43	1,698.62	2,029.32	2,490.60	3,058.28	3,757.16	4,617.84	22.1%
Cell-based Immunotherapy Products	614.55	797.67	1,007.10	1,206.78	1,485.52	1,829.58	2,254.40	2,779.11	22.5%
Total	1,657.32	2,147.10	2,705.72	3,236.10	3,976.12	4,887.86	6,011.56	7,396.95	22.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 90 US: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	375.52	487.30	615.09	736.87	906.86	1,116.62	1,375.55	1,695.27	22.5%
Cell Transplantations	667.25	862.13	1,083.53	1,292.45	1,583.74	1,941.66	2,381.61	2,922.56	22.0%
Total	1,042.77	1,349.43	1,698.62	2,029.32	2,490.60	3,058.28	3,757.16	4,617.84	22.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 91 US: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	273.50	355.40	449.22	538.89	664.12	818.84	1,010.10	1,246.57	22.6%
Autologous Therapies	102.02	131.90	165.88	197.98	242.75	297.78	365.45	448.70	22.0%
Total	375.52	487.30	615.09	736.87	906.86	1,116.62	1,375.55	1,695.27	22.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 92 US: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	1,050.63	1,321.86	1,618.38	1,982.48	2,429.71	2,979.26	3,654.75	4,485.27	22.6%
Oncology	860.29	1,096.70	1,360.53	1,688.74	2,097.24	2,605.86	3,239.38	4,028.70	24.2%
Dermatology & Wound Care	571.97	723.66	891.00	1,097.68	1,353.06	1,668.74	2,059.10	2,541.99	23.3%
Ophthalmology	177.68	228.16	285.07	356.32	445.57	557.38	697.49	873.11	25.1%
Cardiovascular Diseases	216.16	275.23	341.05	422.83	524.51	650.97	808.32	1,004.17	24.1%
Other Applications	464.21	588.77	726.72	897.52	1,109.08	1,371.23	1,696.22	2,099.22	23.6%
Total	3,340.93	4,234.38	5,222.75	6,445.58	7,959.17	9,833.44	12,155.26	15,032.47	23.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.2.2 CANADA

8.2.2.1 Rising government initiatives for regenerative medicine research to drive market

The growth of the Canadian market can primarily be attributed to the rising investments in R&D for regenerative medicine and the growing focus of market players on establishing or strengthening their presence in the country. Bristol Myers Squibb & Bluebird Bio collaboratively worked on Abecma – a prescription medicine based on cell-gene therapy for treating multiple myeloma in patients, which received approval in May 2021 by Health Canada. Moreover, several companies like BlueBird Bio and Krystal Bio in Canada have their products in the pipeline.

The availability of funding for research on regenerative medicine is also a key factor supporting market growth. For instance, in March 2020, the Government of Canada granted USD 6.9 million through the Stem Cell Network's competitive research funding, a body that supports Canada's most promising stem cell research. This funding is channelized to support nine translational projects and four clinical trials across Canada's regenerative medicine market, which comprises ~200 researchers and trainees at 27 Canadian research institutes in Alberta, British Columbia, Quebec, and Ontario, as well as direct partnerships with several emerging biotechnology companies.

TABLE 93 CANADA: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	179.77	237.94	306.12	393.65	505.97	650.07	834.89	1,071.87	28.5%
Cell Therapy	269.83	364.25	477.72	625.96	819.46	1,071.89	1,400.99	1,829.78	30.8%
Tissue Engineering	94.23	115.87	137.75	162.64	190.51	221.06	253.60	286.87	15.8%
Total	543.83	718.06	921.60	1,182.25	1,515.94	1,943.02	2,489.48	3,188.52	28.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 94 CANADA: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	169.75	228.85	299.77	392.28	512.89	670.03	874.62	1,140.85	30.6%
Cell-based Immunotherapy Products	100.08	135.39	177.96	233.68	306.57	401.86	526.37	688.93	31.1%
Total	269.83	364.25	477.72	625.96	819.46	1,071.89	1,400.99	1,829.78	30.8%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 95 CANADA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	61.08	82.60	108.52	142.44	186.80	244.77	320.47	419.28	31.0%
Cell Transplantations	108.67	146.26	191.25	249.84	326.09	425.26	554.15	721.57	30.4%
Total	169.75	228.85	299.77	392.28	512.89	670.03	874.62	1,140.85	30.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 96 CANADA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	44.48	60.22	79.22	104.11	136.71	179.35	235.11	307.97	31.2%
Autologous Therapies	16.60	22.37	29.30	38.33	50.10	65.42	85.37	111.31	30.6%
Total	61.08	82.60	108.52	142.44	186.80	244.77	320.47	419.28	31.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 97 CANADA: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	170.64	223.44	284.38	361.73	459.89	584.41	742.29	942.44	27.1%
Oncology	140.09	186.12	240.35	310.22	400.21	516.06	665.19	857.07	29.0%
Dermatology & Wound Care	93.27	122.86	157.32	201.34	257.56	329.34	420.97	537.90	27.9%
Ophthalmology	28.98	38.83	50.58	65.83	85.62	111.30	144.59	187.75	30.0%
Cardiovascular Diseases	35.24	46.82	60.46	78.03	100.66	129.79	167.29	215.54	28.9%
Other Applications	75.62	99.99	128.51	165.10	212.00	272.11	349.14	447.81	28.4%
Total	543.83	718.06	921.60	1,182.25	1,515.94	1,943.02	2,489.48	3,188.52	28.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.3 EUROPE

The market in Europe primarily comprises the UK, Germany, France, and the Rest of Europe (RoE). Factors such as an increase in biobanking facilities and the rising number of research activities in European countries are driving the growth of the regenerative medicine market in the region.

Europe is considered a pioneer in advanced therapy medicinal products (ATMPs). Europe is also the first region to adopt specific regulations for developing and approving a new class of medicines. European Medicines Agency (EMA) has approved the highest number of marketing authorizations for advanced therapy medicinal products (ATMPs) worldwide. Many ATMPs currently approved or in development originate from European research centers. Clinical research in Europe is the gold standard, with centers and healthcare workers well-qualified domestically and abroad. However, prior research has demonstrated that the European Clinical Trial Directive 2001/20/EC has several flaws, notably for studies conducted in multiple Member States. This was linked to a decrease in clinical trials in the European Union.

Patient access to newer regenerative medicines is increasing rapidly in Europe; however, the lack of reimbursements for some products and therapies is expected to restrain the market growth. For Instance, products like Imlygic (Amgen Inc.), Holoclax (Chiesi Farmaceutici S.p.A.), Strimvelis (GSK Trading Services Limited), and Alofisel (Takeda Pharmaceutical Company Limited) are approved in France. However, they are still not covered in reimbursement policies (Source: Alliance for Regenerative Medicine). The cell and gene therapy products approved in Europe are Imlygic (Amgen Inc.), Holoclax (Chiesi Farmaceutici S.p.A.), Strimvelis (GSK Trading Services Limited), Alofisel (Takeda Pharmaceutical Company Limited), Kymriah (Novartis AG), Yescarta (Kite Pharma), and Luxturna (Spark Therapeutics)

TABLE 98 PRODUCTS AWAITING APPROVALS IN EUROPE (2022)

COMPANY	PRODUCT	THERAPY	INDICATION
Legend Biotech & Janssen	Carvykti	CAR-T therapy	R/R multiple myeloma
GenSight Biologics	Lumevoq	Gene Therapy	Leber hereditary optic neuropathy
PTC Therapeutics	PTC-AADC	Gene Therapy	AADC deficiency
uniQure & CSL Behring	Etranacogene dezaparvovec	Gene Therapy	Hemophilia B

TABLE 99 PRODUCTS AWAITING APPROVALS IN EUROPE (2023)

COMPANY	PRODUCT	THERAPY	INDICATION
Krystal Biotech	Vyjuvek	Gene Therapy	Dystrophic epidermolysis bullosa
Orchard Therapeutics	OTL-103	Gene Therapy	Wiskott-Aldrich syndrome

Source: Alliance for Regenerative Medicine

TABLE 100 EUROPE: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	525.04	687.58	876.71	1,119.10	1,430.05	1,829.41	2,342.85	3,003.67	27.9%
Cell Therapy	787.84	1,051.97	1,366.95	1,777.34	2,312.43	3,010.60	3,922.21	5,113.39	30.2%
Tissue Engineering	274.11	332.01	389.17	453.42	524.35	600.76	680.25	758.54	14.3%
Total	1,586.99	2,071.56	2,632.83	3,349.86	4,266.84	5,440.78	6,945.31	8,875.59	27.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 101 EUROPE: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	495.41	660.36	856.61	1,111.87	1,444.11	1,876.85	2,440.89	3,176.61	30.0%
Cell-based Immunotherapy Products	292.43	391.60	510.33	665.47	868.32	1,133.75	1,481.32	1,936.77	30.6%
Total	787.84	1,051.97	1,366.95	1,777.34	2,312.43	3,010.60	3,922.21	5,113.39	30.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 102 EUROPE: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	177.85	237.97	309.87	403.74	526.37	686.69	896.43	1,171.03	30.5%
Cell Transplantations	317.56	422.39	546.74	708.14	917.74	1,190.16	1,544.46	2,005.59	29.7%
Total	495.41	660.36	856.61	1,111.87	1,444.11	1,876.85	2,440.89	3,176.61	30.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 103 EUROPE: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	129.46	173.38	226.03	294.86	384.88	502.70	657.04	859.33	30.6%
Autologous Therapies	48.38	64.59	83.84	108.88	141.49	183.98	239.39	311.70	30.0%
Total	177.85	237.97	309.87	403.74	526.37	686.69	896.43	1,171.03	30.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 104 EUROPE: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	494.92	638.93	802.99	1,010.13	1,271.90	1,603.01	2,022.21	2,553.39	26.0%
Oncology	409.24	538.09	688.82	882.71	1,132.38	1,454.19	1,869.42	2,405.76	28.4%
Dermatology & Wound Care	273.47	355.58	450.15	570.48	723.75	919.17	1,168.61	1,487.31	27.0%
Cardiovascular Diseases	84.99	113.17	146.67	190.24	246.93	320.77	417.03	542.61	29.9%
Ophthalmology	103.27	136.20	174.89	224.80	289.25	372.56	480.36	619.99	28.8%
Cell Transplantation	221.10	289.59	369.31	471.50	602.63	771.07	987.69	1,266.54	28.0%
Total	1,586.99	2,071.56	2,632.83	3,349.86	4,266.84	5,440.78	6,945.31	8,875.59	27.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.3.1 UK

8.3.1.1 Incorporating regenerative medicine into commercial healthcare services under NHS to drive market

Various centers are emerging in the UK with experience in developing regenerative medicines across a range of products and therapeutic indications. Funding for developing regenerative medicines, coupled with employing a skilled workforce and advanced technologies, is possible through establishing and coordinating a specialized authority in the UK, such as National Health Service (NHS). According to the National Health Service (NHS), 1 in 50 people above 80 suffers from pressure ulcers in the UK. Additionally, in 2021, more than 4.9 million people in the UK had diabetes (Source: Diabetes.co.uk). It is estimated that 10% of people with diabetes suffer from foot ulcers at least once in their lifetime.

According to Care and Public Health Research Institute (CAPHRI), chronic wounds affect more than 1% of the UK's population and impact the healthcare system. This has propelled the demand for advanced and cost-effective wound care treatments, such as grafts, matrices, and other tissue-engineered products, in the country. Players operating in this market are also focusing on increasing R&D in wound care. According to the clinical trial registry, as of August 26th, 2022, there were 1,868 clinical trials of stem cell therapy, of which 82 studies were specifically targeted toward regenerative medicines in the UK.

TABLE 105 UK: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	140.64	179.61	223.21	277.51	345.15	429.45	534.50	665.45	24.4%
Cell Therapy	211.07	274.90	348.22	441.08	558.69	707.62	896.18	1,134.86	26.7%
Tissue Engineering	73.61	87.20	99.95	113.85	128.73	144.24	159.79	174.51	11.8%
Total	425.31	541.71	671.37	832.44	1,032.57	1,281.30	1,590.48	1,974.82	24.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 106 UK: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	132.76	172.66	218.40	276.24	349.40	441.90	558.85	706.66	26.5%
Cell-based Immunotherapy Products	78.31	102.24	129.82	164.84	209.29	265.72	337.33	428.19	27.0%
Total	211.07	274.90	348.22	441.08	558.69	707.62	896.18	1,134.86	26.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 107 UK: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	132.76	172.66	218.40	276.24	349.40	441.90	558.85	706.66	26.5%
Cell Transplantations	78.31	102.24	129.82	164.84	209.29	265.72	337.33	428.19	27.0%
Total	211.07	274.90	348.22	441.08	558.69	707.62	896.18	1,134.86	26.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 108 UK: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	34.75	45.40	57.68	73.28	93.09	118.25	150.20	190.76	27.0%
Autologous Therapies	12.98	16.88	21.36	27.03	34.20	43.27	54.73	69.23	26.5%
Total	47.73	62.28	79.04	100.31	127.29	161.52	204.94	259.99	26.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 109 UK: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	133.15	168.02	206.29	253.37	311.29	382.56	470.26	578.17	22.9%
Oncology	109.60	140.52	175.30	218.77	273.12	341.08	426.09	532.41	24.9%
Dermatology & Wound Care	73.07	92.80	114.67	141.76	175.33	216.92	268.47	332.36	23.7%
Ophthalmology	22.70	29.41	37.05	46.68	58.84	74.16	93.49	117.86	26.0%
Cardiovascular Diseases	27.60	35.43	44.24	55.27	69.08	86.36	107.99	135.08	25.0%
Other Applications	59.18	75.54	93.82	116.58	144.92	180.21	224.17	278.94	24.3%
Total	425.31	541.71	671.37	832.44	1,032.57	1,281.30	1,590.48	1,974.82	24.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.3.2 GERMANY

8.3.2.1 Rising focus on clinical research and patent approvals to drive market

Since 1990, Germany has prioritized scientific groundwork and has invested over €230 million in regenerative medicine. However, the movement of clinical research to the commercialization of products is at a moderate pace. The new research area of indicators of myocardial infarctions and congestive heart failure use of replacement cells to repair damaged sections of the heart, and creating of cultured heart valves are prevailing in the growth of the regenerative medicine market in Germany and can also provide a turnover of €1 billion in the coming years. The rising number of patent applications indicates the short-term market potential of this research field.

According to the OECD, in 2020, the total expenditure on health accounted for 12.5% of the GDP in Germany. The rising healthcare spending in the country is expected to drive investments in the regenerative medicine market, thereby driving the growth of this market in Germany.

TABLE 110 GERMANY: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	90.23	119.17	153.23	197.23	254.11	327.73	423.11	546.81	29.0%
Cell Therapy	135.41	182.36	238.97	313.34	411.09	539.64	708.82	931.62	31.3%
Tissue Engineering	47.16	57.68	68.28	80.35	93.87	108.71	124.47	140.47	15.5%
Total	272.80	359.21	460.48	590.92	759.07	976.08	1,256.41	1,618.91	28.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 111 GERMANY: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	85.16	114.50	149.81	196.12	256.88	336.67	441.52	579.37	31.1%
Cell-based Immunotherapy Products	50.25	67.86	89.16	117.22	154.20	202.96	267.30	352.26	31.6%
Total	135.41	182.36	238.97	313.34	411.09	539.64	708.82	931.62	31.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 112 GERMANY: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	85.16	114.50	149.81	196.12	256.88	336.67	441.52	579.37	31.1%
Cell Transplantations	50.25	67.86	89.16	117.22	154.20	202.96	267.30	352.26	31.6%
Total	135.41	182.36	238.97	313.34	411.09	539.64	708.82	931.62	31.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 113 GERMANY: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	22.27	30.08	39.55	52.02	68.47	90.17	118.81	156.65	31.7%
Autologous Therapies	8.32	11.20	14.66	19.19	25.14	32.96	43.23	56.74	31.1%
Total	30.59	41.28	54.20	71.21	93.61	123.13	162.04	213.39	31.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 114 GERMANY: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	85.22	111.06	140.89	178.91	227.40	289.28	368.34	469.44	27.2%
Oncology	70.33	93.25	120.37	155.53	201.15	260.42	337.47	437.75	29.5%
Dermatology & Wound Care	46.95	61.60	78.70	100.63	128.81	165.05	211.70	271.81	28.1%
Ophthalmology	14.59	19.57	25.55	33.38	43.63	57.08	74.73	97.91	30.8%
Cardiovascular Diseases	17.73	23.56	30.48	39.47	51.16	66.37	86.19	112.03	29.7%
Other Applications	37.99	50.16	64.49	82.99	106.91	137.87	177.96	229.96	29.0%
Total	272.80	359.21	460.48	590.92	759.07	976.08	1,256.41	1,618.91	28.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.3.3 FRANCE

8.3.3.1 Structured regulatory framework to drive RM product pipeline

According to the Alliance for Regenerative Medicine, as of 2022, France had the third-highest number of total clinical trials conducted in Europe. The amount of clinical trials conducted in the region is prevalent owing to thorough process regulations and a structured regulatory framework. The cell and gene therapy products approved in France are Imlygic (Amgen Inc.), Holoclar (Chiesi Farmaceutici S.p.A.), Strimvelis (GSK Trading Services Limited), Alofisel (Takeda Pharmaceutical Company Limited), Kymriah (Novartis AG), Yescarta (Kite Pharma), and Luxturna (Spark Therapeutics) (Source: Alliance for Regenerative Medicine).

Although therapies are gaining patient response rapidly, the response is still lower than anticipated, primarily due to a lack of reimbursement policies for some of these therapies. In France, the different categories of cell therapies (i.e., ATMPs, hospital exemptions, and minimally manipulated cells) are governed by different regulatory statuses, which lead to different market access pathways. So far, only one ATMP (ChondroSelect) has been assessed by the Transparency Committee (Commission de la Transparence, or CT), the French health technology assessment agency, but reimbursement was not granted. Amendments around such reimbursement issues may expand the use of regenerative medicine in France, strengthening the market growth.

TABLE 115 FRANCE: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	63.01	85.62	113.12	149.44	197.41	260.79	344.54	455.25	32.1%
Cell Therapy	94.55	130.98	176.36	237.32	319.18	429.11	576.70	774.88	34.5%
Tissue Engineering	32.88	41.31	50.15	60.45	72.23	85.42	99.72	114.52	18.0%
Total	190.44	257.91	339.64	447.21	588.82	775.31	1,020.96	1,344.65	31.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 116 FRANCE: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	59.45	82.22	110.51	148.44	199.29	267.46	358.82	481.27	34.2%
Cell-based Immunotherapy Products	35.10	48.77	65.86	88.88	119.89	161.65	217.89	293.61	34.8%
Total	94.55	130.98	176.36	237.32	319.18	429.11	576.70	774.88	34.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 117 FRANCE: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Cell Therapy	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	59.45	82.22	110.51	148.44	199.29	267.46	358.82	481.27	34.2%
Cell Transplantations	35.10	48.77	65.86	88.88	119.89	161.65	217.89	293.61	34.8%
Total	94.55	130.98	176.36	237.32	319.18	429.11	576.70	774.88	34.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 118 FRANCE: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	15.53	21.58	29.16	39.36	53.12	71.65	96.61	130.23	34.9%
Autologous Therapies	5.81	8.04	10.82	14.54	19.53	26.22	35.19	47.22	34.3%
Total	21.34	29.62	39.97	53.90	72.64	97.87	131.80	177.45	34.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 119 FRANCE: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	59.35	79.48	103.48	134.69	175.28	228.08	296.77	386.15	30.1%
Oncology	49.11	67.00	88.88	117.88	156.33	207.32	274.94	364.67	32.6%
Dermatology & Wound Care	32.83	44.28	58.08	76.16	99.86	130.95	171.73	225.23	31.1%
Ophthalmology	10.20	14.10	18.95	25.44	34.14	45.81	61.44	82.40	34.2%
Cardiovascular Diseases	12.40	16.97	22.59	30.05	39.98	53.19	70.75	94.13	33.0%
Other Applications	26.54	36.07	47.67	62.99	83.23	109.97	145.33	192.08	32.1%
Total	190.44	257.91	339.64	447.21	588.82	775.31	1,020.96	1,344.65	31.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.3.4 REST OF EUROPE

The Rest of the Europe market is significantly smaller than other European markets. This regional segment comprises Austria, Belgium, Spain, Portugal, Sweden, Switzerland, Turkey, Italy, and Poland.

The growth of the RoE regenerative medicine market is driven by rising research intensity and availability of funding across RoE countries and public and private initiatives in the field of R&D. In Italy, for example, the government invested USD 161.1 million (EUR 150 million) annually for ten years (starting November 2014) to build a research campus in the Northwest Milan area for the “Human Technopole Italy 2040” project. The project focuses on genomics, personalized medicine, cancer, neurodegenerative diseases, big data, aging, and nutrition. Such initiatives will enhance life science and healthcare research and drive the demand for regenerative medicine.

Several companies have attracted funding, including Aelix Therapeutics, which secured USD 12.7 million (EUR 11.5 million) in series A funding from Ysios Capital (Spain), Johnson & Johnson Innovation, Inc. (US), and Caixa Capital Risc (Spain), for the development of immunotherapies. Various companies are also focusing on developing new therapies, expected to drive market growth in the coming years.

TABLE 120 REST OF EUROPE: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	231.16	303.18	387.15	494.92	633.38	811.44	1,040.69	1,336.15	28.1%
Cell Therapy	346.82	463.73	603.39	785.60	1,023.48	1,334.24	1,740.51	2,272.03	30.4%
Tissue Engineering	120.46	145.83	170.80	198.77	229.51	262.41	296.27	329.03	14.0%
Total	698.43	912.73	1,161.34	1,479.30	1,886.37	2,408.09	3,077.47	3,937.21	27.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 121 REST OF EUROPE: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	218.04	290.98	377.90	491.07	638.54	830.82	1,081.71	1,409.31	30.1%
Cell-based Immunotherapy Products	128.78	172.74	225.49	294.53	384.94	503.42	658.80	862.71	30.8%
Total	346.82	463.73	603.39	785.60	1,023.48	1,334.24	1,740.51	2,272.03	30.4%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 122 REST OF EUROPE: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	78.19	104.79	136.65	178.31	232.82	304.17	397.65	520.20	30.7%
Cell Transplantations	139.85	186.20	241.25	312.76	405.72	526.64	684.06	889.12	29.8%
Total	218.04	290.98	377.90	491.07	638.54	830.82	1,081.71	1,409.31	30.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 123 REST OF EUROPE: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	56.91	76.32	99.65	130.19	170.20	222.63	291.41	381.68	30.8%
Autologous Therapies	21.28	28.47	37.00	48.12	62.62	81.54	106.24	138.52	30.2%
Total	78.19	104.79	136.65	178.31	232.82	304.17	397.65	520.20	30.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 124 REST OF EUROPE: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

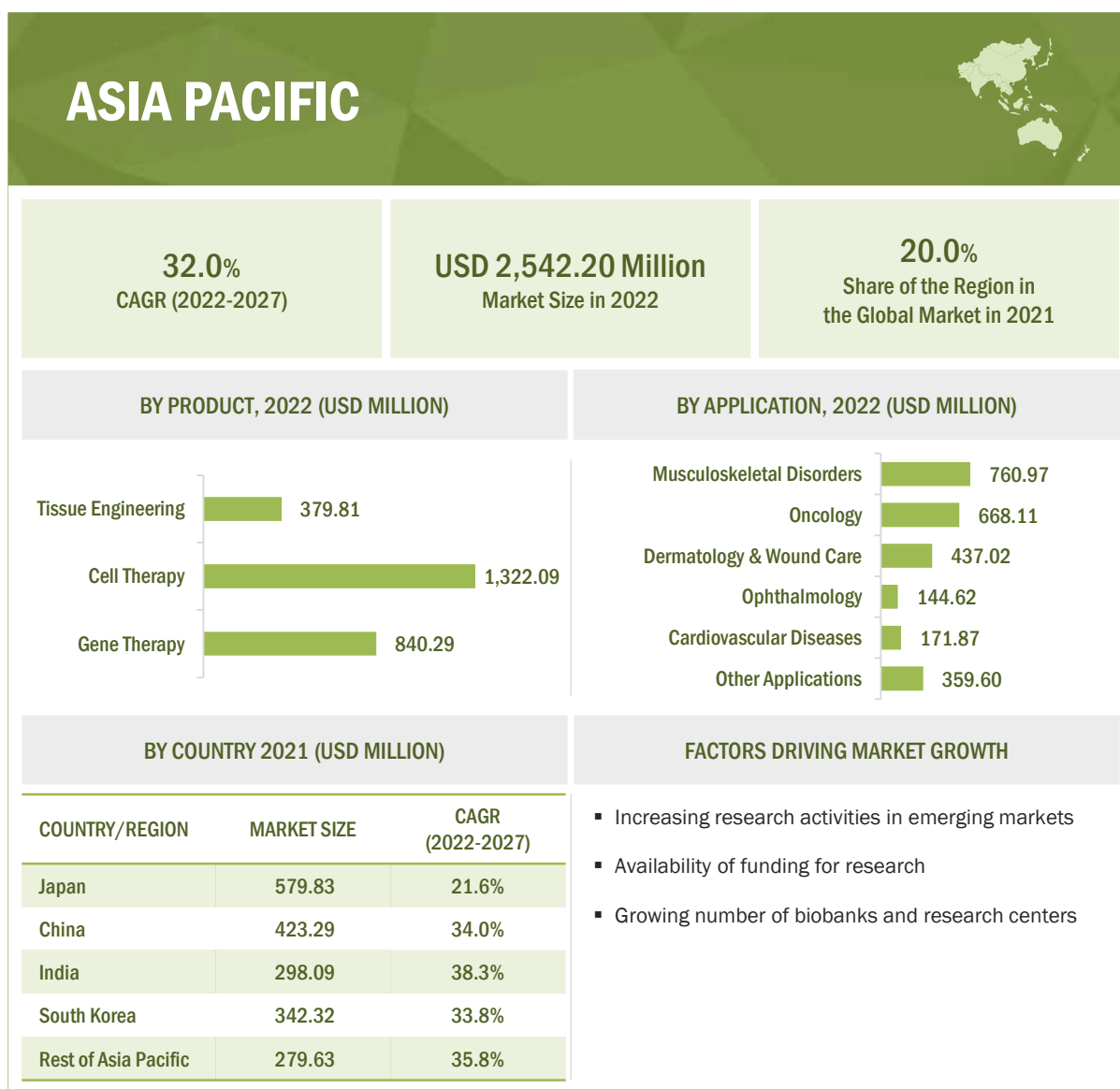
Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	217.19	280.37	352.32	443.16	557.94	703.10	886.84	1,119.63	26.0%
Oncology	180.20	237.31	304.27	390.53	501.77	645.37	830.92	1,070.92	28.6%
Dermatology & Wound Care	120.62	156.90	198.71	251.92	319.74	406.24	516.71	657.91	27.1%
Ophthalmology	37.49	50.09	65.13	84.74	110.32	143.72	187.36	244.43	30.3%
Cardiovascular Diseases	45.54	60.24	77.58	100.00	129.03	166.64	215.42	278.75	29.2%
Other Applications	97.39	127.82	163.34	208.94	267.57	343.02	440.22	565.56	28.2%
Total	698.43	912.73	1,161.34	1,479.30	1,886.37	2,408.09	3,077.47	3,937.21	27.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.4 ASIA PACIFIC

The Asia Pacific (APAC) regenerative medicine market is segmented into China, Japan, India, South Korea, and the Rest of Asia Pacific. The APAC presents significant opportunities for the growth of the regenerative medicine market. Factors such as increasing research activities in Asian countries, the availability of funding for research, and the growing number of biobanks and research centers are driving the growth of the APAC regenerative medicine market.

Starting in the early 2000s, countries in the Asia Pacific have emerged as strong contenders in gene and cell therapy. This region has traditionally been an attractive target for pharmaceutical companies due to its large population and rapidly aging society. Growing at a compound annual growth rate above 40 percent, the gene and cell therapy market in the Asia Pacific region is now outpacing the rest of the world by nearly four percent.

FIGURE 29 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET SNAPSHOT

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 125 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET, BY COUNTRY, 2020–2027 (USD MILLION)

Country/ Region	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022– 2027)
Japan	453.17	579.83	718.17	885.00	1,084.44	1,320.38	1,595.89	1,912.30	21.6%
China	306.26	423.29	568.44	762.75	1,022.79	1,370.74	1,836.24	2,458.97	34.0%
India	207.14	298.09	415.65	577.68	800.58	1,106.66	1,526.33	2,100.98	38.3%
South Korea	248.15	342.32	458.87	614.63	822.76	1,100.81	1,472.24	1,968.40	33.8%
Rest of Asia Pacific	199.23	279.63	381.08	518.57	704.78	956.76	1,297.58	1,758.30	35.8%
Total	1,413.96	1,923.15	2,542.20	3,358.63	4,435.34	5,855.35	7,728.27	10,198.95	32.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 126 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022– 2027)
Gene Therapy	472.69	639.55	840.29	1,102.78	1,445.81	1,893.78	2,478.50	3,241.31	31.0%
Cell Therapy	702.39	977.83	1,322.09	1,785.67	2,409.59	3,248.99	4,377.92	5,895.87	34.9%
Tissue Engineering	238.87	305.76	379.81	470.18	579.95	712.57	871.85	1,061.76	22.8%
Total	1,413.96	1,923.15	2,542.20	3,358.63	4,435.34	5,855.35	7,728.27	10,198.95	32.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 127 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	441.40	613.06	826.95	1,114.25	1,499.96	2,017.59	2,712.02	3,643.36	34.5%
Cell-based Immunotherapy Products	260.99	364.77	495.15	671.42	909.63	1,231.40	1,665.90	2,252.51	35.4%
Total	702.39	977.83	1,322.09	1,785.67	2,409.59	3,248.99	4,377.92	5,895.87	34.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 128 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	157.95	220.45	298.81	404.60	547.32	739.81	999.31	1,349.07	35.2%
Cell Transplantations	283.45	392.61	528.13	709.65	952.64	1,277.78	1,712.70	2,294.29	34.1%
Total	441.40	613.06	826.95	1,114.25	1,499.96	2,017.59	2,712.02	3,643.36	34.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 129 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	112.18	156.49	212.27	287.63	389.38	526.73	712.07	962.09	35.3%
Autologous Therapies	45.77	63.95	86.54	116.97	157.94	213.08	287.25	386.98	34.9%
Total	157.95	220.45	298.81	404.60	547.32	739.81	999.31	1,349.07	35.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 130 ASIA PACIFIC: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	437.86	585.65	760.97	987.76	1,280.99	1,659.89	2,149.29	2,781.10	29.6%
Oncology	365.17	501.04	668.11	890.36	1,185.97	1,579.17	2,102.19	2,797.95	33.2%
Dermatology & Wound Care	244.56	331.61	437.02	575.65	757.96	997.74	1,313.13	1,728.08	31.6%
Ophthalmology	76.27	106.56	144.62	196.07	265.56	359.40	486.07	657.01	35.4%
Cardiovascular Diseases	92.56	127.95	171.87	230.72	309.55	415.13	556.54	745.92	34.1%
Other Applications	197.54	270.35	359.60	478.07	635.31	844.01	1,121.05	1,488.88	32.9%
Total	1,413.96	1,923.15	2,542.20	3,358.63	4,435.34	5,855.35	7,728.27	10,198.95	32.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.4.1 JAPAN

8.4.1.1 Regulatory amendments to pharmaceutical legislation to drive commercialization of therapies

In 2014, Japan enacted new regulations aiming to accelerate and streamline the approval process of regenerative medicine products. The new legislation came from the Pharmaceuticals and Medical Devices (PMD) Act and the Safety of Regenerative Medicine Act. According to new Japanese regulations, manufacturers may undergo a shorter clinical trial before receiving conditional marketing approval, enabling them to market their products.

Japan has also opened up its gene therapy market using an expedited regulatory review process called the Sakigake Designation; it is the equivalent of the FDA breakthrough therapy designation and the EMA Priority Medicine (PRIME) designation. These are also known as conditional, time-limited approval processes. The Japan Pharmaceutical Medical Device Agency (PMDA) approved two gene therapy products simultaneously under the Sakigake designation and is expected to pass at least one therapy annually.

Further, on December 4, 2019, amendments were made to the PMD Act. The amendments were meant to:

- Improve the regulatory process to provide drugs and medical devices to patients more safely, quickly, and efficiently.
- Revise the roles of pharmacists or pharmacies so that patients can use drugs without anxiety.
- Establish a more reliable legal compliance system.

Various Western companies have ventured into Japan because of Japan's increasing focus on regenerative medicine development and the fast marketing process, including Mesoblast, Pluristem, and RepliCel. Mesoblast's TEMCELL HS Injection, a registered trademark of JCR Pharmaceuticals Co. Ltd. of Japan, is licensed and reimbursed in Japan.

TABLE 131 JAPAN: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	150.03	192.72	239.63	296.44	364.66	445.72	540.79	650.50	22.1%
Cell Therapy	225.07	294.71	373.35	470.35	588.92	732.37	903.70	1,105.05	24.2%
Tissue Engineering	78.07	92.41	105.19	118.21	130.86	142.30	151.40	156.75	8.3%
Total	453.17	579.83	718.17	885.00	1,084.44	1,320.38	1,595.89	1,912.30	21.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 132 JAPAN: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	141.48	184.87	233.71	293.82	367.13	455.60	561.00	684.57	24.0%
Cell-based Immunotherapy Products	83.60	109.84	139.64	176.53	221.79	276.77	342.69	420.48	24.7%
Total	225.07	294.71	373.35	470.35	588.92	732.37	903.70	1,105.05	24.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 133 JAPAN: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	50.69	66.54	84.49	106.69	133.90	166.89	206.40	252.96	24.5%
Cell Transplantations	90.79	118.33	149.22	187.13	233.23	288.71	354.60	431.61	23.7%
Total	141.48	184.87	233.71	293.82	367.13	455.60	561.00	684.57	24.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 134 JAPAN: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	36.89	48.45	61.55	77.77	97.65	121.78	150.70	184.79	24.6%
Autologous Therapies	13.80	18.09	22.94	28.92	36.24	45.11	55.70	68.17	24.3%
Total	50.69	66.54	84.49	106.69	133.90	166.89	206.40	252.96	24.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 135 JAPAN: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	140.70	177.53	216.80	263.35	318.04	381.55	454.31	536.16	19.9%
Oncology	116.96	150.87	188.38	233.99	289.00	354.65	432.01	521.68	22.6%
Dermatology & Wound Care	78.31	99.79	123.09	151.07	184.36	223.54	269.07	321.08	21.1%
Ophthalmology	24.37	31.94	40.49	51.05	63.96	79.59	98.28	120.25	24.3%
Cardiovascular Diseases	29.59	38.38	48.19	60.18	74.72	92.16	112.83	136.92	23.2%
Other Applications	63.24	81.32	101.22	125.36	154.37	188.88	229.40	276.22	22.2%
Total	453.17	579.83	718.17	885.00	1,084.44	1,320.38	1,595.89	1,912.30	21.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.4.2 CHINA

8.4.2.1 Favorable government support for development of regenerative medicine to drive market

In China, the geriatric population is increasing much faster than in other low- and middle-income countries. Rising number of people aged over 60 has also pushed the disease prevalence rate, especially for chronic and age-related ailments. Favorable government support for regenerative medicine is also expected to support market growth.

Fourteen years before the FDA approved its first gene therapy product, China had approved Gendicine (Shenzhen SiBiono GeneTech Co. Ltd.) for the treatment of head-and-neck squamous cell carcinoma by replacing mutated copies of the p53 tumor suppressor gene with normal copies.

TABLE 136 CHINA: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	102.81	140.69	187.03	248.43	329.72	437.32	579.71	768.12	32.6%
Cell Therapy	152.11	215.14	295.51	405.37	555.44	760.30	1,039.80	1,420.95	36.9%
Tissue Engineering	51.34	67.46	85.89	108.95	137.63	173.12	216.73	269.89	25.7%
Total	306.26	423.29	568.44	762.75	1,022.79	1,370.74	1,836.24	2,458.97	34.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 137 CHINA: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	95.60	134.91	184.90	253.07	345.98	472.52	644.77	879.13	36.6%
Cell-based Immunotherapy Products	56.51	80.23	110.61	152.30	209.46	287.78	395.03	541.82	37.4%
Total	152.11	215.14	295.51	405.37	555.44	760.30	1,039.80	1,420.95	36.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 138 CHINA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	34.23	48.53	66.82	91.89	126.22	173.18	237.41	325.20	37.2%
Cell Transplantations	61.37	86.38	118.07	161.18	219.76	299.33	407.35	553.93	36.2%
Total	95.60	134.91	184.90	253.07	345.98	472.52	644.77	879.13	36.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 139 CHINA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	24.90	35.33	48.70	67.06	92.22	126.69	173.89	238.48	37.4%
Autologous Therapies	9.32	13.20	18.12	24.84	34.00	46.50	63.53	86.72	36.8%
Total	34.23	48.53	66.82	91.89	126.22	173.18	237.41	325.20	37.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 140 CHINA: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	94.93	129.17	170.74	225.45	297.40	391.99	516.30	679.59	31.8%
Oncology	79.08	110.22	149.27	201.97	273.09	369.00	498.35	672.77	35.1%
Dermatology & Wound Care	52.95	72.93	97.60	130.51	174.39	232.89	310.88	414.83	33.6%
Ophthalmology	16.50	23.40	32.22	44.30	60.84	83.45	114.36	156.59	37.2%
Cardiovascular Diseases	20.03	28.11	38.31	52.17	70.98	96.50	131.11	178.03	36.0%
Other Applications	42.77	59.45	80.29	108.34	146.10	196.90	265.24	357.15	34.8%
Total	306.26	423.29	568.44	762.75	1,022.79	1,370.74	1,836.24	2,458.97	34.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.4.3 INDIA

8.4.3.1 Introduction of novel therapies for CVD and cancer to support market growth

In India, the geriatric population is increasing much faster than other low- and middle-income countries. Development of new therapies like the cell therapy led by Immuneel Therapeutics, backed by Biocon Chairperson Kiran Mazumdar Shaw and US oncologist Siddhartha Mukherjee; Dr. Reddy's Laboratories and Chinese firm Shenzhen Pregene Biopharma; and ImmuniACT, an IIT Bombay spin-off, are among firms working to bring CAR T-cell therapy for cancers, especially blood cancers, to India within the next two years. Moreover, in 2021, the first CAR-T cell therapy was created in India, making more lucrative market growth opportunities.

TABLE 141 INDIA: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	69.56	99.13	136.85	188.27	258.24	353.29	482.18	656.72	36.8%
Cell Therapy	102.90	151.57	216.16	307.13	434.93	614.05	864.61	1,214.50	41.2%
Tissue Engineering	34.68	47.39	62.64	82.28	107.41	139.32	179.54	229.76	29.7%
Total	207.14	298.09	415.65	577.68	800.58	1,106.66	1,526.33	2,100.98	38.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 142 INDIA: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	64.66	95.02	135.19	191.62	270.69	381.26	535.53	750.43	40.9%
Cell-based Immunotherapy Products	38.24	56.55	80.98	115.52	164.23	232.79	329.08	464.08	41.8%
Total	102.90	151.57	216.16	307.13	434.93	614.05	864.61	1,214.50	41.2%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 143 INDIA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	23.13	34.16	48.84	69.58	98.78	139.81	197.35	277.89	41.6%
Cell Transplantations	41.53	60.86	86.34	122.04	171.91	241.44	338.18	472.53	40.5%
Total	64.66	95.02	135.19	191.62	270.69	381.26	535.53	750.43	40.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 144 INDIA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	16.83	24.86	35.59	50.76	72.15	102.25	144.51	203.73	41.8%
Autologous Therapies	6.30	9.30	13.26	18.82	26.63	37.56	52.85	74.16	41.1%
Total	23.13	34.16	48.84	69.58	98.78	139.81	197.35	277.89	41.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 145 INDIA: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	64.11	90.67	124.23	169.59	230.78	313.15	423.82	572.25	35.7%
Oncology	53.51	77.68	109.27	153.20	214.16	298.58	415.32	576.51	39.5%
Dermatology & Wound Care	35.84	51.42	71.49	99.07	136.90	188.69	259.48	356.12	37.9%
Ophthalmology	11.18	16.54	23.68	33.78	48.02	68.04	96.13	135.48	41.7%
Cardiovascular Diseases	13.57	19.85	28.14	39.74	55.96	78.57	110.05	153.79	40.5%
Other Applications	28.95	41.92	58.83	82.29	114.76	159.63	221.54	306.84	39.1%
Total	207.14	298.09	415.65	577.68	800.58	1,106.66	1,526.33	2,100.98	38.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.4.4 SOUTH KOREA

8.4.4.1 Strong research base and availability of funds to support market growth

As a global leader in plastic surgeries and regenerative medicine, the South Korean market has an established base for cell therapies. South Korea also has the largest approved cell therapy product portfolio with a total of 18 products in the world, with products such as Chondron, for treating articular cartilage injury of the knee joint.

In 2020, four cell therapies (Chondron, Kaloderm, Cupistem, and Keraheal-Allo) were formally approved in South Korea. Cupistem and Keraheal-alo were listed based on lower prices than alternative medicines.

The South Korean administration has been involved in pushing gene and cell therapy. The “Act on the Safety and Support of Advanced Regenerative Medical and Advanced Biopharmaceuticals” was passed in 2019. This act allows the conditional approval of drugs that have completed phase two clinical trials for commercial sale. Industry experts expect this legislation to shorten time-to-market by up to four years.

A second proposed legislation will remove significant hurdles to gene therapy and gene editing research by amending the Bioethics Act, which currently limits research to specific and designated diseases, aligning the Act with global norms.

TABLE 146 SOUTH KOREA: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	83.35	113.91	151.17	200.43	265.56	351.64	465.38	615.67	32.4%
Cell Therapy	123.30	174.13	238.73	326.90	447.14	611.02	834.26	1,138.25	36.7%
Tissue Engineering	41.50	54.28	68.97	87.30	110.06	138.15	172.59	214.48	25.5%
Total	248.15	342.32	458.87	614.63	822.76	1,100.81	1,472.24	1,968.40	33.8%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 147 SOUTH KOREA: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	77.47	109.12	149.23	203.82	278.07	379.01	516.15	702.40	36.3%
Cell-based Immunotherapy Products	45.83	65.00	89.50	123.08	169.07	232.01	318.11	435.85	37.2%
Total	123.30	174.13	238.73	326.90	447.14	611.02	834.26	1,138.25	36.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 148 SOUTH KOREA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	27.69	39.21	53.90	74.01	101.50	139.06	190.36	260.39	37.0%
Cell Transplantations	49.78	69.91	95.33	129.81	176.57	239.94	325.79	442.01	35.9%
Total	77.47	109.12	149.23	203.82	278.07	379.01	516.15	702.40	36.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 149 SOUTH KOREA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	17.40	24.57	33.83	46.51	63.88	87.65	120.16	164.59	37.2%
Autologous Therapies	10.29	14.64	20.07	27.50	37.62	51.41	70.21	95.80	36.7%
Total	27.69	39.21	53.90	74.01	101.50	139.06	190.36	260.39	37.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 150 SOUTH KOREA: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	76.67	103.78	136.45	179.21	235.12	308.20	403.65	528.26	31.1%
Oncology	64.12	89.28	120.77	163.25	220.50	297.66	401.63	541.70	35.0%
Dermatology & Wound Care	42.95	59.12	79.06	105.65	141.10	188.35	251.31	335.22	33.5%
Ophthalmology	13.42	19.06	26.28	36.19	49.76	68.34	93.75	128.50	37.4%
Cardiovascular Diseases	16.28	22.87	31.20	42.53	57.92	78.82	107.18	145.66	36.1%
Other Applications	34.70	48.21	65.09	87.80	118.35	159.45	214.72	289.05	34.7%
Total	248.15	342.32	458.87	614.63	822.76	1,100.81	1,472.24	1,968.40	33.8%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.4.5 REST OF ASIA PACIFIC

The Rest of Asia Pacific includes Taiwan, Malaysia, Sri Lanka, Hong Kong, Thailand, Indonesia, Vietnam, Philippines, and Singapore.

Taiwan's Food and Drugs Administration typically defers to the US FDA and Japan's PMDA when adopting new drugs and therapies. This has been the case for gene and cell therapy too. Though it has not officially approved gene therapy, Taiwan became the ninth regulatory agency worldwide to approve clinical cell therapy treatments in 2018. This bill introduced six autologous cell therapy treatments: two in oncology, two in skin cell transplantation, and two in bone and cartilage regeneration. As of January 2020, four hospitals in Taiwan are licensed to provide these treatments, and 26 more have applied for the license to provide cell therapy treatments.

TABLE 151 REST OF ASIA PACIFIC: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	66.94	93.11	125.62	169.21	227.62	305.82	410.43	550.30	34.4%
Cell Therapy	99.01	142.29	198.34	275.91	383.16	531.26	735.55	1,017.11	38.7%
Tissue Engineering	33.28	44.23	57.12	73.45	93.99	119.69	151.59	190.88	27.3%
Total	199.23	279.63	381.08	518.57	704.78	956.76	1,297.58	1,758.30	35.8%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 152 REST OF ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	62.20	89.14	123.92	171.92	238.09	329.21	454.56	626.84	38.3%
Cell-based Immunotherapy Products	36.81	53.15	74.42	104.00	145.07	202.04	280.99	390.28	39.3%
Total	99.01	142.29	198.34	275.91	383.16	531.26	735.55	1,017.11	38.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 153 REST OF ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	22.21	32.01	44.75	62.43	86.93	120.86	167.79	232.63	39.1%
Cell Transplantations	39.98	57.13	79.17	109.49	151.16	208.36	286.78	394.21	37.9%
Total	62.20	89.14	123.92	171.92	238.09	329.21	454.56	626.84	38.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 154 REST OF ASIA PACIFIC: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	16.16	23.29	32.60	45.53	63.48	88.36	122.82	170.50	39.2%
Autologous Therapies	6.05	8.72	12.15	16.90	23.45	32.50	44.96	62.13	38.6%
Total	22.21	32.01	44.75	62.43	86.93	120.86	167.79	232.63	39.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 155 REST OF ASIA PACIFIC: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	437.86	585.65	760.97	987.76	1,280.99	1,659.89	2,149.29	2,781.10	29.6%
Oncology	365.17	501.04	668.11	890.36	1,185.97	1,579.17	2,102.19	2,797.95	33.2%
Dermatology & Wound Care	244.56	331.61	437.02	575.65	757.96	997.74	1,313.13	1,728.08	31.6%
Ophthalmology	76.27	106.56	144.62	196.07	265.56	359.40	486.07	657.01	35.4%
Cardiovascular Diseases	92.56	127.95	171.87	230.72	309.55	415.13	556.54	745.92	34.1%
Other Applications	197.54	270.35	359.60	478.07	635.31	844.01	1,121.05	1,488.88	32.9%
Total	1,413.96	1,923.15	2,542.20	3,358.63	4,435.34	5,855.35	7,728.27	10,198.95	32.0%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.5 LATIN AMERICA

8.5.1 RISING INCIDENCE OF DIABETES AND NEUROLOGICAL DISEASES TO DRIVE ADOPTION OF RM THERAPIES

The Latin American regenerative medicine market is significantly smaller than other regional markets. The growth of this market is mainly driven by ongoing research in stem cells and regenerative medicines due to their positive results in the treatment of Parkinson's disease, multiple sclerosis, diabetes, vision problems, and neurological diseases. Other factors include the establishment of biobanks and research centers, increasing investments in research, and the rising prevalence of diabetes. For instance:

- Diabetes is the second-highest cause of mortality in Mexico; it is estimated that 16.9% of diabetics in the country suffer from DFUs (Source: HealthData.org). According to the IDF, the number of people with diabetes in the age group of 20–79 years in Mexico is expected to reach 17.2 million by 2030 from 14 million in 2021.

- According to the IDF, the cost of diabetes-related health expenditure in Brazil is the third highest in the world, at 42.9 billion USD. In addition, 18 million adults (11.9%) have Impaired Glucose Tolerance (IGT), which places them at high risk of developing type 2 diabetes.

An increase in the incidence of diabetes, Parkinson's disease, and neurological diseases are expected to offer lucrative opportunities to the regenerative medicine players in this market.

TABLE 156 LATIN AMERICA: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	74.33	98.84	127.85	165.41	214.06	277.11	358.84	464.85	29.5%
Cell Therapy	109.93	151.02	201.73	269.39	359.70	480.23	641.13	855.92	33.5%
Tissue Engineering	36.90	46.80	57.57	70.57	86.16	104.74	126.66	152.25	21.5%
Total	221.15	296.66	387.14	505.37	659.93	862.08	1,126.63	1,473.02	30.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 157 LATIN AMERICA: REGENERATIVE MEDICINE MARKET, BY CELL THERAPY, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	69.04	94.58	125.98	167.75	223.34	297.31	395.76	526.81	33.1%
Cell-based Immunotherapy Products	40.88	56.44	75.75	101.65	136.37	182.93	245.37	329.11	34.2%
Total	109.93	151.02	201.73	269.39	359.70	480.23	641.13	855.92	33.5%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 158 LATIN AMERICA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	24.64	33.95	45.48	60.91	81.57	109.21	146.20	195.72	33.9%
Cell Transplantations	44.41	60.64	80.50	106.84	141.77	188.10	249.56	331.09	32.7%
Total	69.04	94.58	125.98	167.75	223.34	297.31	395.76	526.81	33.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 159 LATIN AMERICA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	17.91	24.69	33.10	44.36	59.45	79.65	106.70	142.94	34.0%
Autologous Therapies	6.73	9.26	12.38	16.55	22.12	29.56	39.50	52.78	33.6%
Total	24.64	33.95	45.48	60.91	81.57	109.21	146.20	195.72	33.9%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 160 LATIN AMERICA: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	68.11	89.35	113.96	145.33	185.29	236.19	301.00	383.53	27.5%
Oncology	57.19	77.49	102.13	134.63	177.52	234.14	308.92	407.73	31.9%
Dermatology & Wound Care	38.33	51.35	66.94	87.28	113.84	148.54	193.89	253.21	30.5%
Ophthalmology	12.00	16.64	22.41	30.16	40.57	54.55	73.32	98.52	34.5%
Cardiovascular Diseases	14.55	19.94	26.56	35.38	47.12	62.76	83.60	111.36	33.2%
Other Applications	30.97	41.90	55.15	72.59	95.59	125.90	165.89	218.66	31.7%
Total	221.15	296.66	387.14	505.37	659.93	862.08	1,126.63	1,473.02	30.6%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

8.6 MIDDLE EAST & AFRICA

8.6.1 INCREASE INCIDENCE OF CHRONIC CONDITIONS TO SUPPORT MARKET GROWTH

The regenerative medicine market in the Middle East & Africa is significantly smaller than other regional markets. The growing geriatric population is expected to drive the regenerative medicine market in this region. The increased frequency of genetic and chronic illnesses is one of the primary factors driving the regenerative medicine market's growth. Furthermore, the market is driven by the expanding geriatric population, which is prone to cardiological & oncological diseases.

TABLE 161 MIDDLE EAST & AFRICA: REGENERATIVE MEDICINE MARKET, BY PRODUCT, 2020–2027 (USD MILLION)

Product	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Gene Therapy	93.17	124.17	160.94	208.64	270.51	350.82	455.08	590.51	29.7%
Cell Therapy	137.79	189.68	253.86	339.64	454.28	607.50	812.29	1,086.04	33.7%
Tissue Engineering	46.18	58.61	72.11	88.39	107.89	131.05	158.29	189.93	21.4%
Total	277.14	372.47	486.92	636.67	832.68	1,089.37	1,425.67	1,866.49	30.8%

Source: National Institutes of Health (NIH), National Center for Biotechnology Information (NCBI), Alliance for Regenerative Medicine (ARM), US Food and Drug Administration (FDA), European, Middle Eastern & African Society for Biopreservation and Biobanking (ESBB), International Society for Biological and Environmental Repositories (ISBER), International Society for Stem Cell Research (ISSCR), International Society for Cellular Therapy (ISCT), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), American Society for Cell Biology (ASCB), World Bank, World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 162 MIDDLE EAST & AFRICA: REGENERATIVE MEDICINE MARKET, BY CELL THERAPY, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy	86.53	118.76	158.46	211.36	281.83	375.73	500.85	667.58	33.3%
Cell-based Immunotherapy Products	51.26	70.92	95.40	128.29	172.45	231.77	311.44	418.46	34.4%
Total	137.79	189.68	253.86	339.64	454.28	607.50	812.29	1,086.04	33.7%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 163 MIDDLE EAST & AFRICA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Stem Cell Therapy Products	30.85	42.60	57.19	76.75	102.96	138.09	185.17	248.28	34.1%
Cell Transplantations	55.68	76.16	101.27	134.61	178.88	237.65	315.68	419.30	32.9%
Total	86.53	118.76	158.46	211.36	281.83	375.73	500.85	667.58	33.3%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 164 MIDDLE EAST & AFRICA: REGENERATIVE MEDICINE MARKET FOR STEM CELL THERAPY PRODUCTS, BY TYPE, 2020–2027 (USD MILLION)

Type	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Allogeneic Therapies	22.42	30.98	41.61	55.87	74.99	100.63	135.01	181.13	34.2%
Autologous Therapies	8.43	11.62	15.58	20.88	27.97	37.46	50.16	67.15	33.9%
Total	30.85	42.60	57.19	76.75	102.96	138.09	185.17	248.28	34.1%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 165 MIDDLE EAST & AFRICA: REGENERATIVE MEDICINE MARKET, BY APPLICATION, 2020–2027 (USD MILLION)

Application	2020	2021	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Musculoskeletal Disorders	85.21	111.80	142.61	181.82	231.71	295.19	375.91	478.52	27.4%
Oncology	71.70	97.36	128.60	169.86	224.41	296.53	391.92	518.14	32.1%
Dermatology & Wound Care	48.06	64.55	84.33	110.21	144.05	188.35	246.36	322.34	30.8%
Ophthalmology	15.07	20.96	28.33	38.25	51.61	69.59	93.78	126.33	34.8%
Cardiovascular Diseases	18.26	25.10	33.55	44.82	59.87	79.96	106.78	142.60	33.6%
Other Applications	38.84	52.68	69.51	91.71	121.03	159.75	210.92	278.57	32.0%
Total	277.14	372.47	486.92	636.67	832.68	1,089.37	1,425.67	1,866.49	30.8%

Source: World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), US Food and Drug Administration (FDA), Alliance for Regenerative Medicine (ARM), International Society for Cell & Gene Therapy (ISCT), International Society for Stem Cell Research (ISSCR), American Society of Gene & Cell Therapy (ASGCT), International Cell Research Organization (ICRO), European Society of Gene and Cell Therapy (ESGCT), Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

9 COMPETITIVE LANDSCAPE

9.1 OVERVIEW

The competitive landscape includes the analysis of key growth strategies adopted by major players between January 2020 and October 2022. Players in the global regenerative medicine market have employed various strategies to expand their global footprint and increase their market shares. The key strategies adopted by the top players in this market include service launches, collaborations, agreements, acquisitions, and partnerships.

Prominent players in this market are Novartis AG (Switzerland), Biogen (US), Sarepta Therapeutics, Inc. (US), Gilead Sciences, Inc. (US), Amgen, Inc. (US), Smith & Nephew plc (UK), Medipost Co., Ltd. (South Korea), JCR Pharmaceuticals Co., Ltd (Japan), Takeda Pharmaceutical Company (Japan), CORESTEM, Inc (South Korea), Vericel Corporation (US), MIMEDX (US), Organogenesis Inc. (US), Medtronic plc (US), Orthocell Ltd. (Australia), Mesoblast Ltd (Australia), BioRestorative Therapies, Inc (US), Pluri Inc. (Israel), Tegoscience (South Korea), Anterogen Co., Ltd. (South Korea), bluebird bio, Inc. (US), APAC Biotech (India), Shenzhen SiBiono Genetech Co., Ltd. (China), Aspect Biosystems (Canada), and Athersys, Inc. (US)

9.2 STRATEGIES ADOPTED BY KEY PLAYERS

The market for regenerative medicine is moderately consolidated, with many players strategizing to capture the market. Players provide gene therapies, cell therapies, and tissue engineering products aiming to penetrate various markets and expand their geographical presence.

FIGURE 30 KEY DEVELOPMENTS IN REGENERATIVE MEDICINE MARKET
(JANUARY 2020 – NOVEMBER 2022)

COMPANY NAME	ORGANIC GROWTH STRATEGIES	INORGANIC GROWTH STRATEGIES	
	COLLABORATIONS	ACQUISITIONS	REGULATORY APPROVALS
NOVARTIS (SWITZERLAND)	Novartis collaborated Alnylam to explore targeted therapy to restore liver function.	Novartis acquired CellforCure to expand global production capacities for innovative cell and gene therapies.	- The US Food and Drug Administration (FDA) granted Regenerative Medicine Advanced Therapy (RMAT) designation to Kymriah for an investigational new indication to treat patients with relapsed or refractory (r/r) follicular lymphoma (FL). - The chimeric antigen receptor T-cell (CAR-T) therapy, Kymriah, obtained NHI Price Listing for the treatment of CD19-positive relapsed or refractory B-cell acute lymphoblastic leukemia (ALL) and CD19-positive relapsed or refractory diffuse large B-cell lymphoma (DLBCL).
ORGANOGENESIS (US)			- Organogenesis ReNu received FDA Regenerative Medicine Advanced Therapy (RMAT) designation for osteoarthritis of the knee. - The US Food and Drug Administration (FDA) granted Regenerative Medicine Advanced Therapy (RMAT) designation to Kymriah for an investigational new indication to treat patients with relapsed or refractory (r/r) follicular lymphoma (FL).

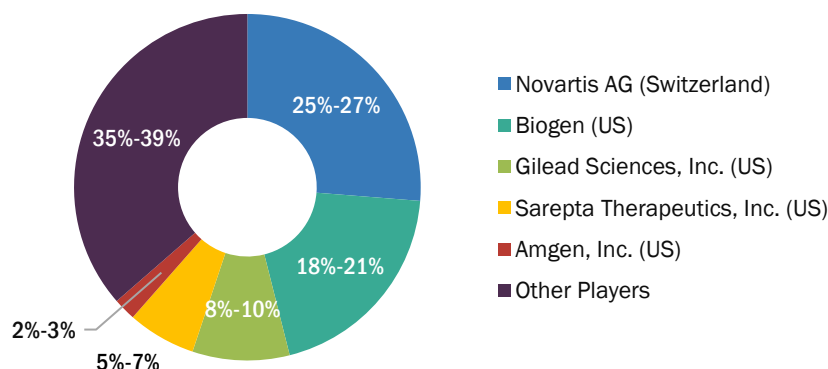
Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

9.3 MARKET SHARE ANALYSIS

The global regenerative medicine market is moderately consolidated with the top five players- Novartis AG (Switzerland), Biogen (US), Sarepta Therapeutics, Inc. (US), Gilead Sciences, Inc. (US), Amgen, Inc. (US) together accounting for above 60-65% of the global regenerative medicine market.

Players in the regenerative medicine market were identified through secondary research, and their market shares (in terms of value) were evaluated based on several aspects, products provided, segmental revenue analysis, geographic presence, recent market developments, and brand image. Primary research interviews were conducted with key opinion leaders in the industry, such as chief executive officers, directors, industry experts, and other executives, to validate revenues and market shares.

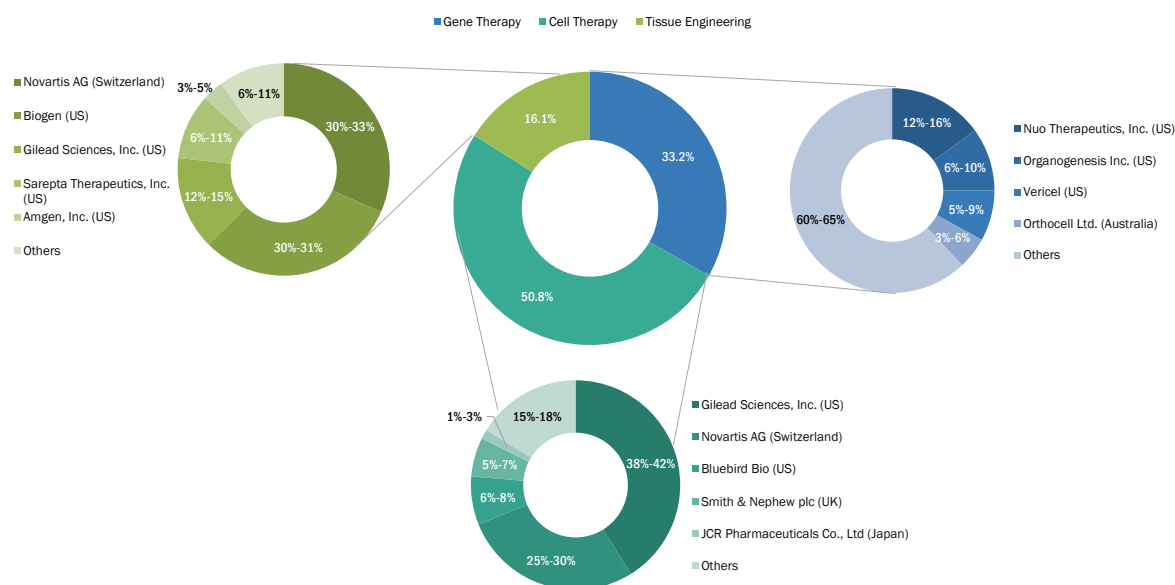
FIGURE 31 REGENERATIVE MEDICINE MARKET: MARKET SHARE ANALYSIS OF KEY PLAYERS (2021)



Source: Annual Reports, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

Note: Other players include Smith & Nephew plc (UK), Medipost Co., Ltd. (South Korea), JCR Pharmaceuticals Co., Ltd (Japan), Takeda Pharmaceutical Company (Japan), CORESTEM Inc (South Korea), Vericel Corporation (US), MIMEDX (US), Organogenesis Inc. (US), Medtronic plc (US), Orthocell Ltd. (Australia), Mesoblast Ltd (Australia), BioRestorative Therapies, Inc (US) Pluri Inc. (Israel), Tegoscience. (South Korea), Anterogen Co., Ltd. (South Korea), bluebird bio, Inc. (US), APAC Biotech (India), Shenzhen SiBiono GeneTech Co., Ltd. (China), Aspect Biosystems (Canada), and Athersys, Inc. (US).

FIGURE 32 REGENERATIVE MEDICINE MARKET: MARKET SHARE ANALYSIS OF KEY PLAYERS, BY PRODUCT (2021)



Source: Annual Reports, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

Novartis AG (Switzerland) was the leading player in the regenerative medicine market, with a share of ~25-27% in 2021. The company offers two gene therapy products—KYMRIAH and ZOLGENSMA. Globally, the company has strong distribution channels and a good brand image. As a result, KYMRIAH registered significant growth in 2018—KYMRIAH registered sales of USD 587 million in 2021, a rise from 24% in 2020. ZOLGENSMA registered sales of USD 1,351 million in 2021, up from 47% in 2020. Novartis has strong distribution channels in the US. As a result, in 2021, the company generated 32.6% of its total revenue from the US alone.

Biogen (US) is the second-leading player in the regenerative medicine market. However, the revenue of SPINRAZA is declining year on year. A decline in demand primarily brought on the decline in SPINRAZA revenue because of heightened competition in the US and Germany and a decline in pricing in the US and the RoW markets, which were only partially offset by an uptick in sales volumes in Latin America and some distributor markets.

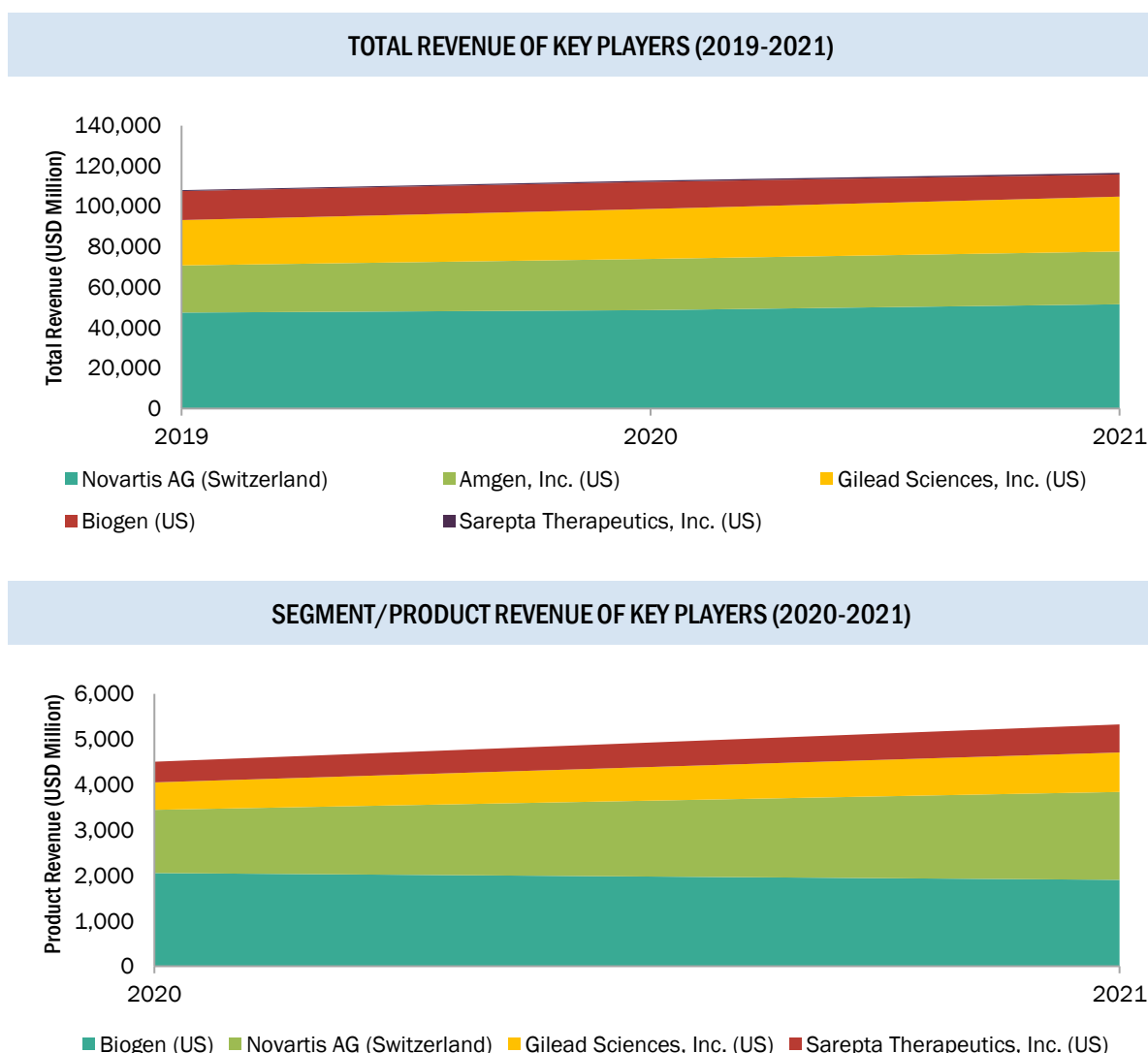
Gilead Sciences Inc. (US) was the third-leading player in the regenerative medicine market. The company offers YESCARTA, a gene therapy product for treating non-Hodgkin's lymphoma, and TECARTUS for mantle cell lymphoma or acute lymphoblastic leukemia. These products generated sales revenue of USD 871 million in 2021. The company is also focusing on R&D to introduce products into the market. For instance, the company spent USD 5.36 billion on research and development activities in 2021.

9.4 REVENUE SHARE ANALYSIS

Players in the regenerative medicine market were identified through secondary research and further validated via primary sources. In terms of value, each market player's share was inferred from annual reports, presentations, and secondary research sources. Primary research interviews were conducted with key opinion leaders, such as CEOs, directors, industry experts, and other executives, to validate revenue and market share.

The size of the regenerative medicine market in terms of value (USD million)—based on region, product, and application has been derived using forecasting techniques based on demand patterns and product trends. The first step in preparing the report was to define the approach and the methodology to estimate market sizes and define the class of variables that can be utilized to obtain estimates and projections. Primary interviews and feature mapping followed this on a regional basis from the MarketsandMarkets repository. The bottom-up approach has been followed to derive the market size in value. In-depth interviews have validated the market size with industry experts (excerpts from the discussion guide in the Appendix) and through secondary research.

The revenue analysis was calculated based on the performance of major players in the regenerative medicine market during the last two years (2020-2022). This gave an estimation of the overall market size, structure, and recent developments. Novartis AG (Switzerland), Biogen (US), Sarepta Therapeutics, Inc. (US), Gilead Sciences, Inc. (US), and Amgen, Inc. (US) are the leading players in the market.

FIGURE 33 REVENUE ANALYSIS FOR KEY COMPANIES

Note: The shares presented in the right-hand side figure above are for the closest business segment (including regenerative medicine) in the total revenues or regenerative medicine product revenues of key players.

Source: Company Website and Annual Reports

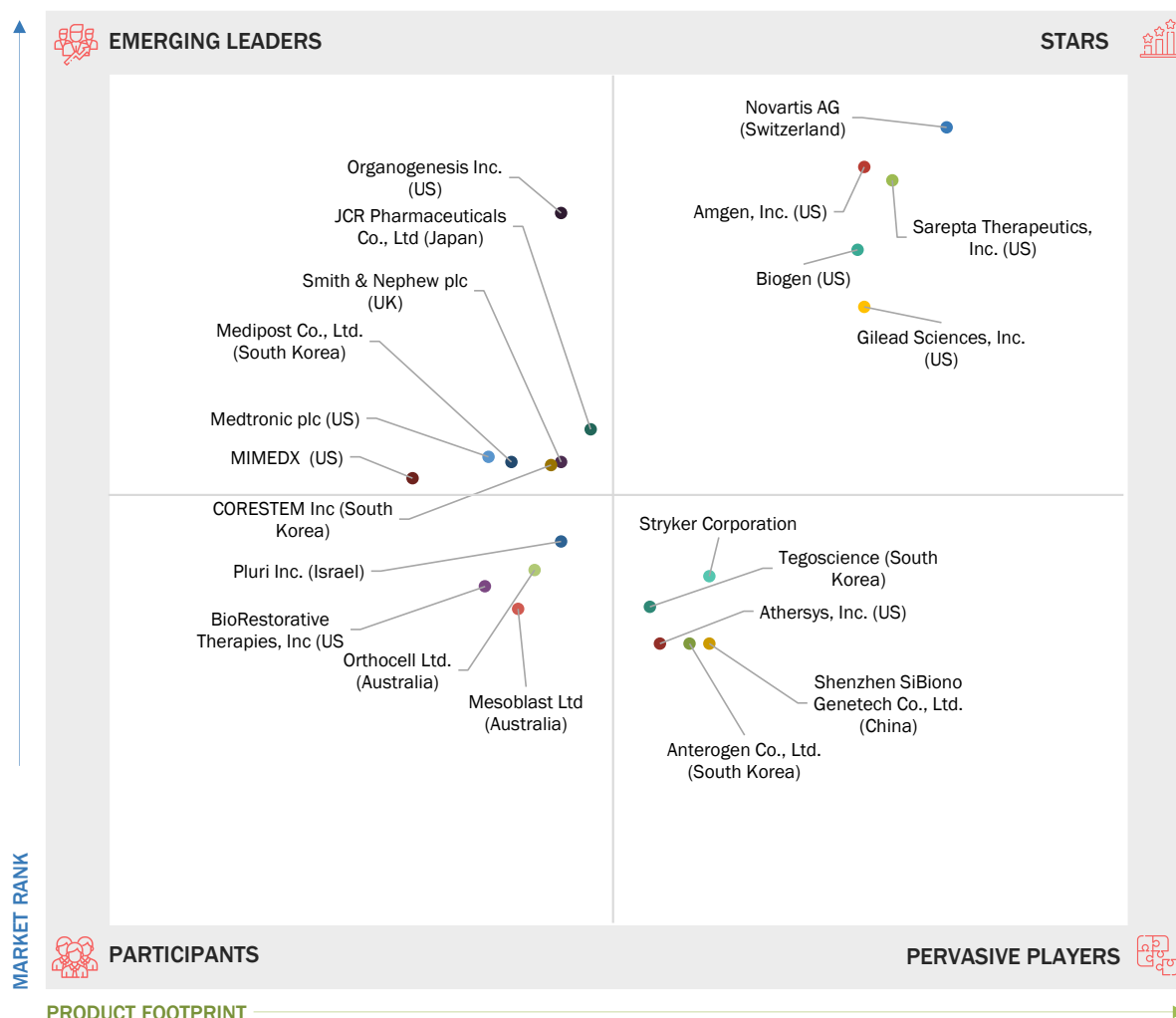
9.5 COMPANY EVALUATION QUADRANT

The company evaluation matrix provides information on the top 25 players offering regenerative medicine and outlines findings on the performance of each vendor. The industry matrix evaluations are based on two broad categories—product footprint and market share in the regenerative medicine market. Each category carries various criteria based on which the companies have been evaluated. The evaluation criteria considered under product footprint include products catering to different regenerative medicine therapies. The evaluation criteria considered under market share include reach (based on market and geographic presence), industry coverage, viability, and inorganic growth strategies. The top 25 companies are placed into four categories—Stars, Emerging Leaders, Pervasive Players, and Participants—based on their performance in each criterion.

The prominent players in the regenerative medicine market are Novartis AG (Switzerland), Biogen (US), Sarepta Therapeutics, Inc. (US), Gilead Sciences, Inc. (US), Amgen, Inc. (US), Smith & Nephew plc (UK), MEDIPOST Co., Ltd. (South Korea), JCR Pharmaceuticals Co., Ltd (Japan), Takeda Pharmaceutical Company (Japan), CORESTEM Inc (South Korea), Vericel Corporation (US), MiMedx (US), Organogenesis Inc. (US),

Medtronic plc (US), Orthocell Ltd. (Australia), Mesoblast Ltd (Australia), BioRestorative Therapies, Inc (US), Pluri Inc. (Israel), Tegoscience (South Korea), Anterogen Co., Ltd. (South Korea), bluebird bio, Inc. (US), APAC Biotech (India), Shenzhen SiBiono Genetech Co., Ltd. (China), Aspect Biosystems (Canada), and Athersys, Inc. (US).

FIGURE 34 REGENERATIVE MEDICINE MARKET: COMPANY EVALUATION MATRIX (2021)



Source: Expert Interviews and MarketsandMarkets Analysis

9.5.1 STARS

Players in this category receive high scores for most of the evaluation criteria. They have strong and established product portfolios, high market shares, wide application coverage, and a strong geographical footprint. These players provide mature, innovative, and reputable services. Novartis AG (Switzerland), Biogen (US), Sarepta Therapeutics, Inc. (US), Gilead Sciences, Inc. (US), and Amgen, Inc. (US) are the star players in this market.

9.5.2 EMERGING LEADERS

Emerging leaders have demonstrated substantial product innovations compared to their competitors (resulting in an evolved product portfolio) and significant market presence. However, they do not have strong growth strategies for geographical penetration and overall business growth. Companies such as Smith & Nephew plc (UK), Medipost Co., Ltd. (South Korea), JCR Pharmaceuticals Co., Ltd (Japan), Takeda Pharmaceutical Company (Japan), CORESTEM Inc (South Korea), Vericel Corporation (US), MIMEDX (US), Organogenesis Inc. (US), Medtronic plc (US) are recognized as emerging leaders in the regenerative medicine market.

9.5.3 PERVASIVE PLAYERS

The companies that fall in this category have a strong product reach across a particular market segment, assisting them in maintaining market position among established players or key competitors. Compared to other established vendors, they do not have strong business strategies spread throughout the market. These companies could be new entrants in the market and require more time to get traction. Stryker Corporation (US), Orthocell Ltd. (Australia), Mesoblast Ltd. (Australia), and BioRestorative Therapies, Inc (US) are pervasive players that fall under this category.

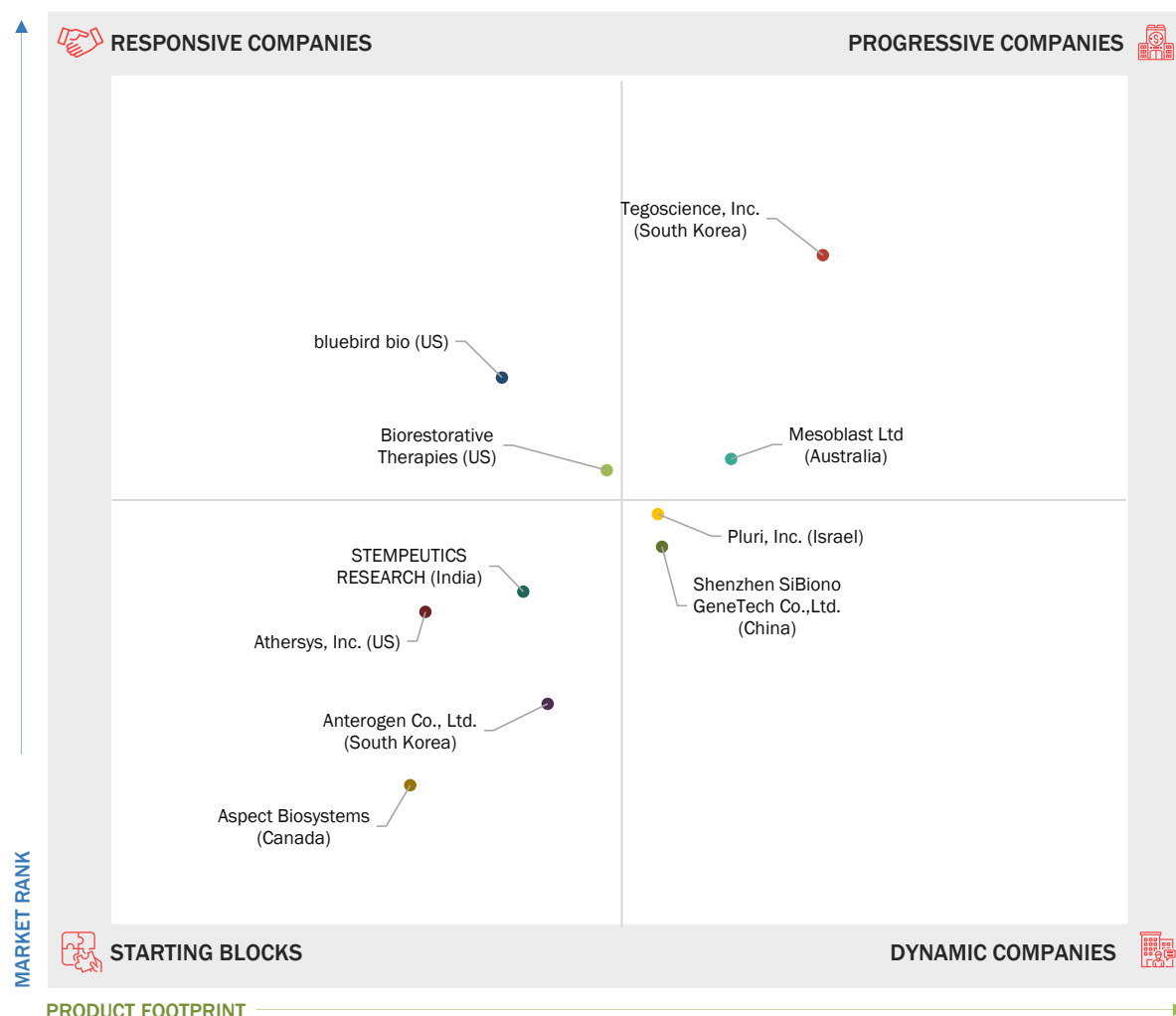
9.5.4 PARTICIPANTS

The companies that fall in this category have niche product offerings, owing to which they are likely to gain a significant position in the market. These companies could be emerging players, new entrants, or established players targeting new market segments. Pluri Inc. (Israel), Tegoscience Inc. (South Korea), Anterogen Co., Ltd. (South Korea), bluebird bio, Inc. (US), APAC Biotech (India), Shenzhen SiBiono Genetech Co., Ltd. (China), Aspect Biosystems (Canada), and Athersys, Inc. (US) are the participants in this market.

9.6 COMPANY EVALUATION QUADRANT: STARTUPS/SMES

The ten emerging companies are placed into four categories based on their performance in each criterion: “Progressive Companies,” “Responsive Companies,” “Starting Blocks,” and “Dynamic Companies.”

FIGURE 35 REGENERATIVE MEDICINE MARKET: COMPANY EVALUATION QUADRANT (2021)



Source: MarketsandMarkets Analysis

9.6.1 PROGRESSIVE COMPANIES

These companies perform exceedingly well in both service excellence and business excellence parameters. They have a robust partner ecosystem and well-developed marketing channels and have received extensive funding to develop their product portfolios. Tegoscience (South Korea) and Mesoblast Ltd (Australia) are progressive companies in this market.

9.6.2 STARTING BLOCKS

These companies with niche offerings are starting to gain their position in the market. They do not have a substantial business or product excellence compared to other established start-ups. However, they are always looking for an opportunity to consolidate their market position. They might be new entrants and require more time before gaining significant market traction. Anterogen Co., Ltd. (South Korea), APAC Biotech (India), Aspect Biosystems (Canada), and Athersys, Inc. (US) are the starting blocks in the regenerative medicine market.

9.6.3 RESPONSIVE COMPANIES

These companies are technology disruptors with highly innovative product offerings compared to their competitors. They focus on constantly developing product/service portfolios and bringing innovations to the market. However, they lag in the business excellence parameter compared to other start-ups. With appropriate marketing activities, suitable partnerships, and VC funding, these start-ups are known as responsive companies. BioRestorative Therapies, Inc (US) and bluebird bio, Inc. (US) are responsive companies in this market.

9.6.4 DYNAMIC COMPANIES

These companies have performed substantially well in the business excellence parameter. However, they score slightly low in product excellence compared to progressive and responsive companies. They generally focus on a specific technology related to the product or service offering. PLURI Inc. (Israel) and Shenzhen SiBiono Genetech Co., Ltd. (China) are the dynamic companies in this market.

9.7 COMPETITIVE BENCHMARKING

9.7.1 REGENERATIVE MEDICINE MARKET: LIST OF STARTUPS/SMES

TABLE 166 REGENERATIVE MEDICINE MARKET: LIST OF STARTUPS/SMES

COMPANY NAME	CATEGORY	OWNERSHIP STATUS	HQ LOCATION	YEAR FOUNDED	PRODUCTS	REVENUE
Tegoscience	Manufacturer	Private	Seoul, South Korea	2001	Cell Therapy Products	2-5 million
CORESTEM Inc.	Manufacturer	Private	Kyonggi-do, South Korea	2001	Stem Cell and Gene Therapy Products	3-4 billion
Anterogen Co., Ltd	Manufacturer	Private	Kyonggi-do, South Korea	2000	Stem Cell-Based Treatments	8-10 billion
bluebird bio, Inc.	Manufacturer	Private	Massachusetts, US	2010	Severe Genetic Diseases and Cancer	3-5 million
Spark Therapeutics	Manufacturer	Private	Philadelphia, US	2013	Genetic mutation in both copies of the RPE65 gene	2-3 billion
Kite Pharma (Gilead)	Manufacturer	Public	California, US	2009	CAR-T cell therapy	8-9 billion
APAC Biotech	Manufacturer	Private	Haryana, India	2009	Distribution of APCEDEN	5-7 million
Shenzhen SiBiono Genetech Co., Ltd.	Manufacturer	Private	Shenzhen, China	1998	Gene Therapy Products	25-30 million
Aspect Biosystems	Manufacturer	Private	Vancouver, Canada	2013	3D bioprinting of	10-15 million

						living human tissue	
Athersys, Inc.	Manufacturer	Private	Ohio, US	1995		Research And Development Activities In The Field Of Regenerative Medicine	5-10 million

9.7.2 REGENERATIVE MEDICINE MARKET: COMPETITIVE BENCHMARKING OF KEY PLAYERS (STARTUPS/SMES)

TABLE 167 REGENERATIVE MEDICINE MARKET: COMPETITIVE BENCHMARKING OF KEY PLAYERS (STARTUPS/SMES)

COMPANY NAME	PRODUCT			GEOGRAPHY				
	GENE THERAPY	CELL THERAPY	TISSUE ENGINEERING	NORTH AMERICA	EUROPE	ASIA PACIFIC	LATIN AMERICA	MIDDLE EAST & AFRICA
Tegoscience	N	Y	Y	N	N	Y	N	Y
CORESTEM Inc.	Y	Y	N	N	N	Y	N	N
Anterogen Co., Ltd	N	Y	N	N	N	Y	N	N
bluebird bio, Inc.	Y	N	N	Y	Y	N	Y	N
Spark Therapeutics	Y	N	N	Y	N	N	N	N
Kite Pharma (Gilead)	N	Y	N	Y	Y	Y	Y	Y
APAC Biotech	N	Y	N	N	N	Y	N	N
Shenzhen SiBiono Genetech Co., Ltd.	Y	N	N	N	N	Y	Y	Y
Aspect Biosystems	N	Y	Y	Y	N	N	N	N
Athersys, Inc.	N	N	Y	Y	Y	N	N	N

9.8 COMPETITIVE SCENARIO AND TRENDS

This section studies the growth strategies adopted by market players from January 2020 to November 2022. Several players in this market have adopted various strategies to expand their global presence and increase their market share. Collaborations, partnerships, and acquisitions are a few major strategies market players adopt to grow in the regenerative medicine market.

TABLE 168 REGENERATIVE MEDICINE MARKET: DEALS (2020–2022)

YEAR	APPROACH	COMPANY 1	COMPANY 2	DESCRIPTION
2022	Collaboration	Novartis AG	Alnylam	Novartis announces collaboration with Alnylam to explore targeted therapy to restore liver function
2020	Acquisition	Stryker Corporation	Wright Medical	Stryker acquired Wright Medical to strengthen its biologics products offerings
2020	Acquisition	AbbVie (US)	Allergan (Ireland)	AbbVie acquired Allergan to enhance its product portfolio in immunology, hematologic oncology, neuroscience, and Allergan aesthetics.
2021	Collaboration	AbbVie (US)	REGENXBIO Inc.	AbbVie and REGENXBIO Inc. (REGENXBIO) entered into a collaboration to develop and commercialize RGX-314, an investigational gene therapy for wet age-related macular degeneration, diabetic retinopathy, and other chronic retinal diseases.

Source: Company Websites and Secondary Sources

9.8.1 PRODUCT LAUNCHES AND REGULATORY APPROVALS (2020–2022)

TABLE 169 REGENERATIVE MEDICINE MARKET: PRODUCT LAUNCHES AND REGULATORY APPROVALS

YEAR	COMPANY	COUNTRY	DESCRIPTION
2022	MiMedx	US	MiMedx announced the launch of Axiofill, an Extracellular Matrix (ECM) Particulate product derived from human placental tissue. Axiofill is a human placental-derived particulate product available for surgical recovery procedures.
2020	Novartis AG	US	The US Food and Drug Administration (FDA) granted Regenerative Medicine Advanced Therapy (RMAT) designation to Kymriah for an investigational new indication to treat patients with relapsed or refractory (r/r) follicular lymphoma (FL)
2021	Organogenesis	US	Organogenesis ReNu received FDA Regenerative Medicine Advanced Therapy (RMAT) designation for osteoarthritis of the knee
2020	Organogenesis	US	The US Food and Drug Administration (FDA) granted Regenerative Medicine Advanced Therapy (RMAT) designation to Kymriah for an investigational new indication to treat patients with relapsed or refractory (r/r) follicular lymphoma (FL)
2022	MEDIPOST	US	MEDIPOST announced the beginning of phase 3 clinical trials of CARTISTEM, a treatment for knee osteoarthritis

Source: Company Websites and Secondary Sources

10 COMPANY PROFILES

10.1 KEY PLAYERS

10.1.1 NOVARTIS AG

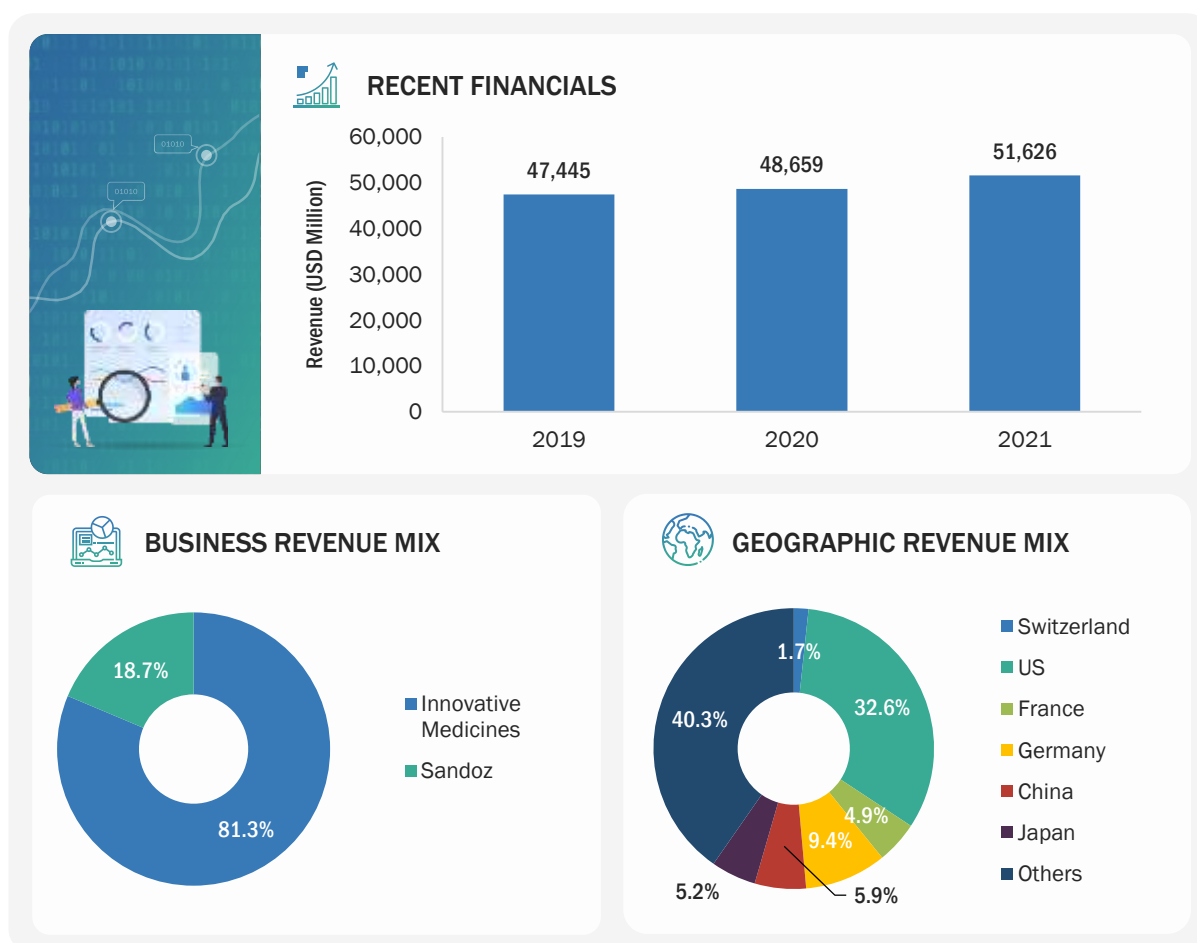
10.1.1.1 Business overview

Novartis AG focuses on discovering, developing, manufacturing, and marketing prescription, generic pharmaceutical, and eye care products. The company operates in two segments—Innovative Medicines and Sandoz (Generics). Novartis offers gene therapy products through its Innovative Medicines segment.

The company operates through a network of subsidiaries and offices in the Americas, Europe, the Middle East, Africa, and the Asia Pacific. Novartis has a presence in 155 countries worldwide. Some of its major subsidiaries include Novartis Argentina S.A., Novartis Pharma S.A. (Belgium), Novartis Biociências S.A. (Brazil), Novartis Pharmaceuticals Canada, Inc., Novartis Deutschland GmbH (Germany), Novartis India Limited, Novartis Holding Japan K.K., and Novartis Corporation (US).

TABLE 170 **NOVARTIS AG: BUSINESS OVERVIEW**

NOVARTIS	
Year Founded	1996
Country	Switzerland
City	Basel
Ownership	Public

FIGURE 36 NOVARTIS AG: COMPANY SNAPSHOT (2021)


Note: Information for the recent financials, business revenue mix, and geographic revenue mix was unavailable for the Innovative Medicine segment. Hence, this section has been provided for the overall company.

Source: Annual Reports and Company Website

10.1.1.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Cell Therapy	B-cell precursor acute lymphoblastic leukemia	Kymriah	Kymriah is a genetically modified autologous T cell immunotherapy indicated for treating patients up to 25 years of age with B-cell precursor acute lymphoblastic leukemia (ALL) that is refractory or in second or later relapse.
Gene Therapy	Spinal muscular atrophy	Zolgensma	Zolgensma is a one-time intravenous gene therapy designed to address the genetic root cause of spinal muscular atrophy (SMA) by replacing the function of the missing or nonworking SMN1 gene.

10.1.1.3 Recent developments

Regulatory approvals

MONTH & YEAR	DESCRIPTION
April 2020	The US Food and Drug Administration (FDA) granted Regenerative Medicine Advanced Therapy (RMAT) designation to Kymriah for an investigational new indication to treat patients with relapsed or refractory (r/r) follicular lymphoma (FL).

Source: Press Releases

Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
January 2022	Collaboration	Novartis AG (Switzerland)	Alnylam (US)	Novartis collaborated with Alnylam to explore targeted therapy to restore liver function

Source: Press Releases

10.1.1.4 MnM view

10.1.1.4.1 Key strengths

Novartis is one of the leading players in the regenerative medicine market. The company offers two gene therapy products globally, Kymriah and Zolgensma. The company has targeted its research activities toward product approvals & launches. As a result, KYMRIAH registered significant growth, with registered sales of USD 587 million in 2021, up from 24% in 2020. ZOLGENSMA registered sales of USD 1,351 million in 2021, up from 47% in 2020. Novartis has strong distribution channels in the US and generated 32.6% of its 2021 revenue from the US alone.

10.1.1.4.2 Strategic choices made

The company has focused on organic and inorganic strategies in this market, including collaborations and acquisitions. For instance, in 2021, the company underwent a definitive agreement to acquire the UK-based ocular gene therapy company, Gyroscope Therapeutics. This acquisition helped Novartis gain rights of GT005 to the Novartis portfolio, an investigational, one-time gene therapy currently in Phase 2 for treating people with geographic atrophy.

10.1.1.4.3 Weaknesses and competitive threats

While Novartis has a well-established product portfolio, new entrants in the market could create competition in the portfolio for gene and CAR-T therapy. The company may face increasing competition from different local companies. Moreover, the ongoing consolidation of pharmaceutical companies and the rise of major competitors such as Sarepta and Gilead, who have been adopting various inorganic growth strategies such as collaborations and acquisitions, are further expected to threaten the company's hold in the market.

10.1.2 BIOGEN, INC.

10.1.2.1 Business overview

Biogen is a global biopharmaceutical company involved in developing, manufacturing, and marketing therapies for neurological & neurodegenerative diseases. The company focuses on multiple sclerosis (MS), neuroimmunology, Alzheimer's, dementia, Parkinson's disease, and neuromuscular disorders. Biogen's gene therapy product, SPINRAZA, treats spinal muscular atrophy.

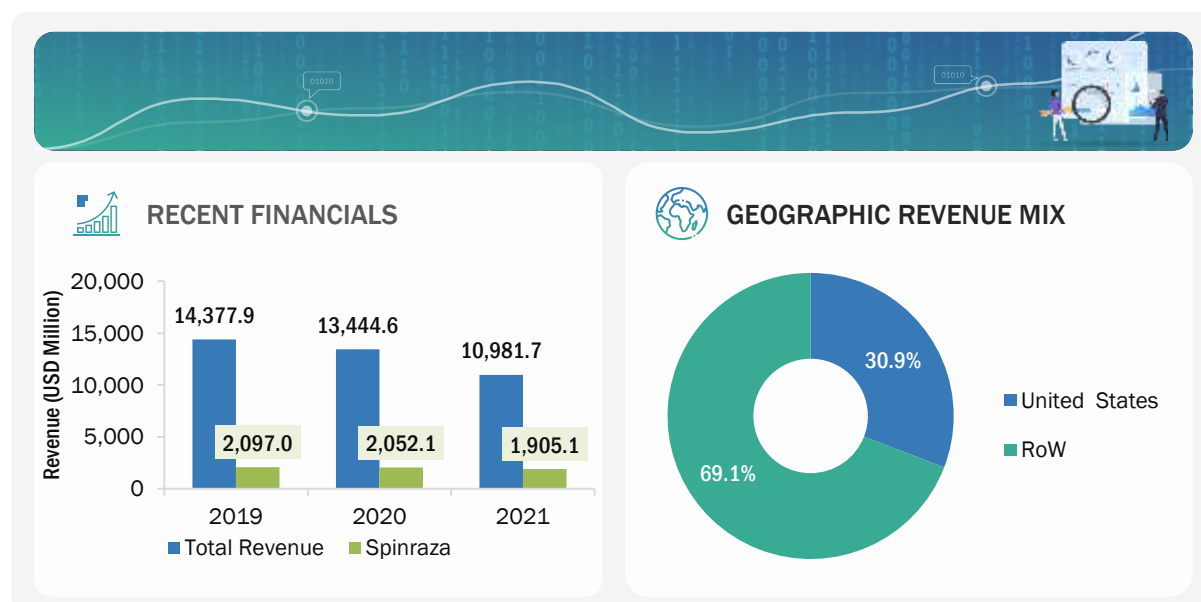
Some of the company's subsidiaries include Biogen Biotechnology (Shanghai) Co., Ltd. (China), Biogen Colombia S.A.S., Biogen France S.A.S., and Biogen Japan Ltd.

TABLE 171 BIOGEN: BUSINESS OVERVIEW

YEAR FOUNDED	1978
Country	US
City/State	Massachusetts
Ownership	Public
No. of Employees	~9,100
R&D Expenditure (2021)	USD 2.50 Billion

Source: Company Website and Annual Reports

FIGURE 37 BIOGEN: COMPANY SNAPSHOT (2021)



Note: Information for the business revenue mix was unavailable in the public domain. Hence, this section has not been provided here. Source: Company Website, Annual Reports, and SEC Filings

10.1.2.2 Products offered

CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Gene Therapy	Spinal muscular atrophy (SMA)	SPINRAZA	SPINRAZA is approved for the treatment of SMA in children, infants, and adults. This therapy is available in more than 50 countries.

Source: Company Website

10.1.2.3 Recent developments

Other developments

YEAR	DEVELOPMENT TYPE	DESCRIPTION
2021	Expansion	The company built a new gene therapy manufacturing facility at its Research Triangle Park (RTP) manufacturing campus in North Carolina. This development aims to support the firm's growing gene therapy pipeline across multiple therapeutic areas.

Source: Press Releases

10.1.2.4 MnM view

10.1.2.4.1 Key strengths

The company is projected to witness strong growth in gene therapy during the forecast period. It has also shown plans to ensure capacity expansion—in 2021; the company announced plans to build a new gene therapy manufacturing facility at its Research Triangle Park (RTP) manufacturing campuses in North Carolina.

10.1.2.4.2 Strategic choices made

The company aims to expand its facilities to gain a competitive edge. For instance, the company announced the expansion of a new therapy manufacturing facility in 2021. Moreover, Biogen also acquired Nightstar Therapeutics in 2019 for around USD 800 million. This development helped the firm establish its presence in the regenerative medicine market by incorporating Nightstar Therapeutics' product portfolio and capabilities.

10.1.2.4.3 Weaknesses and competitive threats

Biogen is a leading player in the regenerative medicine market. However, the revenue of SPINRAZA is declining yearly. The decline in revenue was primarily brought on by a decline in demand due to heightened competition in the US and Germany, as well as a pricing decline in the US and other markets. An upscale could partially offset this decline with sales volumes in Latin America and some distributor markets.

10.1.3 SAREPTA THERAPEUTICS, INC.

10.1.3.1 Business overview

Sarepta Therapeutics is a commercial-stage biopharmaceutical company focused on discovering, developing, and marketing unique RNA-targeted therapeutics to treat rare diseases. The company offers two products—EXONDYS 51, VYONDYS 53, and AMONDYS 45—in the regenerative medicine market. These products are used for the treatment of patients with Duchenne Muscular Dystrophy.

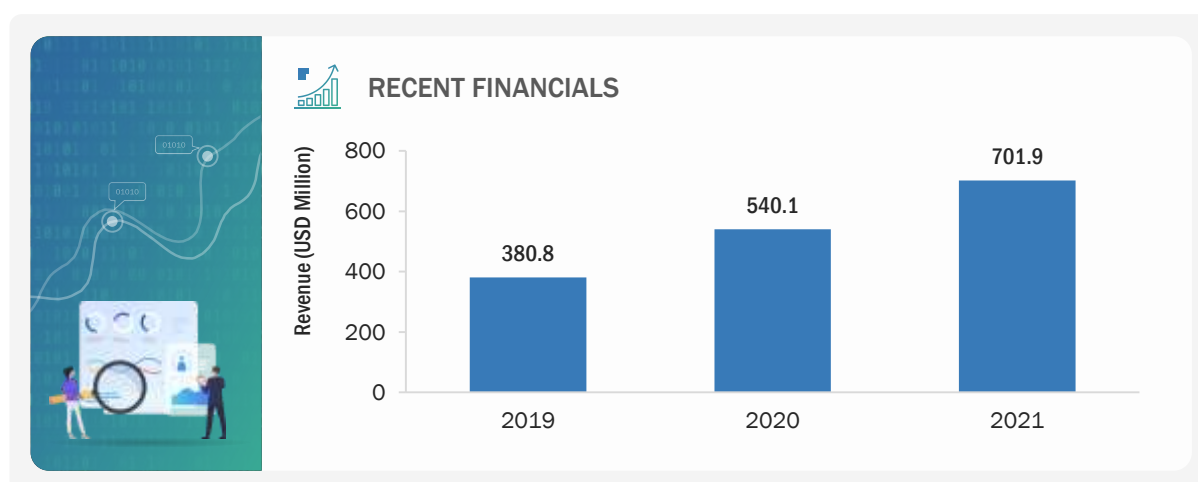
Currently, the company has many pipeline candidates focusing on Duchenne Muscular Dystrophy, Limb-Girdle Muscular Dystrophies, Charcot-Marie-Tooth Neuropathies, and Pompe disease. The company majorly operates its business channels in the US. Some of its subsidiaries are Sarepta Securities Corp (US), ST International Holdings, Inc. (US), and Sarepta International Holdings GmbH (Germany).

TABLE 172 SAREPTA THERAPEUTICS: BUSINESS OVERVIEW

Year Founded	1980
Country	US
City/State	Massachusetts
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 38 SAREPTA THERAPEUTICS: COMPANY SNAPSHOT (2021)



Note: Information for the business & geographic revenue mix of the company was unavailable in the public domain. Hence, these sections are not provided here.

Source: Company Website, Annual Reports, and SEC Filings

10.1.3.2 Products offered

CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Gene Therapy	Duchenne muscular dystrophy (DMD)	- EXONDYS 51 - VYONDYS 53 - AMONDYS 45	AMONDYS 45, VYONDYS 53, and EXONDYS 51 are all gene therapy drugs used to treat DMD patients with a confirmed dystrophin gene mutation. It can be treated by skipping exon 45, exon 53, and exon 51, respectively.

Source: Company Website

10.1.3.3 Recent developments

Product approvals

YEAR	SEGMENT	DESCRIPTION
2021	Gene Therapy	The company received the US FDA approval for AMONDYS 45; Amondys 45 (casimersen) is an antisense oligonucleotide indicated for treating Duchenne Muscular Dystrophy (DMD) in patients.

Source: Press Releases

Deals

YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
2022	Collaboration	Sarepta Therapeutics (US)	GenEdit Inc (US)	Sarepta collaborated with GenEdit to use GenEdit's NanoGalaxy platform and Sarepta's gene-editing technology to develop gene-editing therapeutics to treat neuromuscular diseases.	NA
2020	Licensing	Sarepta Therapeutics (US)	Hansa Biopharma (Sweden)	Sarepta received a global license to create and market IMLIFIDASE as a pre-treatment to gene therapy administration in patients with Duchenne muscular dystrophy (DMD) and Limb-girdle muscular dystrophy (LGMD).	NA
2020	Collaboration	Sarepta Therapeutics (US)	Codiak BioSciences Inc. (US)	The companies collaborated for R&D in exosome-based therapeutics for rare diseases.	USD 72.5 Million
2020	Agreement	Sarepta Therapeutics (US)	Selecta Biosciences, Inc. (US)	Selecta granted rights to Sarepta for developing and commercializing Selecta's immune tolerance platform, ImmTOR, for use in DMD and certain LGMDs.	NA

2020	Agreement	Sarepta Therapeutics (US)	Dyno Therapeutics, Inc. (US)	The companies signed an agreement to develop next-generation adeno-associated virus (AAV) vectors for muscle diseases using Dyno's CapsidMap platform.	USD 40 Million
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Source: Press Releases

Other developments

YEAR	DEVELOPMENT TYPE	DESCRIPTION
2022	Expansion	The company opened a new research facility, Genetic Therapies Center of Excellence (GTCOE), in Ohio, US. The 85,000-square-foot state-of-the-art facility expands Sarepta's research and development capabilities and footprint. The Center is focused on the discovery, pre-clinical, and clinical development of genetic medicines, which include RNA, gene therapy, and gene editing programs.

Source: Press Releases

10.1.3.4 MnM view

10.1.3.4.1 Key strengths

The company offers EXONDYS, VYONDYS, AND AMONDYS (FDA approved in 2021) for treating DMD. Sarepta also has a strong focus on R&D—in 2021, the company invested USD 0.77 billion in R&D. Three approved gene therapy products and a strong focus on R&D will propel the company's growth in the coming years.

10.1.3.4.2 Strategic choices made

The company has focused on expansions, agreements, partnerships, and collaborations for market growth. It opened a new research facility, the Genetic Therapies Center of Excellence (GTCOE), in the US in 2022.

10.1.3.4.3 Weaknesses and competitive threats

The company needs to focus on indications other than DMD, which accounts for 9.2% of the overall regenerative medicine market. There are various opportunities in other applications, such as neurological diseases, cancer, and hematological diseases, which could boost the firm's growth instead of making it succumb to competition.

10.1.4 GILEAD SCIENCES, INC.

10.1.4.1 Business overview

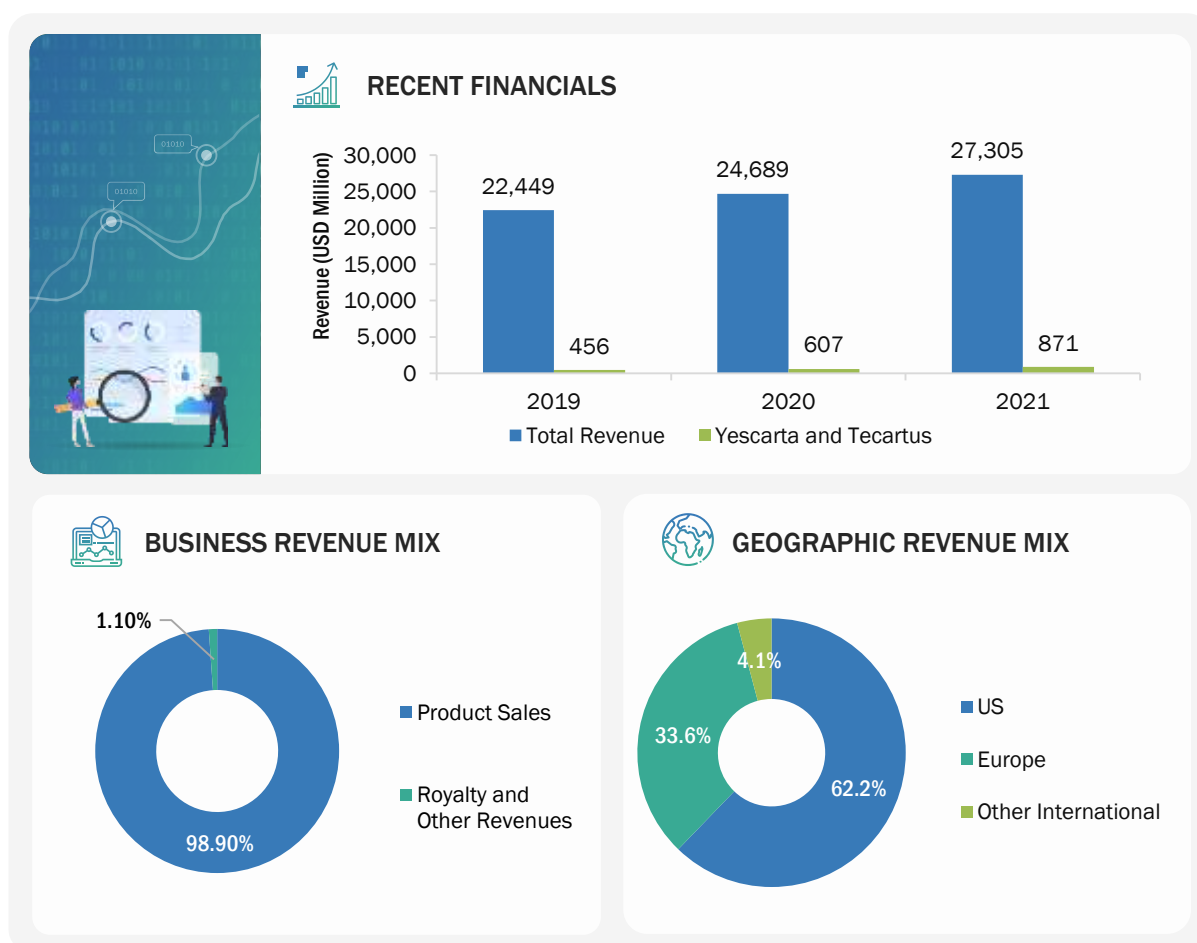
Gilead Sciences is a research-based biopharmaceutical company involved in developing, manufacturing, and marketing innovative medicines to fulfill unmet medical needs. The company operates through two segments: Product Sales and Royalty & Other Revenues. It offers gene therapy products through its Product Sales segment.

In 2017, the company acquired Kite Pharma to expand its gene therapy portfolio. Currently, Gilead offers its gene therapy product—YESCARTA and TECARTUS—through Kite Pharma. Globally, the company operates its business channels in more than 35 countries. Some of its subsidiaries include Gilead Biopharmaceutics Ireland Corporation, Gilead Holdings, LLC (UK), Gilead Sciences Holding LLC (US), Gilead Sciences GmbH (Germany), and Gilead Sciences, S.L. (Spain).

TABLE 173 GILEAD SCIENCES INC.: BUSINESS OVERVIEW

Year Founded	1987
Country	US
City/State	California
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 39 GILEAD SCIENCES, INC: COMPANY SNAPSHOT (2021)


Note: Information for the recent financials, business revenue mix, and geographic revenue mix was unavailable for the Product Sales segment. Hence, these sections are provided for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

10.1.4.2 Products offered

CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Gene Therapy	Non-Hodgkin Lymphoma	YESCARTA	YESCARTA is used for the treatment of patients with non-Hodgkin lymphoma. It is a prescription medicine that uses a patient's white blood cells by modifying recognition and restricting attacks on the patient's lymphoma cells.
	Mantle Cell Lymphoma		
Gene Therapy	Acute Lymphoblastic Leukemia	TECARTUS	This gene therapy treats adults suffering from mantle cell lymphoma or acute lymphoblastic leukemia using a patient's white blood cells.

Source: Company Website

10.1.4.3 Recent developments

Product approvals

YEAR	SEGMENT	DESCRIPTION
2020	Gene Therapy	The company received the US FDA approval for TECARTUS.
2017	Gene therapy	The company received the US FDA approval for YESCARTA.

Source: Press Releases

Deals

YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
2018	Collaboration	Gilead Sciences	Pfizer, Inc. (US)	Both companies collaborated to evaluate the safety and efficacy of the investigational combination of YESCARTA and Pfizer's Utomilumab.	-

Source: Press Releases

Other developments

YEAR	DEVELOPMENT TYPE	DESCRIPTION
2019	Expansion	The company opened its new production facility in Maryland, US. Through this facility, the company is expected to manufacture a variety of chimeric antigen receptor T (CAR T) therapies, including YESCARTA.

Source: Press Releases

10.1.4.4 MnM view

10.1.4.4.1 Key strengths

Gilead offers YESCARTA, a gene therapy product for treating non-Hodgkin's lymphoma, and TECARTUS for mantle cell lymphoma or acute lymphoblastic leukemia. These products generated sales revenue of USD 871 million in 2021. The company is also focusing on R&D to introduce innovative products. For instance, the company spent USD 5.36 billion on R&D activities in 2021.

10.1.4.4.2 Strategic choices made

Gilead focuses on acquisitions to strengthen its position in the regenerative medicine market. For instance, the company acquired Kite Pharma (US) and Cell Design Labs, Inc. (US) in 2017 to expand its gene therapy portfolio in the regenerative medicine market.

10.1.4.4.3 Weaknesses and competitive threats

While Gilead has performed well in the regenerative medicine market, it may face threats from emerging entrants. Rapid product approvals stemming from a robust product pipeline of other competitors may affect the company's position in the market.

10.1.5 AMGEN, INC.

10.1.5.1 Business overview

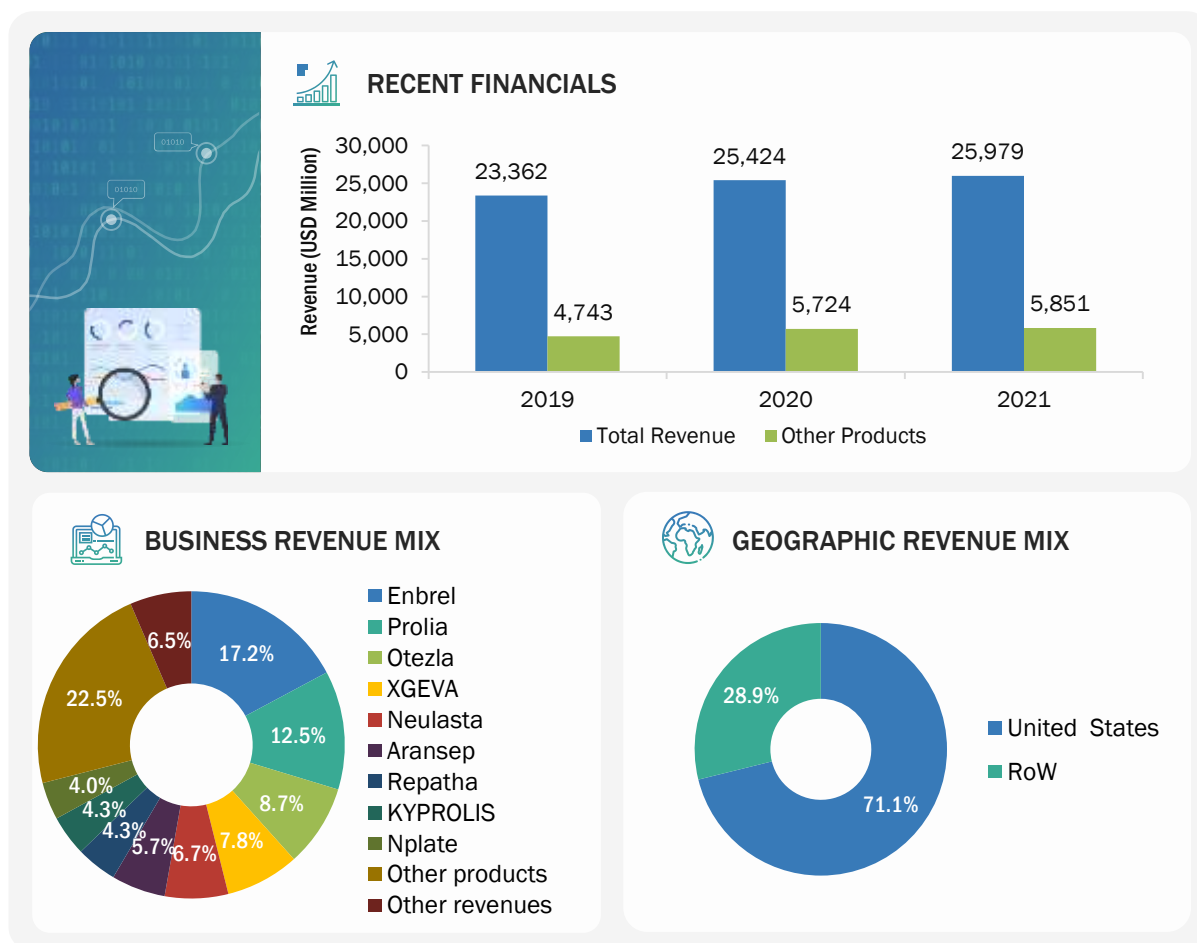
Amgen is a biotechnology medicines company that discovers, develops, manufactures, and markets medicines. The company focuses solely on human therapeutics and innovates in cellular and molecular biology. The company operates through two business segments: Product Sales and Other Revenues. The company's gene therapy product—IMLYGIC—is offered through its Other Products segment (a part of the Product Sales Division).

Amgen primarily operates in the US and Europe. However, the company focuses on enhancing its presence in Latin America and the Middle East. Currently, the company operates its business channels in over 100 countries worldwide. Amgen has three large wholesalers, namely, AmerisourceBergen Corporation (US), McKesson Corporation (US), and Cardinal Health (US).

TABLE 174 AMGEN INC: BUSINESS OVERVIEW

Year Founded	1980
Country	US
City/State	California
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 40 AMGEN, INC: COMPANY SNAPSHOT (2021)


Note: Information for the recent financials, business revenue mix, and geographic revenue mix sections are provided for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

10.1.5.2 Products offered

CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Gene Therapy	Skin cancer/lesions	IMLYGIC	IMLYGIC is a genetically modified oncolytic viral therapy used to treat melanoma patients.

Source: Company Website

10.1.5.3 Recent developments

Deals

YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
2021	Acquisition	Amgen (US)	Five Prime Therapeutics (US)	Amgen acquired Five Prime Therapeutics. The acquisition will improve Amgen's oncology portfolio, focusing on immuno-oncology and targeted cancer therapies.	USD 1.9 Billion

Source: Press Releases

10.1.5.4 MnM view

10.1.5.4.1 Key strengths

Amgen's gene therapy product—IMLYGIC—treats unresectable cutaneous, subcutaneous, and nodal lesions in patients with recurrent melanoma after the initial surgery. The company has marketed IMLYGIC in the US and 28 other European countries since the FDA and EC approved it in 2015. To introduce innovative products into the market, the company focuses on R&D activities. For instance, the company spent USD 4.82 billion on R&D in 2021.

10.1.5.4.2 Strategic choices made

The company is expected to show strong growth in the regenerative medicine market. It has also announced plans to expand its capacity for growth—for instance, in 2021, the company announced a new gene therapy manufacturing facility at its Research Triangle Park (RTP) manufacturing campuses in North Carolina. The company also acquired Five Prime Therapeutics for approximately USD 1.9 billion. This acquisition will improve Amgen's oncology portfolio, focusing on immuno-oncology and targeted cancer therapies.

10.1.5.4.3 Weaknesses and competitive threats

The company has a strong presence in developed markets like the US. As a result, in 2021, the company generated 71.1% of its total revenue from the US alone. The company can focus on improving its presence outside the US in high-growth markets such as Canada, Japan, and China.

10.1.6 SMITH & NEPHEW PLC

10.1.6.1 Business overview

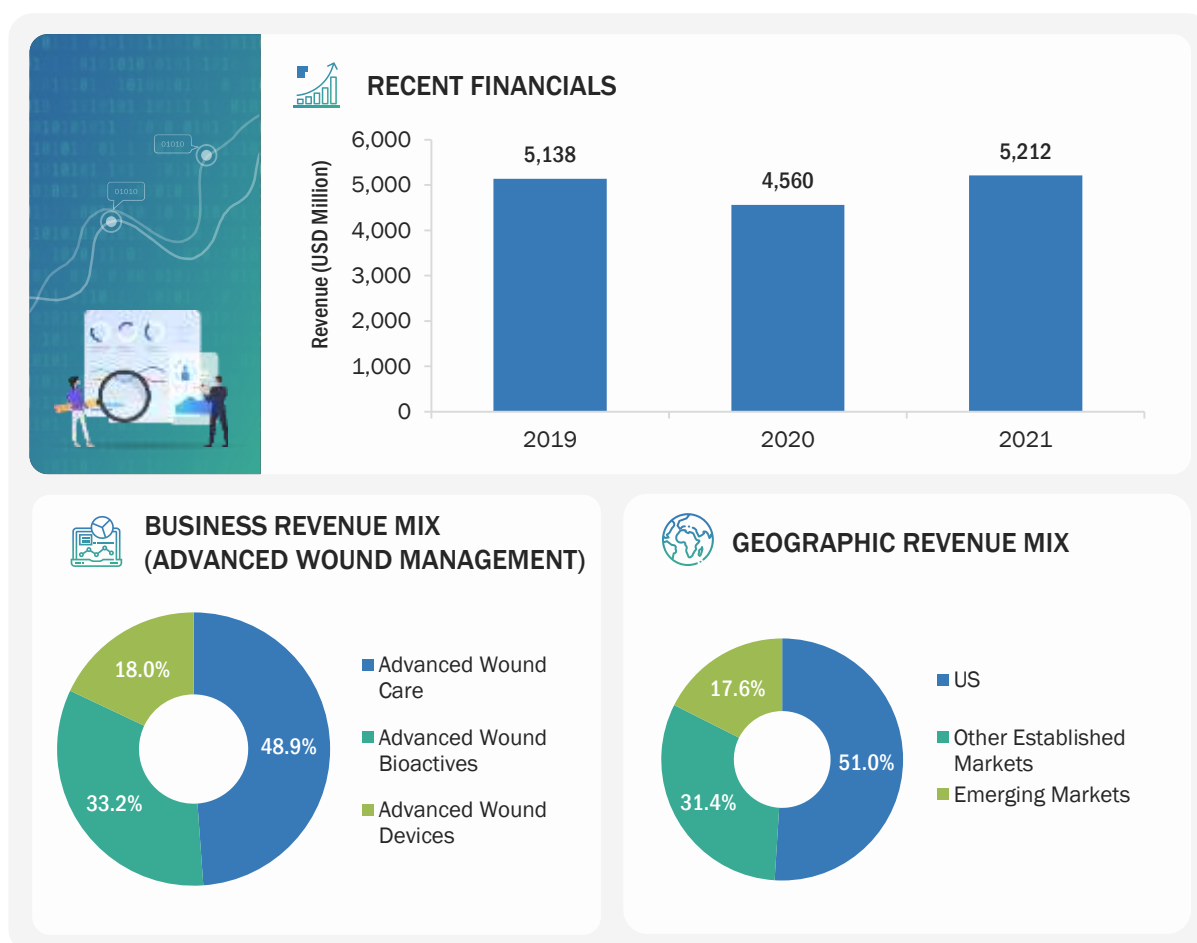
Smith & Nephew plc is a diversified company that develops and manufactures medical devices for advanced wound care management, orthopedics, and endoscopy. The company operates through three business segments—Sports Medicine, Orthopedics & Trauma, and Advanced Wound Management. The company offers regenerative medicine products through the Advanced Wound Management business segment.

Smith & Nephew plc acquired Osiris Therapeutics to strengthen its advanced wound care bioactive segment in 2019. Osiris researches, develops, manufactures, markets, and distributes regenerative medicine products in the US. Its products include Grafix and Stravix for treating chronic wounds in diabetic foot ulcers, venous leg ulcers, pressure ulcers, arterial ulcers, severe burns, and surgical and trauma wounds.

TABLE 175 SMITH & NEPHEW PLC: BUSINESS OVERVIEW

Year Founded	1856
Country	UK
City	London
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 41 SMITH & NEPHEW PLC: COMPANY SNAPSHOT (2021)


Note: Information for the recent financials and geographic revenue mix was unavailable for the Advanced Wound Management segment. Hence, these sections are provided for the overall company.

Source: Annual Reports and Company Website

10.1.6.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Tissue-engineered Products	Treatment of acute and chronic wounds	- Stravix - Grafix - Grafix PL - Grafix PL PRIME - Cartiform - BIO4 Viable Bone Matrix	This category provides products designed to manage exudate and infection, protect the skin, and prevent pressure injuries.

Note: The company reported no significant developments in the regenerative medicine market between January 2020 and October 2022.

10.1.7 MEDIPOST CO., LTD.

10.1.7.1 Business overview

MEDIPOST Co., Ltd. is a leading biotechnology company that develops stem cell technology-based products. The company operates through four business segments: Cord Blood Banking, Stem Cell Therapy, Nutritional Supplements, and Others (including the marketing of cosmetic products and healthcare services). MEDIPOST offers allogeneic stem cell products through its Stem Cell Therapy business segment.

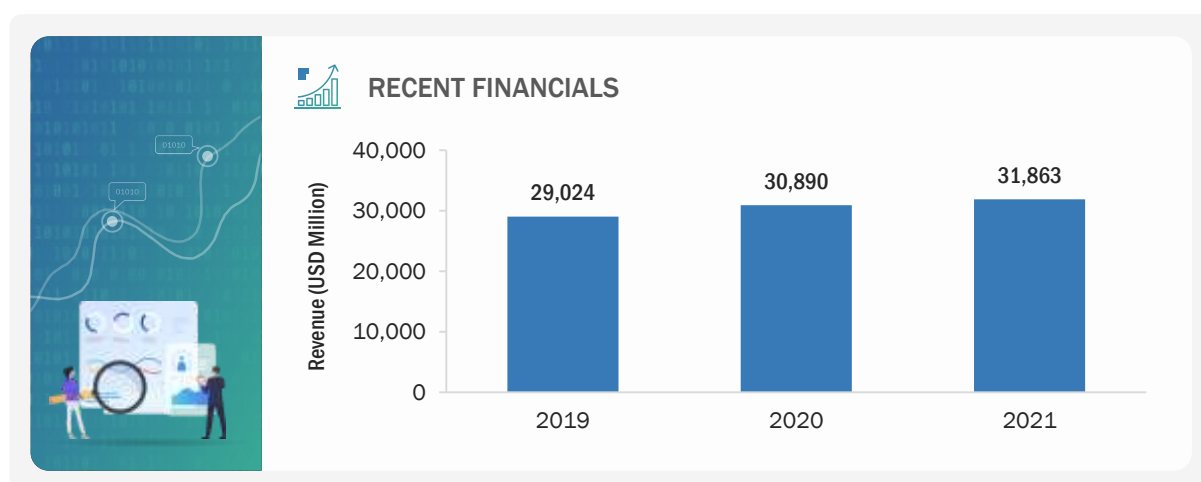
The company operates in major healthcare markets worldwide and has a strong presence in the Americas, Asia Pacific, Europe, and the Middle East. It has research and development facilities located in the US and South Korea. The company operates its global business through MEDIPOST America (US) and MEDIPOST HK (Hong Kong).

TABLE 176 MEDIPOST: BUSINESS OVERVIEW

Year Founded	2000
Country	South Korea
City	Gyeonggi-do
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 42 MEDIPOST: COMPANY SNAPSHOT (2021)



Note: Information for the business & geographic revenue mix sections was unavailable in the public domain. Hence, these sections are not provided here.

Source: Annual Reports, SEC Filings, and Company Website

10.1.7.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Mesenchymal Stem Cells	Knee Osteoarthritis (OA) and Traumatic Cartilage Degeneration	CARTISTEM	These allogeneic human umbilical cord blood-derived mesenchymal stem cells treat knee OA and traumatic cartilage degeneration.

Note: The company reported no significant developments in the regenerative medicine market between January 2020 and October 2022.

10.1.8 JCR PHARMACEUTICALS CO., LTD.

10.1.8.1 Business overview

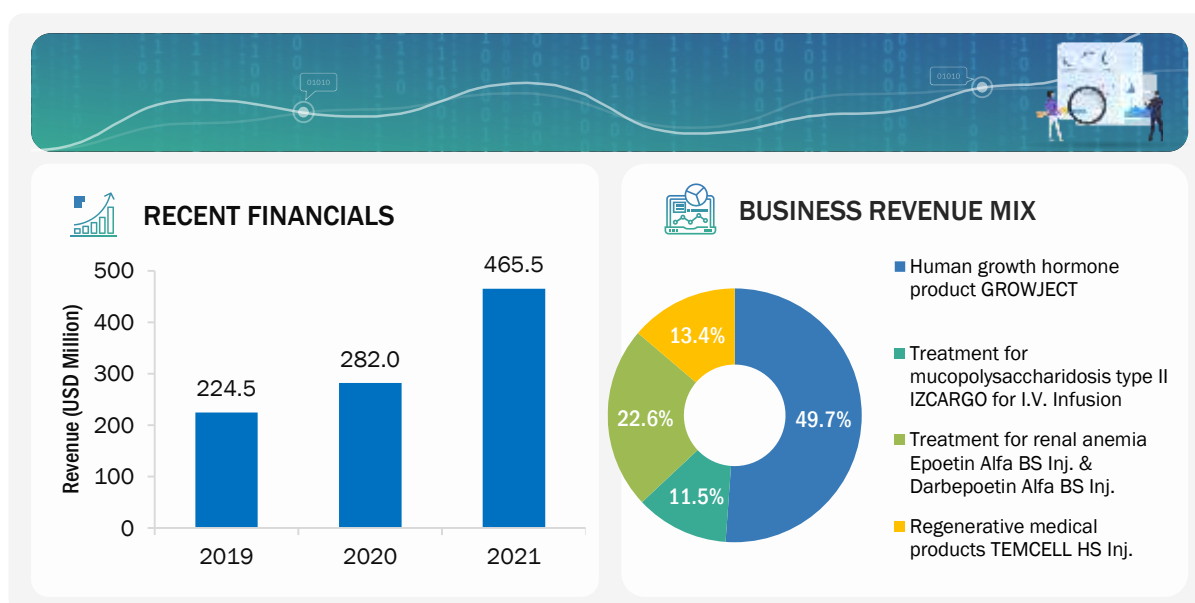
JCR Pharmaceuticals Co., Ltd. is a Japan-based pharmaceutical company that develops, manufactures, and markets biopharmaceuticals, regenerative medicines, APIs, and raw materials for drug manufacturing. The company operates through two segments: Pharmaceuticals and Medical Devices & Laboratory Equipment. The company offers TEMCELL HS Inj- a stem cell therapy, through its Pharmaceuticals division.

JCR Pharmaceuticals has a strong presence in the Asia Pacific (especially Japan and China). The company's subsidiaries include He Bei Jie Xi Bio-products Co., Ltd. (China), Chromatech Co., Ltd. (Japan), JCR Engineering Co., Ltd. (Japan), Family Health Rental Co., Ltd. (Japan), and JCR International SA (Switzerland).

TABLE 177 JCR PHARMACEUTICALS CO., LTD.: BUSINESS OVERVIEW

Year Founded	1975
Country	Japan
City	Hyogo
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 43 JCR PHARMACEUTICALS CO., LTD: COMPANY SNAPSHOT (2021)

Note 1: The company operates in only one business segment (Pharmaceuticals). Hence, the business revenue mix section includes marketed products under this segment in the snapshot.

Note 2: Information for the geographic revenue mix was unavailable in the public domain. Hence, this section has not been provided.

Source: Company Website and Annual Reports

10.1.8.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Bone marrow-derived mesenchymal stem cells	Acute graft versus host disease (aGVHD).	TEMCELL HS Inj.	This stem cell therapy product is developed to treat children and adults in Japan

Source: Company Website

Note: JCR Pharmaceuticals Co., Ltd., reported no recent developments in the regenerative medicine market from January 2020 to October 2022.

10.1.9 TAKEDA PHARMACEUTICAL COMPANY LIMITED

10.1.9.1 Business overview

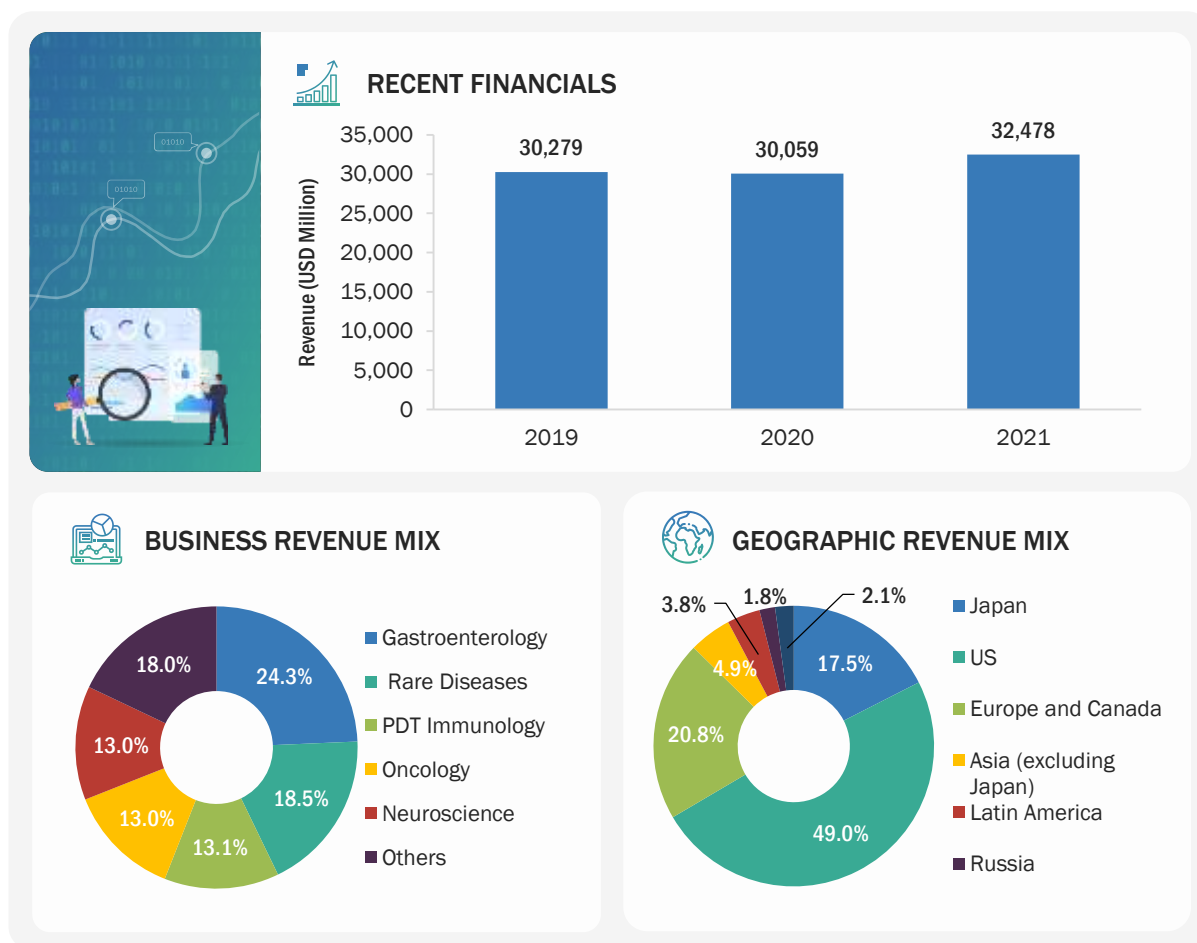
Takeda is a patient-focused and R&D-driven biopharmaceutical company. The company operates under two business divisions: Pharmaceutical Products and Royalty and Services income.

Takeda focuses on its five key segments in the Pharmaceutical Products division: Gastroenterology ("GI"), Rare Diseases, Plasma-Derived Therapies ("PDT"), Immunology, Oncology, and Neuroscience. The products researched, developed, manufactured, and marketed by the company are used to treat cancers, gastrointestinal disorders, and central nervous system diseases. The company has an extensive presence in 80 countries, with a leading position in Japan.

TABLE 178 TAKEDA PHARMACEUTICAL COMPANY: BUSINESS OVERVIEW

Year Founded	1781
Country	Japan
City	Tokyo
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 44 TAKEDA PHARMACEUTICAL COMPANY LIMITED: COMPANY SNAPSHOT (2021)


Note: Information for the recent financials, business revenue mix, and geographic revenue mix was unavailable for the Pharmaceutical Products segment. Hence, these sections are provided for the overall company.

Source: Annual Reports and Company Website

10.1.9.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT	DESCRIPTION
Cell Therapy	Treatment of complex perianal fistulae	Alofisel	Alofisel is indicated for treating complex perianal fistulae in adult patients with nonactive/mildly active luminal Crohn's disease when fistulae have shown an inadequate response to at least one conventional or biologic therapy.

Note: The company reported no significant developments in the regenerative medicine market between January 2020 and October 2022.

10.1.10 CORESTEM, INC.

10.1.10.1 Business overview

Founded in 2003 and headquartered in South Korea, CORESTEM is engaged in developing and distributing stem cell & gene therapy products. These products are developed for applications in neurodegenerative disorders and autoimmune diseases. The company has a commercial line for NEURONATA-R, a stem cell therapy used in treating Lou Gehrig's disease (a rare and incurable disease). The company has a few products in the pipeline for treating diseases such as lupus, multiple system atrophy, and osteoarthritis. The firm's geographic presence is only concentrated in Korea.

Note: The company is publicly held; however, the financial data of the company is unavailable in the public domain.

TABLE 179 CORESTEM, INC.: BUSINESS OVERVIEW

Year Founded	2003
Country	South Korea
City	Gyeonggi-do
Ownership	Public

Source: Company Website and Annual Reports

10.1.10.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT	PRODUCT TYPE
Autologous bone marrow mesenchymal stem cell	ALS (Amyotrophic Lateral Sclerosis)	ALS (NeuroNata-R) Injection	NeuroNata-R (lenzestemcel) is a mesenchymal stem cell therapy that has received conditional approval in South Korea to treat amyotrophic lateral sclerosis (ALS).

Source: Company Website

10.1.10.3 Recent developments

TABLE 180 CORESTEM, INC.: REGULATORY APPROVALS

MONTH & YEAR	PRODUCT	COUNTRY	DESCRIPTION
July 2022	NEURONATA-R	South Korea	The company is enrolling participants for the Phase 3 clinical trial of NeuroNata-R. This therapy has received conditional approval for treating ALS patients in South Korea.

Source: Company Website and Press Releases

10.1.11 VERICEL CORPORATION

10.1.11.1 Business overview

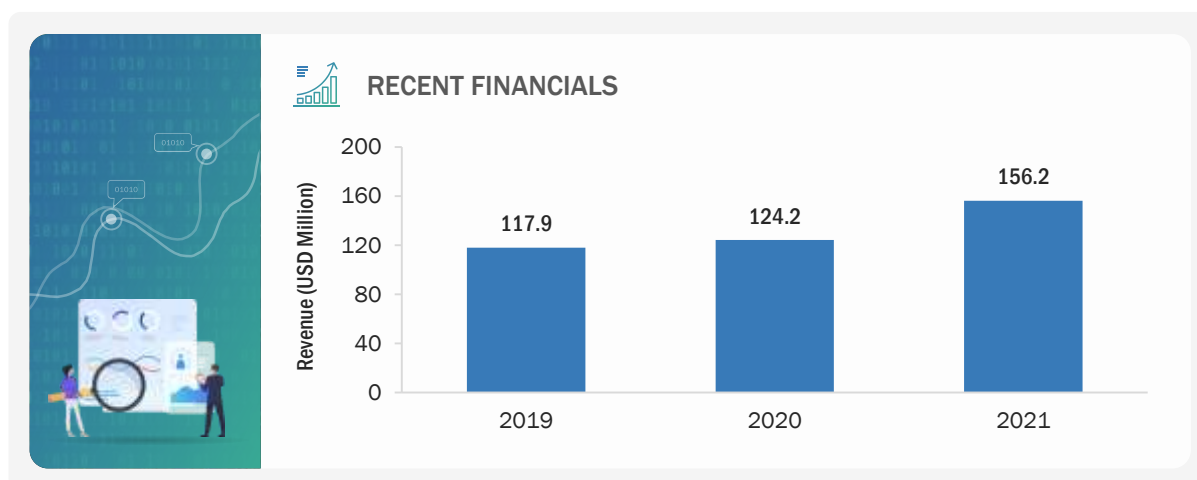
Vericel Corporation develops, manufactures, and markets autologous cell-based therapies for patients with serious diseases and conditions. The company develops patient-specific expanded cell therapies to treat patients with severe wounds and burn injuries. Vericel markets autologous cell therapy products such as MACI (an autologous chondrocyte implant for the treatment of cartilage defects in the knee) and Epicel (cultured epidermal autografts for the treatment of patients with deep dermal or full-thickness burns, which is approximately 30% of the total body surface area) in the US.

TABLE 181 VERICEL CORPORATION: BUSINESS OVERVIEW

Year Founded	1989
Country	US
City	Cambridge
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 45 VERICEL CORPORATION: COMPANY SNAPSHOT (2021)



Note: The company operates in one business segment relevant to regenerative medicine. Information for the business & geographic revenue mix was unavailable in the public domain. Hence, it is not provided here.

Source: Annual Reports, SEC Filings, and Company Website

10.1.11.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCTS	DESCRIPTION
Cell therapy	Cartilage, degenerative joint disease, osteoarthritis	MACI	Maci is a third-generation autologous cellularized scaffold product indicated for the repair of symptomatic, single, or multiple full-thickness cartilage defects of the knee with or without bone involvement in adults.
Tissue-engineered Products	Deep-dermal or full-thickness burns	Epicel	Epicel is a permanent skin replacement for adult and pediatric patients with deep-dermal or full-thickness burns comprising greater than or equal to 30 percent of TBSA.

Note: The company reported no significant developments in the regenerative medicine market between January 2020 and October 2022.

10.1.12 MIMEDX

10.1.12.1 Business overview

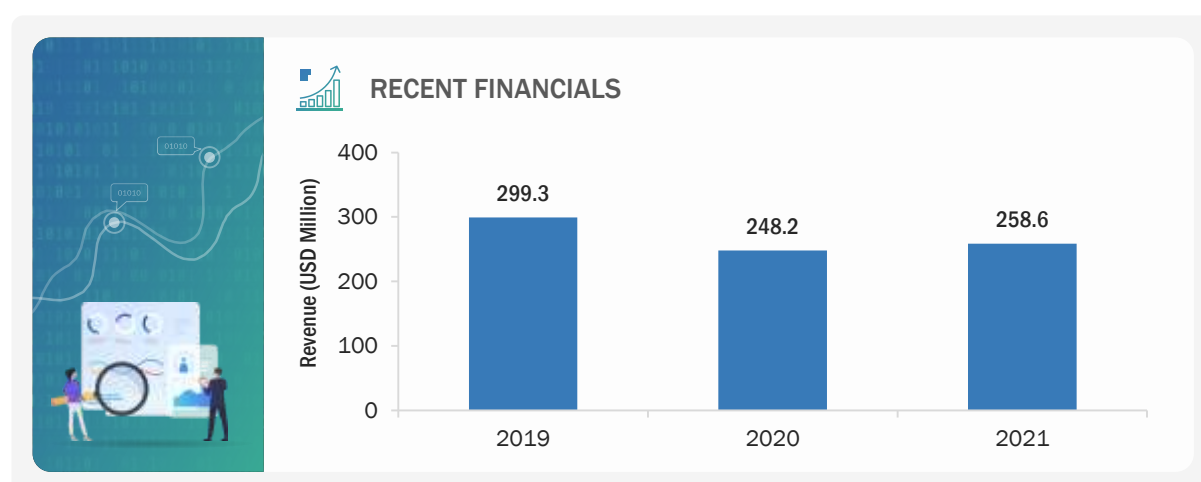
MIMEDX develops and markets regenerative biologics utilizing human placental tissue allografts with patent-protected processes for various healthcare sectors. The company operates through the Regenerative Biomaterials segment. It designs, manufactures, and markets tissue processing services for the wound care, surgical, sports medicine, ophthalmic, and dental markets.

TABLE 182 MIMEDX: BUSINESS OVERVIEW

Year Founded	2008
Country	US
City	Marietta
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 46 MIMEDX: COMPANY SNAPSHOT (2021)



Note: Information for the business & geographic revenue mix of the company was unavailable in the public domain. Hence, it has not been provided here.

Source: Annual Reports, SEC Filings, and Company Website

10.1.12.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Tissue-engineered products	Placenta-based allograft products	▪ Epifix ▪ Epicord ▪ Amniocord ▪ Amniofill ▪ Epiburn	This product category includes traditional dressings such as bandages, gauze, and ointments, used to treat non-severe or non-chronic wounds. The product category also includes advanced wound care products such as medical devices, advanced dressings, xenografts, biological products, and hct/ps, used as skin substitutes to treat severe or chronic wounds.

10.1.12.3 Recent developments

Product launches

MONTH & YEAR	COUNTRY	PRODUCT NAME	PRODUCT
September 2022	us	Axiofill	MIMEDX launched Axiofill, an Extracellular Matrix (ECM) Particulate product derived from human placental tissue. Axiofill is a human placental-derived particulate product available for surgical recovery procedures.

Source: Press Releases

10.1.13 ORGANOGENESIS, INC.

10.1.13.1 Business overview

Organogenesis is a regenerative medicine company focused on developing, manufacturing, and commercializing product solutions for advanced wound care, surgical, and sports medicine markets. The company offers products associated with soft tissue regeneration and bioactive wound healing through the advanced wound care business segment. It aims at providing patients with access to regenerative medicine and enhanced wound management. The company operates in the US and Europe.

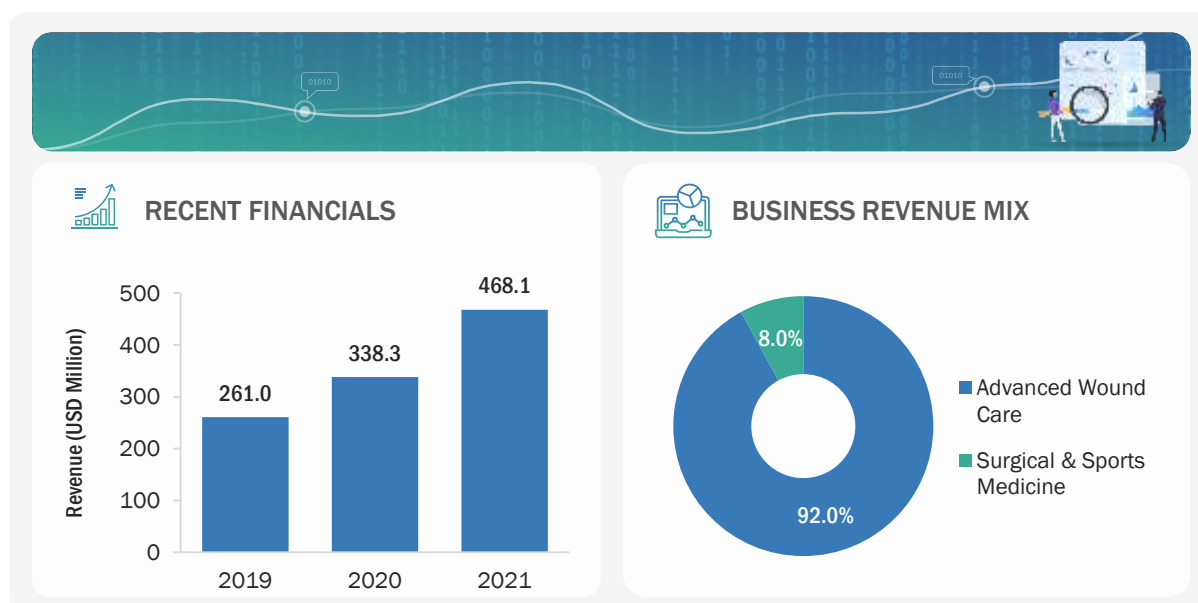
The company developed the PuraPly Antimicrobial collagen matrix, a key innovation in the wound care industry. This matrix utilizes a strong, biocompatible fenestrated sheet of porcine-derived collagen coated with polyhexamethylene biguanide (PHMB). It addresses acute and chronic wound types, including pressure ulcers, trauma wounds, surgical wounds, and diabetic ulcers.

TABLE 183 ORGANOGENESIS INC: BUSINESS OVERVIEW

Year Founded	1985
Country	US
City	Canton
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 47 ORGANOGENESIS: COMPANY SNAPSHOT (2021)



Note: Information for the geographic revenue mix was unavailable in the public domain. Hence, this section has not been provided for the company.

Source: Annual Reports, SEC Filings, and Company Website

10.1.13.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCTS	DESCRIPTION
Tissue-engineered products	Management of acute and chronic wounds	- Affinity - PuraPly AM - PuraPly - NuShield - Dermagraft - FiberOS- Osteoconductive Matrix - OCMP- Osteoconductive Matrix Plus - Re-Nu Suspension Allograft - NuCel Amniotic Suspension Allograft	This product largely focuses on utilizing the body's ability to heal itself and creating a hydrated microenvironment for cells to thrive.

10.1.13.3 Recent developments

REGULATORY APPROVALS

MONTH & YEAR	DESCRIPTION
January 2021	Organogenesis ReNu received FDA Regenerative Medicine Advanced Therapy (RMAT) designation for knee osteoarthritis.
April 2020	The US Food and Drug Administration (FDA) granted Regenerative Medicine Advanced Therapy (RMAT) designation to Kymriah for an investigational new indication to treat patients with relapsed or refractory (r/r) follicular lymphoma (FL).

Source: Press Releases

10.1.14 MEDTRONIC PLC

10.1.14.1 Business overview

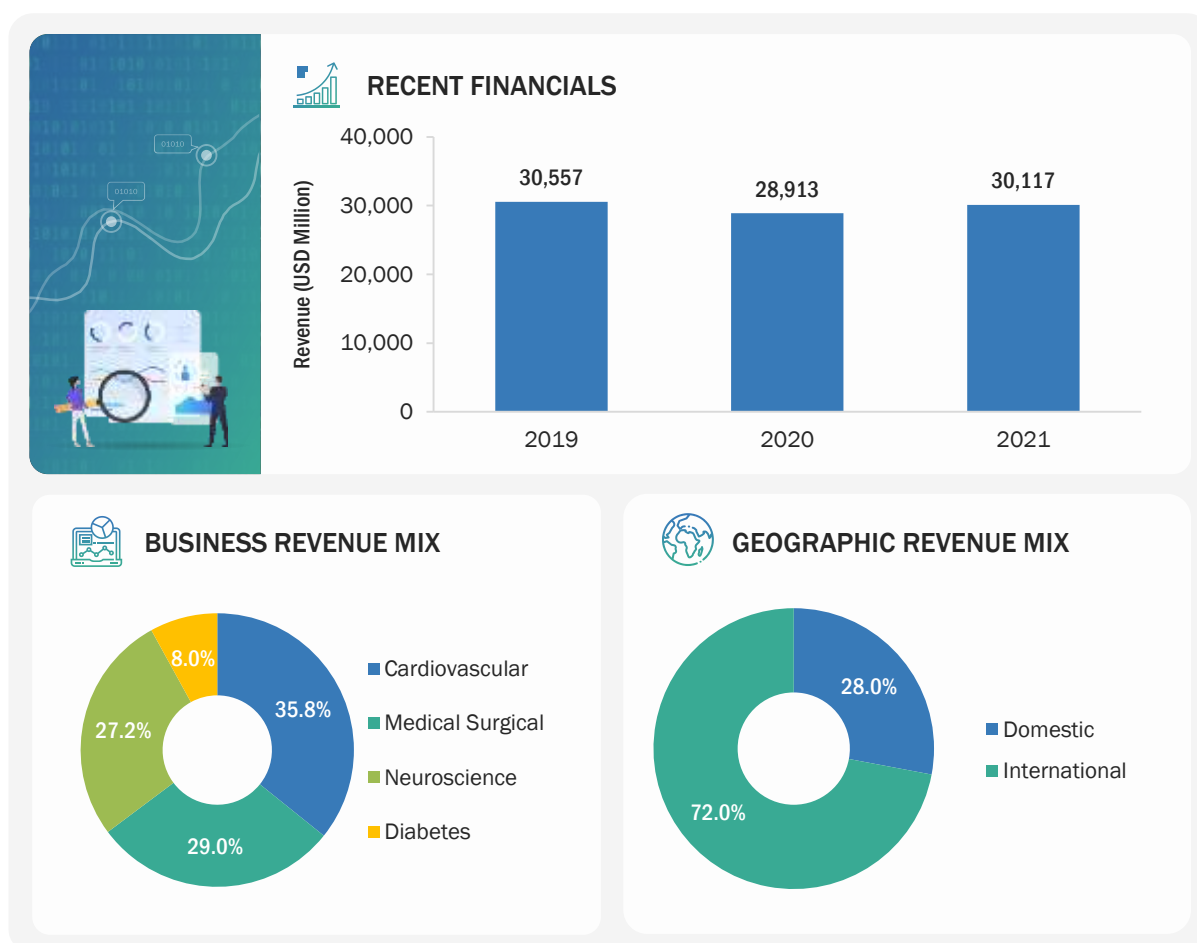
Medtronic plc is a global supplier of healthcare products used in various medical specialties, such as cardiovascular, neurological, spinal & orthopedic, urological & gynecological, and cardiac rhythm management. The major end users of the company's products include hospitals and surgical clinics. The company offers products under four major business segments: Cardiovascular, Medical Surgical, Neuroscience, and Diabetes Group.

The Cardiovascular Group offers several products under the Cardiac Rhythm & Heart Failure, Structural Heart & Aortic, and Coronary and Heart divisions. The Medical Surgical segment offers products under the Surgical Innovations and Respiratory, Gastrointestinal, and Renal divisions. The Neuroscience segment offers neuromodulation, Specialty Therapies, and Cranial & Spinal Technologies, which meet the demands of hospital settings and long-term care patients. The Diabetes segment offers diabetes and intensive insulin management services and solutions. The company offers regenerative medicine products through the Medical-Surgical business segment. Medtronic operates in 150 countries worldwide.

TABLE 184 MEDTRONIC PLC: BUSINESS OVERVIEW

Year Founded	1949
Country	US
City	Minnesota
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 48 MEDTRONIC PLC: COMPANY SNAPSHOT (2021)


Note: Information for the business & geographic revenue mix was unavailable for the Medical Surgical segment. Hence, these sections are provided for the overall company.

Source: Annual Reports and Company Website

10.1.14.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Tissue-engineered Products	Orthopedic or reconstructive bone grafting	- INFUSE BONE GRAFT - Grafton Demineralized Bone Matrix (DBM)	This product category offers products offering autograft replacement and aseptic processing technology.

Note: The company reported no significant developments in the regenerative medicine market between January 2020 and October 2022.

10.1.15 ORTHOCELL LTD.

10.1.15.1 Business overview

Founded in 2006 and headquartered in Australia, Orthocell Ltd. is a regenerative medicine company engaged in developing and distributing novel collagen medical devices and cellular therapies. These products focus on repairing and regenerating human tendons, bones, nerves, and cartilage-related defects.

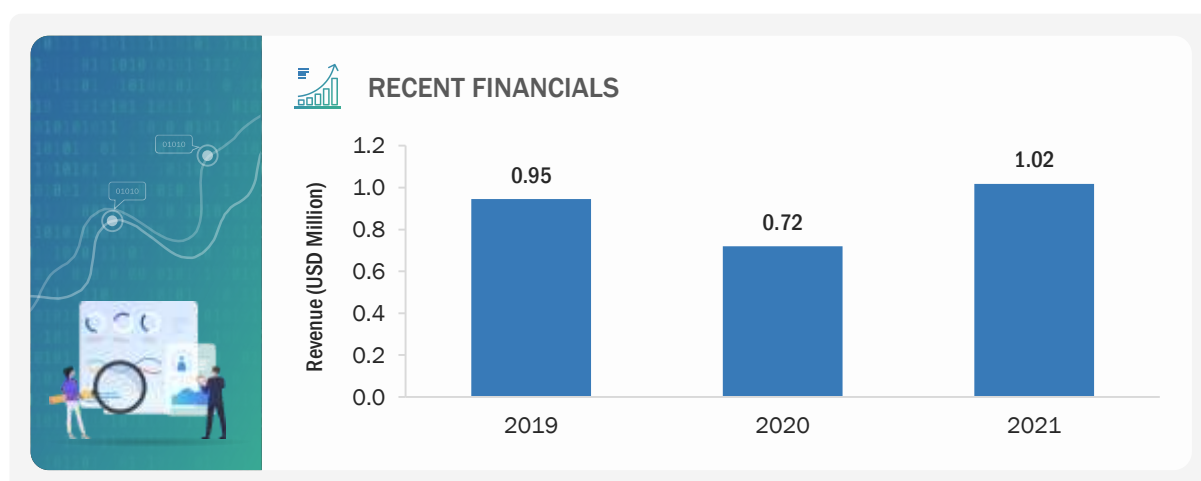
The company provides products under two product lines: CelGro and Ortho-ACI. The CelGro product line is derived from collagen medical devices, and the Ortho-ACI line is developed from autologous tenocytes.

TABLE 185 ORTHOCELL: BUSINESS OVERVIEW

Year Founded	2006
Country	Australia
City	Melville
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 49 ORTHOCELL LIMITED: COMPANY SNAPSHOT (2021)



Note: Information for the business & geographic revenue mix of the company was unavailable in the public domain. Hence, it has not been provided here.

Source: Annual Reports and Company Website

10.1.15.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Tissue-engineered Products	Bone and soft tissue repair	CelGro- Dental	This product category offers a non-reactive resorbable membrane that promotes enhanced bone and soft tissue repair.

Note: The company reported no significant developments in the regenerative medicine market between January 2020 and October 2022.

10.2 OTHER PLAYERS

10.2.1 MESOBLAST LTD.

Established	2004
Headquarters	Melbourne, Australia
Ownership	Public
Total Revenue (2021)	USD 7.46 Million
Business Overview	Mesoblast is a mesenchymal lineage cell technology platform provider that develops and commercializes innovative allogeneic cellular therapies. The company has more than 1,000 patents and patent applications in all major markets. Mesoblast has a global presence in the US, Australia, and Singapore.
Products Offered	▪ RYONCIL – RYONCIL is developed for the treatment of acute graft versus host disease (acute GVHD) and is in Phase 3 ▪ Remestemcel-L – This stem cell therapy is developed for severe acute respiratory distress syndrome (ARDS) due to COVID-19 infection and is in Phase 3 ▪ REVASCOR - This stem cell therapy is developed for chronic heart failure and is in Phase 3 ▪ MPC-06-ID - This stem cell therapy is developed for degenerative disc disease and is in Phase 3

Source: Company Website

10.2.2 BIORESTORATIVE THERAPIES, INC.

Established	2010
Headquarters	NewYork, US
Ownership	Private
Business Overview	BioRestorative Therapies develops and provides regenerative medicine and stem cell therapies. The company has a multi-product pipeline developing a clinical-stage spinal disc program. The company focuses on applications such as chronic back pain caused by disc degeneration and metabolic disorders, including obesity and diabetes.
Products Offered	▪ BRTX-100– BRTX-100is developed for the treatment of disc/spine associated diseases and is in Phase 1 ▪ THERMOSTEM – This stem cell therapy is developed for the treatment of metabolic disorders and is in the pre-clinical phase

Source: Company Website

10.2.3 PLURI INC.

Established	2001
Headquarters	Haifa, Israel
Ownership	Private
Business Overview	Established in 2001 and headquartered in Haifa, Israel, PLURI, Inc is a clinical-stage regenerative medicine firm providing placental cell-based products. The company focuses on inflammation, muscle injuries, and hematological disorders. The company has a presence in the US, Israel, and the EU, with over 120 patents globally.
Products Offered	- PLX-PAD – PLX-PAD is developed to treat hip fractures and COVID-19-associated acute respiratory distress syndrome (ARDS). The former is in Phase 3, whereas the latter is in Phase 2. - PLX-R18– This stem cell therapy is developed for the treatment of acute radiation syndrome (ARS) and has received an orphan drug designation

Source: Company Website

10.2.4 TEGOSCIENCE

TABLE 186 TEGOSCIENCE: BUSINESS OVERVIEW

Year Founded	2001
Country	South Korea
City	Seoul
Ownership	Public
Business overview	The company researches, develops, and markets cell therapy products.
Product	Holoderm and Kalorderm
Geographical Presence	Asia Pacific

Source: Company Website

10.2.5 ANTEROGEN CO., LTD.

TABLE 187 ANTEROGEN CO., LTD.: BUSINESS OVERVIEW

Year Founded	2000
Country	South Korea
City	Kyonggi-do
Ownership	Public
Business overview	The company is engaged in developing and distributing stem cell-based treatments for therapeutic and cosmetic purposes.
Product	Cupistem and Queencell
Geographical Presence	Asia Pacific

Source: Company Website

10.2.6 BLUEBIRD BIO

TABLE 188 BLUEBIRD BIO: BUSINESS OVERVIEW

Year Founded	2010
Country	US
City	Massachusetts
Ownership	Public
Business overview	bluebird bio is a clinical-stage biotechnology company involved in developing gene therapies for severe genetic diseases and cancer.
Product	Zynteglo
Geographical Presence	North America, Europe, and the Asia Pacific

Source: Company Website

10.2.7 STEMPEUTICS RESEARCH

TABLE 189 STEMPEUTICS RESEARCH: BUSINESS OVERVIEW

Year Founded	2006
Country	India
City	Bangalore
Ownership	Private
Business overview	Stempeutics Research is a late-stage life science firm that develops and commercializes novel therapeutics based on adult stem cells. The company has three novel products: Stempeucel, Stempeucare, and Stempeutron. The company offers stem cell therapy offerings through Stempeucel. Stempeucel patents have been granted in the US, China, the UK, Germany, France, Italy, Australia, New Zealand, South Africa, Japan, and Singapore.
Product	Stempeucel- This therapy is approved in India by DCGL for treating Critical Limb Ischemia (CLI).
Geographical Presence	Asia Pacific

Source: Company Website

10.2.8 SHENZHEN SIBIONO GENETECH CO., LTD.

TABLE 190 SHENZHEN SIBIONO GENETECH CO., LTD.: BUSINESS OVERVIEW

Year Founded	1998
Country	China
City	Shenzhen
Ownership	Private
Business overview	Shenzen SiBiono Genetech Co. Ltd. Is involved in developing gene therapy products.
Product	Gendicine
Geographical Presence	Asia Pacific

Source: Company Website

10.2.9 ASPECT BIOSYSTEMS

TABLE 191 ASPECT BIOSYSTEMS: BUSINESS OVERVIEW

Year Founded	2013
Country	Canada
City	Vancouver
Ownership	Private
Business overview	Aspect Biosystems is a biotechnology company pioneering the microfluidics 3D bioprinting of living human tissue.
Product	3D Bioprinted Meniscus Tissue, and 3D Bioprinted Pancreatic Tissue
Geographical Presence	North America and Europe

Source: Company Website

10.2.10 ATHERSYS, INC.

TABLE 192 ATHERSYS, INC.: BUSINESS OVERVIEW

Year Founded	1995
Country	US
City	Cleveland
Ownership	Public
Business overview	Athersys, Inc. focuses on research and development activities in regenerative medicine. Its clinical development programs treat neurological conditions, cardiovascular diseases, inflammatory and immune disorders, and pulmonary and other conditions.
Product	MultiStem
Geographical Presence	North America and Asia Pacific

Source: Company Website

11 APPENDIX

11.1 DISCUSSION GUIDE

- Q. 1.** What are your views on the growth prospects of the regenerative medicine market? What is the current scenario of this market, and at what rate is the market projected to grow in the coming five years (2022–2027)?
- Q. 2.** Kindly provide the key drivers, restraints, opportunities, and challenges impacting the growth of the regenerative medicine market.

PARAMETER	DESCRIPTION
Drivers	
Restraints	
Opportunities	
Challenges	

- Q. 3.** Kindly provide the market shares and projected growth rates for the regenerative medicine market product segments. What key factors will drive the fastest-growing market segment in the coming five years (2022–2027)? (Kindly validate the market segmentation and let us know if we are missing anything or if there is any overlap between these segments)

PRODUCT	MARKET SHARE (2021)	CAGR (2022–2027)
Gene Therapy		
Cell Therapy		
Tissue Engineering		

- Q. 4.** Kindly provide the market shares and projected growth rates of the following product segments in the cell therapy market. What key factors will drive the fastest-growing market segment in the coming five years (2022–2027)? (Kindly validate the market segmentation and let us know if we are missing anything or if there is any overlap between these segments)

PRODUCT	MARKET SHARE (2021)	CAGR (2022–2027)
Stem Cell Therapy		
Cell-based Immunotherapy Product		

Q. 5. Kindly provide the market shares and projected growth rates of the following product segments in the stem cell therapy market. What key factors will drive the fastest-growing market segment in the coming five years (2022-2027)? (Kindly validate the market segmentation and let us know if we are missing anything or if there is any overlap between these segments)

PRODUCT	MARKET SHARE (2021)	CAGR (2022-2027)
Stem Cell Therapy Products		
Cell Transplantations		

Q. 6. Kindly provide the market shares and projected growth rates of the following application segments in the regenerative medicine market. What key factors will drive the fastest-growing market segment in the coming five years (2022-2027)? (Kindly validate the market segmentation and let us know if we are missing anything or if there is any overlap between these segments)

APPLICATION	MARKET SHARE (2021)	CAGR (2022-2027)
Musculoskeletal Disorders		
Dermatology & Wound Care		
Cardiovascular Diseases		
Ophthalmology		
Other Applications		

Q. 7. Kindly provide the market shares and projected growth rates of the following regions in the regenerative medicine market.

REGION	MARKET SHARE (2021)	CAGR (2022-2027)
North America		
Europe		
Asia Pacific		
Latin America		
Middle East and Africa		

Q. 8. Provide market shares as well as projected growth rates of the following countries for North America in regenerative medicine.

COUNTRY	MARKET SHARE (2021)	CAGR (2022–2027)
US	%	%
Canada	%	%

Q. 9. Provide market shares and projected growth rates of the following countries for Europe in the regenerative medicine market.

COUNTRY/REGION	MARKET SHARE (2021)	CAGR (2022–2027)
Germany	%	%
France	%	%
UK	%	%
Rest of Europe	%	%

Q. 10. Provide market shares and projected growth rates of the following countries for the Asia Pacific in the regenerative medicine market.

COUNTRY/REGION	MARKET SHARE (2021)	CAGR (2022–2027)
China	%	%
Japan	%	%
India	%	%
Rest of Asia Pacific	%	%

Q. 11. Kindly validate the list of key market players in the regenerative medicine market and provide their market shares for 2021.

COMPANY NAME	MARKET SHARE (2021)
Novartis AG (Switzerland)	
Biogen (US)	
Gilead Sciences, Inc. (US)	
Sarepta Therapeutics, Inc. (US)	
Amgen, Inc. (US)	

11.2 KNOWLEDGESTORE: MARKETSandMARKETS' SUBSCRIPTION PORTAL

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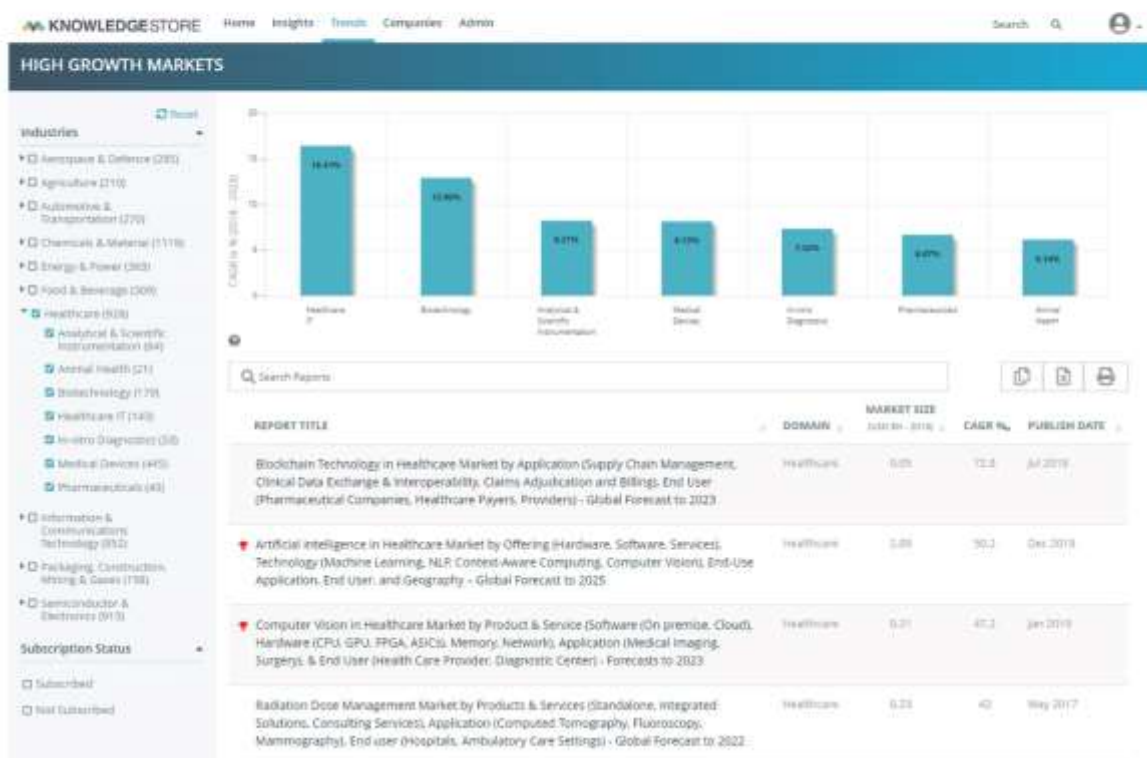
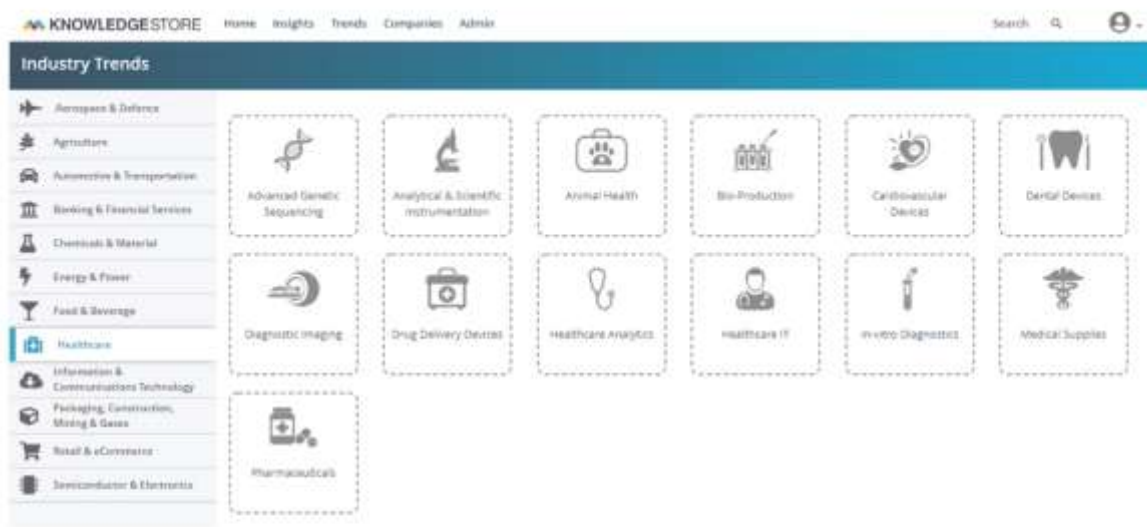
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11.3 CUSTOMIZATION OPTIONS

With the given market data, MarketsandMarkets offers customizations per the company's specific needs. The following customization options are available for the report:

Geographical Analysis

- Further breakdown of the rest of Europe's regenerative medicine market by country
- Further breakdown of the rest of the Asia Pacific regenerative medicine market by country
- Further breakdown of the Latin American and Middle Eastern & African regenerative medicine market by country

Company Information

- Detailed analysis and profiling of additional market players (Up to five)

11.4 RELATED REPORTS

SR. NO.	REPORT TITLE	PUBLISHED DATE
1	ADVANCED WOUND CARE MARKET – GLOBAL FORECAST TO 2026 By Product (Dressings (Foam, Hydrocolloid, Alginate, Film), NPWT, Debridement Devices, Grafts, Matrices, Topical Agents), Wound Type (Surgical, Traumatic, Ulcers, Burns), End User (Hospital, Homecare) https://www.marketsandmarkets.com/Market-Reports/advanced-wound-care-market-88705076.html	October 2021
2	WOUND CARE BIOLOGICS MARKET - GLOBAL FORECAST TO 2023 By Product (Biological Skin Substitutes (Biosynthetic Products), Topical Agents), Wound Type (Diabetic Foot Ulcer, Pressure Ulcers, Surgical & Traumatic Wound), End User (Hospitals, Burn Centers) https://www.marketsandmarkets.com/Market-Reports/wound-care-biologic-market-259878425.html	June 2022
3	STEM CELL THERAPY MARKET – GLOBAL FORECAST TO 2027 By Type (Allogeneic, Autologous), Therapeutic Application (Musculoskeletal, Wound & Injury, CVD, Surgery, and aGVHD), Cell Source (Adipose tissue, Bone Marrow, Neural, Embryo/Cord Blood derived, iPSCs) https://www.marketsandmarkets.com/Market-Reports/stem-cell-technologies-and-global-market-48.html	September 2022
4	GENE THERAPY MARKET – GLOBAL FORECAST TO 2025 By Vectors (Non-viral (Oligonucleotides)), Viral (Retroviral, Adeno-associated), Indication (Cancer, Neurological, Hepatological Diseases, Duchenne Muscular Dystrophy), Delivery Method (In Vivo, Ex Vivo), Region https://www.marketsandmarkets.com/Market-Reports/gene-therapy-market-122857962.html	July 2022

11.5 AUTHOR DETAILS

Rajiv Kalia

*Associate Director
Healthcare*

Rajiv has over 15 years of experience in market research and consulting in the healthcare vertical. He has worked across diverse areas in the pharma/biotech, medical device, and healthcare IT industries.

He has been instrumental in tracking high-growth and niche markets in the healthcare vertical. He has worked on several projects requiring expertise in market estimations, primary & secondary research techniques, strategic & competitive assessment of companies, investment opportunity analysis, and product opportunity evaluations. His current role involves tracking industry developments to develop research areas in collaboration with clients.

Rajiv has diligently worked on addressing clients' problems regarding market and product strategy and formulated detailed solutions in terms of research and consulting outputs. He has completed his master's degree in Pharmacy with a major in Pharmacology from Manipal University and a Postgraduate Diploma in Financial Management.

Contributors

Ninad Nanadikar

Senior Research Manager

Tanvi Sapatnekar

Senior Research Analyst

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